FLUCTUATIONS OF THE FREE ENERGY FOR THE PURE ISING p-SPIN MODEL AT CRITICAL TEMPERATURE

ABSTRACT. We study the free energy of the pure Ising p-spin model at its static critical inverse temperature, for every fixed integer $p \geqslant 3$. The annealed-normalized partition function converges in probability to $1/2$. With a deterministic centering defined by a planted-band probability, its first fluctuations occur on scale $N^{-1/2}$ and have a spectrally positive stable component of index one. For distinct unordered p-tuples, the limit also has a Gaussian component precisely when $p = 3$. For the ordered-tensor covariance NR^p, additional Gaussian contributions occur at $p = 3, 4$. The corresponding total free-energy fluctuations have scale $N^{-1/2}$, and the free energy per spin has scale $N^{-3/2}$. The proof combines quantitative replica symmetry in an auxiliary field, joint local limits for planted bands, concentration of empirical band thresholds, and a Gaussian planting identity. The three-spin argument uses compact characteristic observables and weighted Gaussian divergence estimates. Scalar and finite cavity certificates are supplied with reproducible source code.

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1. Introduction

The fluctuations of the free energy distinguish mean-field spin glasses whose limiting pressures can have the same form. In the pure Ising p-spin model, the critical endpoint raises a further question: how does the partition function fluctuate when the annealed formula first ceases to describe the thermodynamic pressure? We address this question for each fixed integer $p \geqslant 3$, at zero external field. The parameter p remains fixed throughout the limit in the system size.

The main result identifies the critical half-partition limit and the next fluctuation scale. The limiting law contains a spectrally positive stable component of index one; low-order covariance terms contribute an additional Gaussian component in the three-spin model and, for ordered tensors, also in the four-spin model. The exact finite-size covariance therefore enters both the centering and the Gaussian variance. We give the model conventions and the deterministic centering before stating the theorem. A detailed comparison with earlier fluctuation, phase-transition, and inference results follows in Section 1.5.

1.1. Models and normalization.

Fix an integer $p \geqslant 3$ throughout the limit $N \to \infty$. All finite-size assertions involving a hard band are understood for $N \geqslant N_0(p, \delta_p)$, so that the band contains allowed magnetizations; the distinct-index model is used for $N \geqslant p$. Let $\Sigma_N = \{-1, 1\}^N$, with uniform probability measure μ_N, and let

$$R(\sigma, \tau) = \frac{1}{N} \sum_{i=1}^{N} \sigma_i \tau_i.$$

Our principal model uses distinct unordered p-tuples:

$$H_N^{\mathrm{d}}(\sigma) = \sqrt{\frac{N}{\binom{N}{p}}} \sum_{\substack{I \subseteq [N] \\ |I| = p}} g_I \prod_{i \in I} \sigma_i, \qquad g_I \overset{\text{iid}}{\sim} N(0,1). \tag{1.1}$$

Thus $\operatorname{Var} H_N^{\mathrm{d}}(\sigma) = N$. The conventional coefficient $\sqrt{p!/N^{p-1}}$ gives

$$H_N^{\mathrm{std}} = \sqrt{c_{N,p}}\, H_N^{\mathrm{d}}, \qquad c_{N,p} = \frac{(N)_p}{N^p}, \qquad (N)_p = N(N-1)\cdots(N-p+1).$$

We also treat the ordered-tensor convention

$$H_N^{\mathrm{o}}(\sigma) = N^{-(p-1)/2} \sum_{i_1, \ldots, i_p = 1}^{N} g_{i_1 \ldots i_p} \sigma_{i_1} \cdots \sigma_{i_p}, \qquad \mathbb{E}\, H_N^{\mathrm{o}}(\sigma) H_N^{\mathrm{o}}(\tau) = N R(\sigma, \tau)^p. \tag{1.2}$$

All Gaussian variables in each model are independent standard normals. The models have the same thermodynamic critical inverse temperature, but their finite-size fluctuation laws need not coincide.

For any chosen convention, define

$$Z_N(\beta) = \mathbb{E}_{\sigma \sim \mu_N}\, e^{\beta H_N(\sigma)}, \qquad F_N(\beta) = \log Z_N(\beta), \qquad L_N(\beta) = \frac{Z_N(\beta)}{\mathbb{E}\, Z_N(\beta)}.$$

The total log partition function F_N is the free energy used here; the free energy per spin is F_N/N. A counting-measure partition function adds $N \log 2$ to F_N, with no change to the centered laws. The physical convention $-\beta^{-1} \log Z_N$ follows by multiplying the centered limit by $-\beta_c^{-1}$.

1.2. **The static critical point.** The critical inverse temperature is the true annealed endpoint:

$$\beta_c = \sup\left\{\beta \geqslant 0 : \lim_{N\to\infty} \frac{1}{N}\,\mathbb{E}\,F_N(\beta) = \frac{\beta^2}{2}\right\}.$$

It is the first RS–1RSB critical point characterized by Zhou [55], with the same covariance normalization as (1.2). Let $q = q_p \in (0,1)$ and $t = t_p > 0$ be the unique positive critical solution

$$\begin{aligned} q &= \mathbb{E}\tanh^2(t + \sqrt{t}\,G), \\ 2\,\mathbb{E}\log\cosh(t+\sqrt{t}\,G) - t &= \frac{p-1}{p}\,tq, \\ \beta_c^2 &= \frac{t}{pq^{p-1}}, \qquad G \sim N(0,1). \end{aligned} \tag{1.3}$$

In the rest of the manuscript $\beta = \beta_c$, unless specified otherwise. Put

$$\begin{aligned} \xi(r) &= \beta_c^2 r^p, \qquad \theta(q) = q\xi'(q) - \xi(q), \\ v_p &= \mathrm{Var}\big(\log\cosh(t+\sqrt{t}\,G)\big) - \theta(q), \qquad \kappa_p = (2\pi v_p)^{-1/2}. \end{aligned} \tag{1.4}$$

The proof establishes $v_p > 0$ and the equivalent expression $v_p = \int_0^1 \xi(u_\rho)\,d\rho$, with u_ρ defined in the joint field argument.

1.3. **A canonical deterministic centering.** Use the actual finite-N covariance of the chosen model, including temperature:

$$X_N(\sigma) = \beta_c H_N(\sigma) = g\cdot v_\sigma, \qquad C_N(\sigma,\tau) = v_\sigma\cdot v_\tau, \qquad W_\sigma = e^{X_N(\sigma) - C_N(\sigma,\sigma)/2}.$$

For a uniform independent seed $a \in \Sigma_N$, its Gaussian planting shift is $P_a g = g + v_a$. Choose any sufficiently small fixed band width $\delta_p > 0$, as specified in the estimates below, and set

$$\begin{aligned} \widehat{B}_{N,a}(g) &= d_p\ \mathbb{E}_{\sigma\sim\mu_N}\, e^{X_N(\sigma;g) + C_N(\sigma,a) - C_N(\sigma,\sigma)/2}\mathbb{1}_{\{|R(\sigma,a)-q_p|\leqslant\delta_p\}}, \\ d_p &= \begin{cases} 1, & p \text{ odd}, \\ 2, & p \text{ even}. \end{cases} \end{aligned} \tag{1.5}$$

For even p, the factor two accounts exactly for the reflected negative band. Define

$$a_N = \mathbb{P}_{g,a}\{\sqrt{N}\,\widehat{B}_{N,a}(g) \leqslant 1\}. \tag{1.6}$$

This is a deterministic finite Gaussian probability. Gauge invariance allows the seed to be fixed to the all-plus pattern in its definition. It does not use the free energy that is being centered. We prove $a_N \to 1/2$; changing the sufficiently small fixed band width changes a_N by $o(N^{-1/2})$.

1.4. **Main results.** Let S_κ have the probability law

$$\begin{aligned} \mathbb{E}\,e^{izS_\kappa} &= \exp\left\{\kappa\int_0^\infty \frac{e^{izy} - 1 - izy\mathbb{1}_{\{y\leqslant 1\}}}{y^2}\,dy\right\} \\ &= \exp\left\{-\frac{\pi\kappa}{2}|z| + i\kappa z(1 - \gamma_{\mathrm{E}} - \log|z|)\right\}, \qquad z \neq 0. \end{aligned} \tag{1.7}$$

At zero the characteristic function is one. This specifies both the stable law and its location convention. Its Lévy measure is $\kappa y^{-2}\mathbb{1}_{\{y>0\}}\,dy$; *spectrally positive* refers to the jumps, and does not mean that S_κ is a positive random variable.

Theorem 1.1 (Critical fluctuations for each fixed $p \geqslant 3$)**.** *For either distinct-index convention above, at $\beta = \beta_c$,*

$$L_N \longrightarrow \frac{1}{2} \qquad \text{in probability,} \tag{1.8}$$

$$\sqrt{N}(L_N - a_N) \Longrightarrow Y_p, \qquad Y_p = S_{\kappa_p} + \tau_p G_0, \tag{1.9}$$

$$\sqrt{N}\big(F_N - \log \mathbb{E}\, Z_N - \log a_N\big) \Longrightarrow 2Y_p. \tag{1.10}$$

Here $G_0 \sim N(0,1)$ is independent of S_{κ_p}, and

$$\tau_p^2 = \begin{cases} 3\beta_c^4/4, & p = 3, \\ 0, & p \geqslant 4. \end{cases}$$

For the ordered-tensor model (1.2), the same statements hold, with its own center (1.6), and

$$\tau_p^2 = \begin{cases} 15\beta_c^4/8, & p = 3, \\ 3\beta_c^2/4, & p = 4, \\ 0, & p \geqslant 5. \end{cases}$$

In the variance-N distinct and ordered models, $\log \mathbb{E}\, Z_N = N\beta_c^2/2$. In the conventional distinct model, $\log \mathbb{E}\, Z_N = N c_{N,p} \beta_c^2/2$, exactly.

Corollary 1.2 (Median centering)**.** *Let m_N be any deterministic median of $F_N(\beta_c)$. For each model in Theorem 1.1,*

$$\sqrt{N}(F_N - m_N) \Longrightarrow 2\big(Y_p - \operatorname{med}(Y_p)\big).$$

Equivalently, for the free energy per spin $f_N = F_N/N$,

$$N^{3/2}\left(f_N - \frac{m_N}{N}\right) \Longrightarrow 2\big(Y_p - \operatorname{med}(Y_p)\big).$$

Proof. The characteristic function of Y_p has modulus $\exp\{-\pi\kappa_p|z|/2 - \tau_p^2 z^2/2\}$. Fourier inversion therefore gives a real-analytic nonnegative density. It cannot vanish on a nonempty interval, since analyticity would make it identically zero. Its distribution function is continuous and strictly increasing, so its median is unique. Convergence in distribution in (1.10) implies convergence of the corresponding medians, by bounding a median between any two continuity points on opposite sides of the limiting median. Subtract those medians in (1.10). ∎

Corollary 1.3 (A consequence for the variance)**.** *For every model in Theorem 1.1,*

$$\lim_{N\to\infty} N \operatorname{Var}(F_N) = +\infty.$$

This statement does not identify an asymptotic equivalent for the variance.

Proof. Let F_N' be an independent copy. The theorem gives $\sqrt{N}(F_N - F_N') \Rightarrow 2(Y_p - Y_p')$. The difference of the two stable components has characteristic function $e^{-\pi\kappa_p|z|}$, so it is a nondegenerate Cauchy law; adding an independent Gaussian preserves its infinite second moment. For each M, apply weak convergence to the bounded continuous function $x \mapsto \min(x^2, M)$. Since $\mathbb{E}[N(F_N - F_N')^2] = 2N \operatorname{Var}(F_N)$, first taking the lower limit in N, and then $M \to \infty$, proves the claim. ∎

Centering and moments. The deterministic probability (1.6) fixes the location of the limiting law without requiring a finite-size expansion of a_N. Corollary 1.2 gives the corresponding median-centered formulation. The stable limit has infinite variance. Expectation centering and an asymptotic equivalent for $\operatorname{Var}(F_N)$ require additional tail information; they are not consequences of the weak convergence proved here.

1.5. **Related work.**

1.5.1. *The variational framework and the static transition.* The mean-field model of Sherrington and Kirkpatrick [47] and Parisi's order-parameter construction [42,43] initiated the study of the pressure and Gibbs structure considered here. Gaussian interpolation gave the thermodynamic-limit argument of Guerra and Toninelli [30] and Guerra's replica-symmetry-breaking bound [29]. The extended cavity variational principle of Aizenman, Sims, and Starr [2], Talagrand's proof of the Parisi formula [52], and Panchenko's extension to general mixed interactions [41] provide the rigorous variational setting. In particular, the last extension includes odd interactions. We use the standard conventions and methods developed in the monographs [40, 53, 54].

For higher-spin interactions, Gardner's replica analysis [28] describes a first transition to one-step replica symmetry breaking and a further instability at lower temperature. The exact first static transition for each integer $p \geqslant 3$ is characterized by Zhou [55]. Its two scalar contact equations, in covariance convention $N\beta^2 R^p$, are precisely (1.3). The strict stability consequences used in this paper are derived in Section 2; the fluctuation law does not follow from the thermodynamic characterization alone. Zhou's subsequent preprint [56] announces the remaining Gardner phase diagram; its first version states that the appendix proofs await final authorial verification and revision. Those lower-temperature conclusions are not needed here.

There are several distinct notions of stability in these arguments. The de Almeida–Thouless condition originated as a local criterion for the SK replica-symmetric solution [20]. Strict convexity of the Parisi functional and uniqueness of its minimizer were proved by Auffinger and Chen [3]. Directional-derivative and first-order criteria are developed in [12, 31]. We retain both the local stability margin and the global contact-sign condition in the auxiliary-field argument. A stationary scalar solution, a positive contact point, and a positive support point of the minimizing Parisi measure play different roles.

1.5.2. *High-temperature Ising fluctuations and covariance conventions.* Aizenman, Lebowitz, and Ruelle [1] proved order-one Gaussian fluctuations of the total SK free energy in the high-temperature phase. Comets and Neveu [19] developed the stochastic-calculus approach. Bovier, Kurkova, and Löwe [7] proved higher-spin Gaussian fluctuation theorems and explicitly distinguished distinct-index and ordered-tensor Hamiltonians, whose thermodynamic limits agree while their fluctuation scales can differ. The distinction in Theorem 1.1 is therefore essential already in comparison with the high-temperature literature. Their ordered-tensor theorem is stated for even p; the broader covariance extension discussed in their appendix is conjectural.

The most directly comparable published refinement is Bovier–Schertzer [8]. In the variance-N distinct-index convention, their leading theorem gives

$$N^{(p-2)/2}\big(F_N(\beta) - \log \mathbb{E}\, Z_N(\beta)\big) \Longrightarrow \mathcal{N}\left(0, \frac{\beta^4 p!}{2}\right)$$

for a strict temperature range below the stated Talagrand bound. After subtracting a random center formed from the squared couplings, they identify a second, parity-dependent Gaussian scale. The temperature bound comes from the truncated-second-moment analysis of Talagrand [51]. These fluctuation theorems do not include equality at the true static endpoint used in this paper. Kurkova [34] studies free-energy and overlap fluctuations in the high-temperature p-spin and Hopfield models. Knöpfel and Löwe [33] treat non-Gaussian couplings and exhibit the role of their fourth moment in the leading variance. Our disorder remains Gaussian throughout.

Dey and Wu [24] develop hypergraph-counting and multivariate-Stein methods for mixed models with weak external fields. Their zero-field higher-order theorem concerns the cluster residual after extracting the product of hyperbolic-cosine factors; the temperature condition is $\beta < \beta_f$ strictly, where β_f is their second-moment threshold. These results help explain the parity-sensitive polynomial terms but do not supply an endpoint theorem. Banerjee and Belius [5] obtain order-one Gaussian fluctuations in a high-temperature mixed model with a nonzero two-spin component. That assumption is explicit and cannot be removed by setting the two-spin coefficient to zero in their conclusion.

1.5.3. *Auxiliary fields, planting, and band geometry.* The quantitative field method is closely related to Kusuoka–Nakajima [35]. Their SK theorem gives $O(N^{-1})$ mean-square overlap concentration and $O(N^{-1})$ normalized pressure error, uniformly on compact subsets of the strict AT region with positive deterministic field. Section 3 develops the higher-spin specialization needed here. Chen's two-dimensional Guerra–Talagrand bound [12] provides the interpolation framework. Its matrix-positivity and covariance-convexity hypotheses are retained; for a pure odd covariance, the positive-magnetization restriction is proved explicitly.

Chen, Dey, and Panchenko [14] prove extensive free-energy variance in a field and a univariate Gaussian limit when odd interactions of order at least three are absent. Their covariance/Stein route is related to the Gaussian identities in [10, 11]. Section 4 establishes the joint theorem for independent field patterns and treats pure odd powers. The relevant asymptotic independence thus comes from the cross-pattern calculation in this paper.

The planting identity has a direct statistical interpretation. For a Gaussian tensor with a Rademacher spike, the likelihood ratio is an annealed-normalized Ising partition function. Chen [13] identifies the detection threshold with the pure model's thermodynamic annealed endpoint and proves total-variation limits strictly below and strictly above it. In that paper's convention, the exponent is $\beta H_N/\sqrt{2}$, so its critical parameter equals $\sqrt{2}\beta_c$ here. The equality case is excluded from those two detection statements. Related variational and conditioned-second-moment analyses appear in [37, 44]; Chen, Handschy, and Lerman [15] treat centered bounded priors and multiple independent spikes. The exact critical likelihood-ratio law requires the additional finite-size estimates developed below.

The generalized TAP free energy of Chen, Panchenko, and Subag [17] gives a thermodynamic description of replicated Ising bands. Gamarnik, Jagannath, and Kızıldağ [27] prove shattering for sufficiently large p in a specified high-temperature interval, and El Alaoui [26] extends the range toward the predicted shattering threshold. The TAP results concern thermodynamic band corrections, while the cited shattering theorems concern Gibbs geometry in specified large-p regimes. The present proof fixes an arbitrary $p \geqslant 3$ and needs local limits, weighted densities, and polynomial moment estimates at its exact static critical point.

1.5.4. *The critical REM benchmark.* Derrida's random energy model [21, 22] and Ruelle's Poisson formulation [45] provide a useful comparison. In the variance-N REM, let $\beta_{\mathrm{REM}} = \sqrt{2\log 2}$ and $L_N^{\mathrm{REM}} = Z_N^{\mathrm{REM}}/\,\mathbb{E}\,Z_N^{\mathrm{REM}}$. The published Theorem 1.6(i) of [7], equation (1.30), gives

$$a_N^{\mathrm{REM}} = \frac{1}{2} - \frac{\log(4\pi N \log 2)}{4\sqrt{\pi N \log 2}}, \qquad \sqrt{4\pi N \log 2}\,(L_N^{\mathrm{REM}} - a_N^{\mathrm{REM}}) \Longrightarrow S_1,$$

where S_1 uses the compensation convention in (1.7). Thus the half limit and index-one stable component have a precise independent-energy precedent. For fixed p, the informative-band geometry and finite covariance corrections must also be controlled. In particular, we do not transfer the REM expansion of its center to a_N in (1.6). Ben Arous, Bogachev, and Molchanov [6] give a broader limit theory for sums of independent random exponentials, while

Subag–Zeitouni [49] study freezing and decorated Poisson processes. Neither framework alone justifies independence of fixed-p Ising configurations or cluster weights.

1.5.5. *Critical SK and spherical comparisons.* For the two-spin Ising model, the critical variance problem is discussed in [54]. Chen and Lam [16] prove a squared-logarithmic upper bound. Recent work treats the approach from high temperature [23] and sharpens endpoint upper and lower bounds [46]. Du and Huang's critical preprint [25] identifies the variance coefficient and a Gaussian limit. In their covariance convention $NR^2/2$, at $\beta = 1$,

$$\operatorname{Var}(F_N) = \frac{1}{6}\log N + O(1), \qquad \frac{F_N - N/4 + (\log N)/12}{\sqrt{(\log N)/6}} \Longrightarrow \mathcal{N}(0,1).$$

This is a two-spin transition with a different fluctuation scale from Theorem 1.1. More generally, covariance interpolation relates fluctuation bounds to disorder chaos [9]; a variance bound alone does not identify the critical distribution.

For spherical two-spin models, Baik and Lee [4] prove the high- and low-temperature fluctuation theorems, with Gaussian and Tracy–Widom limits in the respective regimes. Landon [36] proves the critical Gaussian theorem; for the total logarithm and covariance $NR^2/2$, his Gaussian limit uses the centering and scale in the displayed Gaussian limit above. Johnstone, Klochkov, Onatski, and Pavlyshyn [32] identify the Gaussian/Tracy–Widom crossover in a critical window. Collins-Woodfin and Le [18] obtain a related transition in the bipartite spherical model. These quadratic models admit spectral representations that are unavailable for a general fixed higher-spin Ising Hamiltonian.

For pure spherical $p \geqslant 3$, Subag and Zeitouni [50] prove convergence of the extremal process of stationary values and a centered ground-state limit. Subag [48] identifies low-temperature Gibbs bands, a logarithmic correction to the free energy, and tightness after centering. These results concern sufficiently low temperature; they do not provide the static critical law for the Ising cube. Throughout these comparisons, a theorem about stationary points, an estimate at fixed low temperature, and a theorem at critical temperature are kept distinct.

1.5.6. *The OpenAI manuscripts and the scope of the present result.* The two specific OpenAI manuscripts [38, 39] announce a limiting law and a fluctuation scale for the zero-field Gaussian SK model at fixed $\beta > 1$. They concern the two-spin low-temperature problem. Their title pages and stated regimes were checked, and they are cited as research manuscripts. The present proof does not invoke either as a higher-spin critical theorem.

The comparison literature therefore supplies the variational and Gaussian-interpolation foundations, established high-temperature fluctuation mechanisms, and exact benchmarks in related models. The critical-band local limits, threshold concentration, finite covariance corrections, and the exceptional three- and four-spin Palm estimates used in Theorem 1.1 are supplied in the sections that follow. The stable/Gaussian decomposition in the conclusion is a factorization of the limiting law; no stronger finite-size component decomposition is asserted.

1.6. **Organization of the proof.** Section 2 records the critical scalar data and proves strict stability of the auxiliary field. Sections 3 and 4 give the quantitative field estimates and the joint Gaussian limit for independent field patterns. The planting and truncation argument in Section 5 then establishes the critical half limit.

Sections 6 and 7 provide the three-spin band moments and the cross-band estimates. Monotone Gaussian projection upgrades the band central limit theorems to local limits in Section 8. Together with empirical-threshold concentration and oscillatory tail bounds, these identify the band term in the characteristic equation. Section 9 tracks the finite covariance

corrections, and Section 10 supplies the weighted overlap and density estimates used in the exceptional low-order cases.

The final step is the exact planting identity

$$\mathbb{E}[L_N f(L_N)] = \mathbb{E}_{a,g}\, f(L_N(P_a g)).$$

For $p \geqslant 5$, the low-overlap remainder is negligible at the required scale. Section 11 identifies the four-spin constant mode and removes an auxiliary Gaussian mode in the distinct-index model. Section 12 treats the nonlinear three-spin correction using compact characteristic observables. The positivity estimate

$$V_a \geqslant \frac{T_a^2}{2L_N} \quad \Longrightarrow \quad L_N + \widehat{B}_a + V_a + \vartheta T_a \geqslant \widehat{B}_a + \frac{1}{2}L_N, \qquad 0 \leqslant \vartheta \leqslant 1,$$

keeps these observables away from the singular lower boundary. Gaussian divergence and weighted two-band densities remove the odd term; the quadratic term determines the Gaussian variance. Section 13 completes the main theorem, and Appendix A supplies the scalar certificate.

2. The critical scalar contact

Fix an integer $p \geqslant 3$. Initially use the exact covariance convention

$$H_N(\sigma) = N^{-(p-1)/2} \sum_{i_1,\dots,i_p=1}^{N} g_{i_1\cdots i_p}\sigma_{i_1}\cdots\sigma_{i_p}, \qquad \mathbb{E}\, H_N(\sigma)H_N(\tau) = NR(\sigma,\tau)^p,$$

where $\sigma \in \Sigma_N = \{-1,1\}^N$ and $R(\sigma,\tau) = N^{-1}\sum_i \sigma_i\tau_i$. The reference measure μ_N on Σ_N is uniform. Let β_c be the true thermodynamic transition out of the annealed phase. Set

$$\begin{aligned} X_N &= \beta_c H_N, & \xi(r) &= \beta_c^2 r^p, \\ W_\sigma(g) &= \exp\{X_N(\sigma) - N\xi(1)/2\}, & L_N(g) &= \mathbb{E}_{\sigma\sim\mu_N} W_\sigma(g). \end{aligned}$$

Thus $L_N = Z_N(\beta_c)/\,\mathbb{E}\, Z_N(\beta_c)$, where $Z_N = \mathbb{E}_{\mu_N} e^{\beta_c H_N}$, and $\mathbb{E}\, L_N = 1$.

Write

$$\Gamma(t) = \mathbb{E}\tanh^2(t+\sqrt{t}Z), \qquad \Psi(t) = \mathbb{E}\log\cosh(t+\sqrt{t}Z), \qquad Z \sim N(0,1),$$

and $\theta(q) = q\xi'(q) - \xi(q)$. The pure-spin critical-point characterization of Zhou gives a unique nonzero contact $q_* = q \in (0,1)$, with $t_* = t = \xi'(q)$, such that

$$\Gamma(t) = q, \tag{2.1}$$

$$C(q) := \Psi(\xi'(q)) - \frac{\xi'(q)}{2} - \frac{\theta(q)}{2} = 0. \tag{2.2}$$

For $c > 0$, write

$$\xi_c(u) = cu^p, \qquad D_c(u) = \Gamma(\xi_c'(u)) - u, \qquad C_c(u) = \Psi(\xi_c'(u)) - \frac{\xi_c'(u)}{2} - \frac{u\xi_c'(u) - \xi_c(u)}{2}.$$

The existence and uniqueness of the positive static contact $(c_0, q_0) = (\beta_c^2, q)$ are supplied by Zhou [55], Section 2.2 and Theorem 1. The following argument derives the strict inequalities needed below from Zhou's convexity Lemma 6.

Lemma 2.1 (Strict static contact and continuation of the larger root)**.** *Fix an integer $p \geqslant 3$, and let $c_0 > 0$, $q_0 \in (0,1)$ satisfy*

$$D_{c_0}(q_0) = 0, \qquad C_{c_0}(q_0) = 0.$$

Set $t_0 = \xi'_{c_0}(q_0)$. Then

$$\lambda_0 := \xi''_{c_0}(q_0)\Gamma'(t_0) < 1, \qquad D_{c_0}(u) < 0 \quad (u > q_0),$$

and

$$C_{c_0}(u) < 0 \qquad (0 < u \leqslant 1,\ u \neq q_0).$$

There are precisely two positive zeros of D_{c_0}, and q_0 is the larger one.

Moreover, on a sufficiently small closed interval $I \subset (0,\infty)$ containing c_0 in its interior, the larger root continues to a smooth function $q(c) \in (0,1)$. With

$$t(c) = cpq(c)^{p-1}, \qquad \lambda(c) = cp(p-1)q(c)^{p-2}\Gamma'(t(c)),$$

one has

$$\sup_{c\in I} \lambda(c) < 1, \qquad D_c(u) < 0 \quad (c \in I,\ u > q(c)).$$

For every fixed $\varepsilon > 0$, the latter gap is uniform on $\{(c,u) : c \in I,\ q(c) + \varepsilon \leqslant u \leqslant 1\}$, whenever this set is nonempty. On this branch,

$$\frac{dq}{dc} = \frac{pq(c)^{p-1}\Gamma'(t(c))}{1-\lambda(c)}, \qquad \frac{dt}{dc} = \frac{pq(c)^{p-1}}{1-\lambda(c)}, \qquad \frac{d}{dc}C_c(q(c)) = \frac{q(c)^p}{2}. \tag{2.3}$$

Proof. First record the scalar identities and their normalization. If $Y_s = s + \sqrt{s}\,Z$, its density is the density of $\sqrt{s}\,Z$ multiplied by $e^{x-s/2}$. Symmetrizing this identity shows that, for every bounded even function a,

$$\mathbb{E}[a(Y_s)\tanh Y_s] = \mathbb{E}[a(Y_s)\tanh^2 Y_s].$$

In particular, $\mathbb{E}\tanh Y_s = \Gamma(s)$. Gaussian differentiation gives

$$\Psi'(s) = \mathbb{E}\left[\tanh Y_s + \frac{1}{2}\operatorname{sech}^2 Y_s\right] = \frac{1+\Gamma(s)}{2}.$$

Writing $v = \tanh Y_s$, the same differentiation and the preceding identity with $a(x) = \operatorname{sech}^2 x$ give

$$\begin{aligned}\Gamma'(s) &= \mathbb{E}\left[2v(1-v^2) + (1-v^2)(1-3v^2)\right] \\ &= \mathbb{E}(1-v^2)^2 = \mathbb{E}\operatorname{sech}^4(s+\sqrt{s}\,Z) > 0.\end{aligned}$$

All differentiations are justified by bounded derivatives of tanh and $\log\cosh$, or by the Gaussian heat semigroup. Since $\Psi(0) = 0$, these identities imply

$$\mathbb{E}\tanh(s+\sqrt{s}\,Z) = \Gamma(s), \qquad \Gamma'(s) = \mathbb{E}\operatorname{sech}^4(s+\sqrt{s}\,Z), \qquad 2\Psi(s) - s = \int_0^s \Gamma(v)\,dv. \tag{2.4}$$

The estimate $\tanh^2 x = x^2 + O(x^4)$, with a global bound for the remainder, yields $\Gamma(s) = s + O(s^2)$ as $s \downarrow 0$. Consequently

$$C_c(u) = -\frac{c(p-1)}{2}u^p + O(u^{2p-2}) < 0 \quad \text{for all sufficiently small } u > 0. \tag{2.5}$$

Here $2p-2 > p$ for every $p \geqslant 3$; the expansion is uniform when c ranges over a compact subset of $(0,\infty)$. Differentiating C_c using (2.4) gives the exact identity

$$C'_c(u) = \frac{\xi''_c(u)}{2}D_c(u). \tag{2.6}$$

For $u > 0$, introduce the function

$$T_c(u) = \frac{u}{\Gamma(cpu^{p-1})} - 1.$$

Zhou [55], Lemma 6, states that T_c is convex on $(0,\infty)$. Its relation to the cosh-weighted expression in that lemma follows from the Gaussian density identity above. Furthermore,

$$T_c(u) \longrightarrow +\infty \quad (u \downarrow 0), \qquad \operatorname{sgn} T_c(u) = -\operatorname{sgn} D_c(u).$$

At the static contact, $T_{c_0}(q_0) = 0$. By (2.5), C_{c_0} is negative at some point in $(0, q_0)$, whereas $C_{c_0}(q_0) = 0$. The mean value theorem and (2.6) therefore give $s \in (0, q_0)$ with $D_{c_0}(s) > 0$, equivalently $T_{c_0}(s) < 0$. Convexity now implies

$$T'_{c_0}(q_0) \geqslant \frac{T_{c_0}(q_0) - T_{c_0}(s)}{q_0 - s} > 0.$$

At a fixed point, direct differentiation gives

$$T'_{c_0}(q_0) = \frac{1 - \lambda_0}{q_0}.$$

This proves $\lambda_0 < 1$, and convexity also gives $T_{c_0}(u) > 0$, hence $D_{c_0}(u) < 0$, for all $u > q_0$.

There is a zero $q_- \in (0, s)$ because $T_{c_0}(u) \to +\infty$ at zero. Convexity and $T_{c_0}(s) < 0$ imply that this zero is unique to the left of s, that $T_{c_0} > 0$ on $(0, q_-)$, and that $T_{c_0} < 0$ on (q_-, q_0). There are no other positive zeros. Thus C_{c_0} decreases on $(0, q_-)$, increases on (q_-, q_0), and decreases on (q_0, ∞). Together with $C_{c_0}(0) = C_{c_0}(q_0) = 0$, this proves the claimed strict contact inequality.

Finally,

$$\partial_u D_c(u)\big|_{(c,u)=(c_0,q_0)} = \lambda_0 - 1 < 0.$$

The implicit function theorem therefore supplies a unique smooth root $q(c)$ near q_0. Shrink I so that $q(c) > s$, $D_c(s) > 0$, and $\lambda(c) \leqslant 1 - \eta$ for some $\eta > 0$. The same convexity argument then shows that $q(c)$ is the larger positive root and $D_c(u) < 0$ above it. Compactness gives the uniform gap away from this root. Differentiating its defining equation proves the first two formulas in (2.3). Since $C'_c(q(c)) = 0$, the last formula follows from

$$\partial_c C_c(q(c)) = \frac{\Gamma(t(c))}{2} p q(c)^{p-1} - \frac{p-1}{2} q(c)^p = \frac{q(c)^p}{2}.$$

The continuation is a branch of fixed points; away from c_0, its contact height $C_c(q(c))$ is allowed to be nonzero. ∎

Lemma 2.2 (A uniform scalar margin)**.** *For every fixed $p \geqslant 3$, the above critical parameters satisfy*

$$q > 3/4 > 1/\sqrt{2}, \qquad \beta_c^2 \leqslant 2\log 2, \qquad \beta_c^2 (1-q)^{p-2} < 1/2.$$

Consequently there exist fixed positive δ, γ such that

$$b := 1 - q + \delta + \gamma < 1, \qquad \beta_c^2 b^{p-2} < 1/2. \tag{2.7}$$

Proof. Define

$$A_p(s) = \int_0^s \Gamma(v)\, dv - (1 - 1/p) s \Gamma(s).$$

The positive zero $s = t_p$ of A_p is unique, by the critical-point characterization, because every positive zero gives $q_p = \Gamma(s)$, $\beta_p^2 = s/[p\Gamma(s)^{p-1}]$, and conversely. The expansion $\Gamma(s) = s + O(s^2)$ shows that $A_p < 0$ near zero. As $s \to \infty$, $A_p(s) = s/p - 2\log 2 + o(1) > 0$. Hence A_p is negative before its unique positive zero. For $p \geqslant 3$, $A_p \leqslant A_3$, so $t_p \geqslant t_3$ and $q_p \geqslant q_3$.

The outward-rounded elementary integral certificate in Appendix A gives

$$\Gamma(2) > 0.7689, \qquad \Psi(2) < 1.5001.$$

Thus $A_3(2) = 2\Psi(2) - 2 - \frac{4}{3}\Gamma(2) < -0.025$. It follows that $t_3 > 2$, and $q_p \geqslant q_3 > 0.7689 > 3/4$. Nonnegativity of the Gibbs entropy at annealed contact gives $\log 2 - \beta_c^2/2 \geqslant 0$. Finally

$$\beta_c^2(1-q)^{p-2} < (2\log 2)(1/4) = \tfrac{1}{2}\log 2 < 1/2.$$

Continuity gives (2.7). The same certificate gives $q_p > 0.9313$ for $p \geqslant 4$, and $q_p > 0.9781$ for $p \geqslant 5$, by applying the monotonicity argument to $A_4(4) < 0$ and $A_5(6) < 0$. ∎

3. Quantitative replica symmetry in the auxiliary field

Method and assumptions. We extend the finite-step Guerra–Talagrand and one-site cavity method of Kusuoka–Nakajima [35] to the auxiliary field used here. The finite interpolation inequality comes from the Guerra–Talagrand framework [12, 29]; we specify its covariance and correction terms below. The published and preprint SK results do not themselves give the higher-spin extension. We first treat nonnegative mixtures of even powers and then establish the pure odd-power case by restricting to positive magnetization.

Let

$$\xi(x) = \sum_{p\in 2\mathbb{N},\ p\geqslant 2} b_p x^p, \qquad b_p \geqslant 0,$$

be a finite polynomial, not identically zero. A finite nonnegative mixture with the same uniform derivative bounds would work identically. The Gaussian Hamiltonian has exact covariance

$$\mathbb{E}\, H_N(\sigma)H_N(\tau) = N\xi(R(\sigma,\tau)), \qquad R(\sigma,\tau) = N^{-1}\sum_i \sigma_i\tau_i.$$

The inverse temperature is included in ξ. Define

$$F_N(h) = \log 2^{-N} \sum_{\sigma\in\{-1,1\}^N} \exp\{H_N(\sigma) + h\sum_i \sigma_i\}.$$

For $t \geqslant 0$, the cosh-transform semigroup is

$$(T_t f)(x) = \frac{\mathbb{E}[f(x+\sqrt{t}Z)\cosh(x+\sqrt{t}Z)]}{e^{t/2}\cosh x}, \qquad Z \sim N(0,1).$$

Theorem 3.1 (Quantitative RS under a pointwise strict criterion)**.** *Suppose $h > 0$ and $q \in (0,1)$ satisfy, with $X = h + \sqrt{\xi'(q)}Z$,*

$$q = \mathbb{E}\tanh^2 X, \tag{3.1}$$

$$\alpha := \xi''(q)\,\mathbb{E}\,\mathrm{sech}^4 X < 1, \tag{3.2}$$

$$g(u) := \mathbb{E}[T_{\xi'(u)-\xi'(q)}(\tanh^2)(X)] < u, \qquad q < u \leqslant 1. \tag{3.3}$$

Then there is a finite constant C, depending on the parameters, such that

$$\mathbb{E}\langle (R_{12}-q)^2\rangle \leqslant C/N, \qquad 0 \leqslant NP_{\mathrm{RS}}(\xi,h,q) - \mathbb{E}\, F_N(h) \leqslant C,$$

where

$$P_{\mathrm{RS}}(\xi,h,q) = \mathbb{E}\log\cosh X + \frac{1}{2}\{\xi(1) - \xi(q) - \xi'(q)(1-q)\}.$$

Also, writing $M(\sigma) = N^{-1}\sum_i \sigma_i$ and $m = \mathbb{E}\tanh X$,

$$\mathbb{E}\langle (M-m)^2\rangle \leqslant C/N, \qquad \mathrm{Var}(F_N'(h)) \leqslant CN.$$

Both conclusions are uniform on any compact family satisfying these conditions with a uniform AT margin and a uniform strict gap in (3.3) away from q. In particular they are uniform for h in a sufficiently small closed neighborhood of a parameter satisfying (3.1)–(3.3), with the continuously continued fixed point $q(h)$.

3.1. **Smart path.** Fix h, q, and set

$$\xi_s(r) = s\xi(r) + (1-s)\xi'(q)r, \quad 0 \leqslant s \leqslant 1,$$
$$H_{N,s}(\sigma) = \sqrt{s}H_N(\sigma) + \sqrt{(1-s)\xi'(q)}\sum_i z_i\sigma_i + h\sum_i \sigma_i.$$

Write $\nu_s(f) = \mathbb{E}\langle f\rangle_s$, $\phi_N(s) = N^{-1}\,\mathbb{E}\log 2^{-N}\sum e^{H_{N,s}}$, and

$$P_s = \mathbb{E}\log\cosh X + \frac{s}{2}\{\xi(1) - \xi(q) - \xi'(q)(1-q)\}.$$

Here $\xi_s'(q) = \xi'(q)$, so the same X, q serve at every point of the path. Define, for $u \geqslant q$,

$$g_s(u) = \mathbb{E}[T_{s(\xi'(u)-\xi'(q))}(\tanh^2)(X)].$$

The generator of T_t is $\frac{1}{2}\partial_{xx} + \tanh x\,\partial_x$. Applied to $\tanh^2 x$, it gives $\operatorname{sech}^4 x \geqslant 0$. Consequently $g_s(u) \leqslant g_1(u) < u$ for $u > q$,

$$g_s(q) = q, \qquad g_s'(q+) = s\alpha < 1.$$

The signs and local gap are uniform for $s \in [0,1]$.

3.2. **Explicit two-replica Guerra–Talagrand construction.** For an allowed overlap $v \in [-1,1]$, let $\iota = \operatorname{sgn} v$, with $\iota = 1$ at zero. Let $0 = a_0 < \cdots < a_M = 1$ be the distinct elements of $\{0, q, |v|, 1\}$. Set

$$Q_j = \begin{pmatrix} a_j & \iota(a_j \wedge |v|) \\ \iota(a_j \wedge |v|) & a_j \end{pmatrix}, \qquad B_j = \xi_s'(Q_j)$$

(the latter evaluated entrywise), and, for $1 \leqslant j \leqslant M$,

$$m_j = \begin{cases} 0, & a_j \leqslant q, \\ 1/2, & q < a_j \leqslant |v|, \\ 1, & a_j > \max(q, |v|). \end{cases}$$

The increments $B_j - B_{j-1}$ are positive semidefinite: before $|v|$ they are

$$s(\xi'(a_j) - \xi'(a_{j-1}))\begin{pmatrix} 1 & \iota \\ \iota & 1 \end{pmatrix},$$

and thereafter they are $s(\xi'(a_j) - \xi'(a_{j-1}))I_2$. Let G_j be independent centered Gaussian vectors with these covariance matrices. Starting with

$$U_M^{\lambda,v}(x_1, x_2) = \log\left(\frac{1}{4}\sum_{\varepsilon_1,\varepsilon_2=\pm1} e^{\varepsilon_1 x_1 + \varepsilon_2 x_2 + \lambda\varepsilon_1\varepsilon_2}\right),$$

recursively define

$$U_{j-1}^{\lambda,v}(x) = \begin{cases} m_j^{-1}\log\mathbb{E}\exp\{m_j U_j^{\lambda,v}(x + G_j)\}, & m_j > 0, \\ \mathbb{E}\,U_j^{\lambda,v}(x + G_j), & m_j = 0. \end{cases}$$

Let $\theta(r) = r\xi'(r) - \xi(r)$. It is even, and increasing on $[0,1]$. The scalar correction is

$$L_s(v) = \frac{1}{2}\sum_{j=1}^{M} m_j(d_j - d_{j-1}), \qquad d_j = \sum_{a,b=1}^{2}\theta_s(Q_{j,ab}), \quad \theta_s = s\theta.$$

For a breakpoint value a,

$$d(a) = \begin{cases} 4s\theta(a), & a \leqslant |v|, \\ 2s\theta(a) + 2s\theta(|v|), & a \geqslant |v|. \end{cases}$$

Thus the factor $1/2$ on $(q, |v|]$ gives the exact cancellation

$$L_s(v) = s\{\theta(1) - \theta(q)\}, \qquad -1 \leqslant v \leqslant 1. \tag{3.4}$$

Put $Y_0 = h + \sqrt{(1-s)\xi'(q)}Z_0$ and

$$\mathcal{P}_s(\lambda, v) = \mathbb{E}\, U_0^{\lambda,v}(Y_0, Y_0) - \lambda v - L_s(v).$$

The finite-step two-replica Guerra–Talagrand interpolation gives, exactly at finite N,

$$\frac{1}{N}\,\mathbb{E}\log\left[2^{-2N}\sum_{R(\sigma,\tau)=v} e^{H_{N,s}(\sigma)+H_{N,s}(\tau)}\right] \leqslant \inf_{\lambda\in\mathbb{R}} \mathcal{P}_s(\lambda, v). \tag{3.5}$$

To check the applicability of that inequality here: the terminal overlap matrix is Q_M, the Gaussian increments are exactly $\xi_s'(Q_j) - \xi_s'(Q_{j-1})$, and the correction is the sum in (3.4). The Gaussian integration-by-parts remainder consists of negative multiples of

$$\xi_s(x) - \xi_s(y) - \xi_s'(y)(x - y),$$

one for each pair of vector-spin coordinates and cascade levels. Every such term is nonnegative because ξ_s is convex on $[-1, 1]$. Thus there is no positivity perturbation or asymptotic error in (3.5). The normalizing factors 2^{-2N} match the terminal factor $1/4$.

3.3. **Evaluation and coercivity of the two-replica bound.** The finite recursion has at most three nontrivial increments. Differentiating the log moment generating function at each level gives

$$|\partial_\lambda U_0^{\lambda,v}| \leqslant 1, \qquad 0 \leqslant \partial_{\lambda\lambda}\mathcal{P}_s(\lambda, v) \leqslant 1 + \sum_j m_j \leqslant 5/2.$$

For $-q \leqslant v \leqslant q$, direct Gaussian averaging gives

$$\mathcal{P}_s(0, v) = 2P_s, \quad \partial_\lambda \mathcal{P}_s(0, v) = \widetilde{g}_s(v) - v,$$

where $(X_1(v), X_2(v))$ has means h, variances $\xi'(q)$, and covariance

$$(1-s)\xi'(q) + s\xi'(v), \qquad \widetilde{g}_s(v) = \mathbb{E}[\tanh X_1(v) \tanh X_2(v)].$$

Price's formula and Cauchy–Schwarz yield

$$\widetilde{g}_s'(v) = s\xi''(v)\,\mathbb{E}[\operatorname{sech}^2 X_1(v) \operatorname{sech}^2 X_2(v)] \leqslant s\xi''(q)\,\mathbb{E}\operatorname{sech}^4 X = s\alpha.$$

We used $\xi''(v) \leqslant \xi''(q)$ on $[-q, q]$, valid for our nonnegative even mixture. Since $\widetilde{g}_s(q) = q$,

$$\widetilde{g}_s(v) - v \geqslant (1 - s\alpha)(q - v) > 0 \quad (v < q).$$

For $q < v \leqslant 1$, the common Gaussian increment from q to v, taken at cascade level $1/2$, changes two log coshes into twice a log cosh. Precisely, its variance is $t = s(\xi'(v) - \xi'(q))$, and

$$2\log\mathbb{E}\cosh(x + \sqrt{t}Z) = 2\log\cosh x + t.$$

Consequently the same direct evaluation gives

$$\mathcal{P}_s(0, v) = 2P_s, \quad \partial_\lambda \mathcal{P}_s(0, v) = g_s(v) - v < 0.$$

Near $v = q$, the derivative with respect to v of the last expression is $s\alpha - 1 \leqslant \alpha - 1 < 0$, on both sides of q. Thus for some uniform $a, \varepsilon > 0$,

$$(\partial_\lambda \mathcal{P}_s(0, v))(v - q) \leqslant -a(v - q)^2, \qquad |v - q| \leqslant \varepsilon.$$

Using the second-derivative bound and choosing $\lambda = 2a(v - q)/5$ gives

$$\inf_\lambda \mathcal{P}_s(\lambda, v) \leqslant 2P_s - a^2(v - q)^2/5, \qquad |v - q| \leqslant \varepsilon.$$

For $v < -q$, $s > 0$, the same common increment instead gives

$$2 \log \mathbb{E} \sqrt{\cosh(x_1 + \sqrt{t}Z) \cosh(x_2 - \sqrt{t}Z)} \leqslant \log \cosh x_1 + \log \cosh x_2 + t,$$

strictly unless $x_1 = -x_2$. After averaging the increments below q, this equality condition is $Y_0 = 0$, an event of probability zero since $h > 0$. Hence $\mathcal{P}_s(0, v) < 2P_s$. At $s = 0$, $\mathcal{P}_0(0, v) = 2P_0$ and $\partial_\lambda \mathcal{P}_0(0, v) = q - v > 0$, so a small negative λ gives the strict inequality.

For every $v \neq q$ we have therefore found $\lambda \in [-1, 1]$ with $\mathcal{P}_s(\lambda, v) < 2P_s$. Joint continuity, compactness away from q, and the local quadratic bound imply the uniform estimate

$$\inf_\lambda \mathcal{P}_s(\lambda, v) \leqslant 2P_s - c(v - q)^2, \qquad s \in [0, 1], \quad v \in [-1, 1], \tag{3.6}$$

for some $c > 0$.

3.4. **Preliminary concentration, without assuming the Parisi limit.** Let $Q_{ab} = R_{ab} - q$, and fix $0 < \rho < 2c$. Define

$$p_N^{(2)}(s, \rho) = \frac{1}{2N} \mathbb{E} \log \left[2^{-2N} \sum_{\sigma, \tau} e^{H_{N,s}(\sigma) + H_{N,s}(\tau) + \rho N Q_{12}^2/2} \right].$$

There are at most $N + 1$ overlap values. Each constrained log partition function is Gaussian-Lipschitz with squared Lipschitz constant at most CN. The Gaussian maximum bound, (3.5), and (3.6) therefore give

$$p_N^{(2)}(s, \rho) \leqslant P_s + \varepsilon_N, \qquad \varepsilon_N = C\sqrt{\log(N + 1)/N},$$

uniformly in s. Write $D_N(s) = P_s - \phi_N(s)$. Gaussian integration by parts gives the exact identity

$$D_N'(s) = \frac{1}{2} \nu_s\{\xi(R_{12}) - \xi(q) - \xi'(q)(R_{12} - q)\}, \qquad D_N(0) = 0. \tag{3.7}$$

In particular $D_N \geqslant 0$, and, with $K = \sup_{[-1,1]} \xi''$,

$$0 \leqslant D_N'(s) \leqslant \frac{K}{4} \nu_s(Q_{12}^2).$$

By convexity in ρ,

$$\frac{1}{4} \nu_s(Q_{12}^2) \leqslant \frac{p_N^{(2)}(s, \rho) - \phi_N(s)}{\rho} \leqslant \frac{D_N(s) + \varepsilon_N}{\rho}.$$

Gronwall now gives $D_N(s) + \nu_s(Q_{12}^2) \leqslant C\varepsilon_N = o(1)$. Moreover

$$Y_{N,s} := \log \langle e^{\rho N Q_{12}^2/2} \rangle_s \quad \text{satisfies} \quad \mathbb{E}\, Y_{N,s} = o(N)$$

uniformly in s, and is Gaussian-Lipschitz with squared constant CN. Gaussian concentration and exponential Markov inside the Gibbs measure imply, for each fixed $\delta > 0$,

$$\sup_{s \in [0,1]} \nu_s(|Q_{12}| > \delta) \leqslant C_\delta e^{-c_\delta N}. \tag{3.8}$$

For completeness, except on an event of probability $C_\delta e^{-c_\delta N}$, one has $Y_{N,s} \leqslant \rho\delta^2 N/4$, while on that event's complement $\langle \mathbf{1}_{|Q_{12}|>\delta}\rangle_s \leqslant e^{-\rho\delta^2 N/4}$. This proves (3.8).

3.5. **The exact tensor cavity and its cubic error.** Let

$$A = \nu_s(Q_{12}^2), \quad B = \nu_s(Q_{12}Q_{13}), \quad C_0 = \nu_s(Q_{12}Q_{34}),$$

and set

$$r = \mathbb{E}\tanh^4 X, \quad a = 1-q^2, \quad b = q-q^2, \quad c_0 = r-q^2, \quad k = \xi''(q).$$

Write $\sigma = (\rho, \varepsilon)$, $R^-_{ab} = N^{-1}\sum_{i<N}\sigma_i^a\sigma_i^b$, and $Q^-_{ab} = R^-_{ab} - q$. Then $Q_{ab} = Q^-_{ab} + \varepsilon_a\varepsilon_b/N$.

The original exact-covariance Hamiltonian decomposes into independent Gaussian processes $H_N(\rho,\varepsilon) = B_N(\rho) + \varepsilon Z_N(\rho)$. Its cavity-field covariance is exactly

$$K_N(x) = \frac{N}{2}\{\xi(x+N^{-1}) - \xi(x-N^{-1})\} = \xi'(x) + O(N^{-2})$$

uniformly for the possible $x = R^-$. This follows either directly from the Gaussian even/odd decomposition in ε, or by separating tensor terms according to the parity of the number of occurrences of the last index. Interpolate the last-spin field from $\sqrt{s}Z_N(\rho)$ to an independent scalar field of variance $s\xi'(q)$, leaving the baseline B_N, all other fields, and the last spin's field of variance $(1-s)\xi'(q)$ in place. Denote expectations by $\nu_{s,u}$, $0 \leqslant u \leqslant 1$. At $u=1$, $\nu_{s,1} = \nu_s$. At $u=0$, the last-spin total field has distribution X and is independent of the other spins.

For a bounded function f of n replicas, Gaussian integration by parts gives

$$\partial_u \nu_{s,u}(f) = s\Bigg[\sum_{1\leqslant a<b\leqslant n}\nu_{s,u}(f\varepsilon_a\varepsilon_b\Delta_{ab}) - n\sum_{a=1}^{n}\nu_{s,u}(f\varepsilon_a\varepsilon_{n+1}\Delta_{a,n+1}) + \frac{n(n+1)}{2}\nu_{s,u}(f\varepsilon_{n+1}\varepsilon_{n+2}\Delta_{n+1,n+2})\Bigg],$$

where

$$\Delta_{ab} = K_N(R^-_{ab}) - \xi'(q) = kQ^-_{ab} + O((Q^-_{ab})^2 + N^{-2}).$$

The diagonal covariance difference is configuration-independent and cancels in the normalized Gibbs derivative. Since Δ is uniformly bounded, this identity and Gronwall imply

$$\sup_{0\leqslant u\leqslant 1}\nu_{s,u}(|Q^-_{12}|^3) \leqslant C\{\nu_s(|Q_{12}|^3) + N^{-3}\}.$$

For any two replica edges e, f, differentiating $\nu_{s,u}(Q^-_eQ^-_f)$ and applying Hölder bounds every leading term by the displayed third moment; the $O(N^{-2})$ covariance error contributes $O(N^{-2})$. Replacing Q^- by Q costs at most

$$CN^{-1}\nu_s|Q_{12}| + CN^{-2} \leqslant C\nu_s|Q_{12}|^3 + CN^{-3/2}.$$

Thus all cavity second moments differ from A, B, C_0 by $O(\nu_s|Q_{12}|^3 + N^{-3/2})$.

Site exchangeability at $u=1$ gives

$$A = \nu_s[Q_{12}(\varepsilon_1\varepsilon_2 - q)], \quad B = \nu_s[Q_{12}(\varepsilon_1\varepsilon_3 - q)], \quad C_0 = \nu_s[Q_{12}(\varepsilon_3\varepsilon_4 - q)].$$

For each expression, separate the contribution $\varepsilon_1\varepsilon_2/N$ in Q_{12}, and expand the remaining term to first order in u at zero. Its zeroth-order value is zero. Two applications of the streaming

identity bound the Taylor remainder by $O(\nu_s|Q_{12}|^3 + N^{-3/2})$; replacing Δ by kQ^- has the same bound. The elementary last-spin coefficients are

$$\mathbb{E}_0[(\varepsilon_i\varepsilon_j - q)\varepsilon_a\varepsilon_b] = \begin{cases} a, & \{i,j\} = \{a,b\}, \\ b, & \text{the two edges share exactly one endpoint}, \\ c_0, & \text{the two edges are disjoint}. \end{cases}$$

Substitution gives the explicit linear system

$$\begin{aligned} (I - skM)\begin{pmatrix} A \\ B \\ C_0 \end{pmatrix} &= \frac{1}{N}\begin{pmatrix} a \\ b \\ c_0 \end{pmatrix} + O\{\nu_s|Q_{12}|^3 + N^{-3/2}\}, \\ M &= \begin{pmatrix} a & -4b & 3c_0 \\ b & a - 2b - 3c_0 & -3b + 6c_0 \\ c_0 & 4b - 8c_0 & a - 8b + 10c_0 \end{pmatrix}. \end{aligned} \tag{3.9}$$

The big-O in (3.9) is componentwise with a uniform constant.

To verify invertibility without relying on a matrix eigenvalue assertion, put

$$U = A - 4B + 3C_0, \quad V = 2B - 3C_0, \quad D = A - 2B + C_0,$$

$$\kappa = 1 - 4q + 3r, \qquad \zeta = 2q + q^2 - 3r, \qquad \ell = 1 - 2q + r = \mathbb{E}\,\mathrm{sech}^4 X.$$

The three rows of (3.9) give

$$\begin{aligned} (1 - sk\kappa)U &= \kappa/N + O(\nu_s|Q|^3 + N^{-3/2}), \\ (1 - sk\kappa)V - sk\zeta U &= \zeta/N + O(\nu_s|Q|^3 + N^{-3/2}), \\ (1 - sk\ell)D &= \ell/N + O(\nu_s|Q|^3 + N^{-3/2}). \end{aligned}$$

Because $r \leqslant q$, $\kappa \leqslant \ell$; hence both denominators are bounded below by $1 - \alpha > 0$. Solving in the order U, V, D and using $A = -2U - V + 3D$ yields

$$A \leqslant C/N + C\nu_s|Q_{12}|^3.$$

By (3.8), for any fixed $\delta > 0$,

$$\nu_s|Q_{12}|^3 \leqslant \delta A + 8C_\delta e^{-c_\delta N}.$$

Taking δ small enough to absorb $C\delta A$ proves

$$\sup_{s\in[0,1]} \nu_s(Q_{12}^2) \leqslant C/N. \tag{3.10}$$

Finally integrate (3.7), using $0 \leqslant \xi(r) - \xi(q) - \xi'(q)(r-q) \leqslant K(r-q)^2/2$, to obtain

$$0 \leqslant NP_{\mathrm{RS}} - \mathbb{E}\,F_N(h) \leqslant C.$$

3.6. **Uniformity in the deterministic field.** The derivative in q of $\mathbb{E}\tanh^2(h + \sqrt{\xi'(q)}Z)$ at a fixed point is

$$\xi''(q)\,\mathbb{E}[(1 - \tanh^2 X)(1 - 3\tanh^2 X)] = k\kappa < 1.$$

The implicit function theorem therefore continues $q(h)$ smoothly near the initial $h > 0$. The AT inequality and the local derivative gap persist uniformly. The pointwise strict inequality (3.3) persists away from q by compactness and continuity, and near q by the uniform AT gap. Every preceding estimate is then uniform on a sufficiently small compact field interval.

3.7. **Magnetic fluctuations and the derivative variance.** Here is the promised additional estimate, using only (3.10) and the same exact one-site cavity; it requires no separate assumption about a longitudinal susceptibility. Fix s, put $m = \mathbb{E}\tanh X$, and define

$$M_1 = N^{-1}\sum_{i=1}^{N}\sigma_i^1, \quad \widehat{M}_1^- = N^{-1}\sum_{i<N}\sigma_i^1 - m, \quad D_M = \nu_s[(M_1-m)^2].$$

Site exchangeability yields

$$D_M = \nu_s[(M_1-m)(\varepsilon_1-m)] = \nu_s[\widehat{M}_1^-(\varepsilon_1-m)] + N^{-1}\nu_s[\varepsilon_1(\varepsilon_1-m)].$$

The last term has absolute value at most $2/N$. At cavity time zero,

$$\nu_{s,0}[\widehat{M}_1^-(\varepsilon_1-m)] = 0,$$

since the last spin and its scalar field are independent of all remaining spins and $\mathbb{E}_0\,\varepsilon_1 = m$.

For any nonnegative function f of finitely many replicas, the streaming identity and boundedness of Δ imply $|\partial_u \nu_{s,u}(f)| \leqslant C\nu_{s,u}(f)$. Consequently,

$$\sup_u \nu_{s,u}[(\widehat{M}_1^-)^2] \leqslant C\nu_s[(\widehat{M}_1^-)^2] \leqslant C(D_M + N^{-2}),$$

and, by (3.10),

$$\sup_u \nu_{s,u}[(Q_{12}^-)^2] \leqslant C/N, \qquad \sup_u \nu_{s,u}(\Delta_{12}^2) \leqslant C/N.$$

Apply the streaming identity to $f = \widehat{M}_1^-(\varepsilon_1 - m)$. Every term is bounded by Cauchy–Schwarz by

$$C\{\nu_{s,u}[(\widehat{M}_1^-)^2]\}^{1/2}\{\nu_{s,u}(\Delta_{ab}^2)\}^{1/2} \leqslant C\sqrt{(D_M + N^{-2})/N}.$$

Integration from zero to one therefore gives

$$D_M \leqslant 2/N + C\sqrt{(D_M + N^{-2})/N}.$$

Young's inequality absorbs half of D_M, giving $D_M \leqslant C/N$. At $s = 1$,

$$F_N'(h) = N\langle M\rangle, \qquad \operatorname{Var}(F_N'(h)) \leqslant N^2\,\mathbb{E}(\langle M\rangle - m)^2 \leqslant N^2\,\mathbb{E}\langle (M-m)^2\rangle \leqslant CN.$$

The proof and the constants are uniform on the field neighborhood established above. Thus, for the centered field process $\overline{F}_N(h) = F_N(h) - \mathbb{E}\,F_N(h)$,

$$\mathbb{E}\left|\frac{\overline{F}_N(h) - \overline{F}_N(k)}{\sqrt{N}}\right|^2 \leqslant C|h-k|^2$$

throughout that neighborhood, by integrating the centered derivatives and applying Cauchy–Schwarz. This finishes the proof of Theorem 3.1, including its magnetic assertions.

3.8. **Application at the pure even-spin static threshold.** Take $\xi(u) = \beta_c^2 u^p$, with fixed even $p \geqslant 4$. Let $q_* \in (0,1)$ be the nonzero static critical contact and let $t_* = \xi'(q_*)$, $h_* = t_*$. Write

$$Q(t) = \mathbb{E}\tanh^2(t + \sqrt{t}Z).$$

The critical contact equations give $Q(t_*) = q_*$. Lemma 2.1, proved using the convexity result in Zhou [55], Lemma 6, gives

$$Q(\xi'(u)) - u < 0 \quad (u > q_*), \qquad \xi''(q_*)Q'(t_*) < 1.$$

The scalar channel identity $Q'(t) = \mathbb{E}\operatorname{sech}^4(t + \sqrt{t}Z)$ gives the strict AT condition. For every even test function f,

$$\mathbb{E}\,f(t + \sqrt{t}Z) = e^{-t/2}\,\mathbb{E}[f(\sqrt{t}Z)\cosh(\sqrt{t}Z)].$$

The cosh-transform semigroup preserves these symmetric tilted Gaussian laws as its time increases. Consequently

$$\mathbb{E}[T_{\xi'(u)-t_*}(\tanh^2)(t_* + \sqrt{t_*}Z)] = Q(\xi'(u)), \qquad u \geqslant q_*.$$

The hypotheses of Theorem 3.1 therefore hold at (h_*, q_*), and uniformly for h near h_*. In particular,

$$\mathbb{E}\, F_N(h) = NP_{\mathrm{RS}}(\xi, h, q(h)) + O(1)$$

uniformly on that neighborhood. This is a field-model estimate. It does not by itself establish the zero-field critical free-energy limiting law; a separate planted-band and joint-fluctuation argument is still needed.

3.9. **Odd powers: a rigorous restriction to positive mutual overlaps.** The evenness assumption can be removed for a pure power $\xi(r) = \beta^2 r^p$, including odd $p \geqslant 3$, if the same conditions (3.1)–(3.3) hold and additionally

$$m = \mathbb{E}\tanh(h + \sqrt{\xi'(q)}Z) > 1/2. \tag{3.11}$$

The conclusions are then

$$\mathbb{E}\langle (R_{12}-q)^2\rangle \leqslant C/N, \quad |\,\mathbb{E}\, F_N(h) - NP_{\mathrm{RS}}| \leqslant C, \quad \mathbb{E}\langle (M-m)^2\rangle \leqslant C/N, \quad \operatorname{Var} F_N'(h) \leqslant CN,$$

uniformly on a sufficiently small field neighborhood. We do not assert an exact sign for the finite-size pressure deficit in this odd-power extension.

First establish the qualitative RS formula using the general Parisi formula of Panchenko [41], which is valid for mixtures containing odd powers. Only values of ξ'' on $[0,1]$ enter its PDE and first variation. For the candidate δ_q, below q the Parisi derivative is the heat martingale

$$M_u = \mathbb{E}[\tanh X_q \mid \mathcal{F}_u], \qquad X_u = h + B_{\xi_s'(u)}, \quad X_q \overset{d}{=} X.$$

The initial Gaussian field of variance $\xi_s'(0)$ is included in this Brownian notation. Ito's formula and conditional Jensen give

$$\Gamma_s'(u) = \xi_s''(u)\,\mathbb{E}\{\mathbb{E}[\operatorname{sech}^2 X_q \mid \mathcal{F}_u]^2\} \leqslant s\xi''(q)\,\mathbb{E}\operatorname{sech}^4 X = s\alpha < 1.$$

Thus $\Gamma_s(u) - u \geqslant (1-s\alpha)(q-u) > 0$ for $0 \leqslant u < q$. Above q, $\Gamma_s(u) = g_s(u) < u$, as proved in Section 3.1. The first variation toward an atom at $a < q$ is $\frac{1}{2}\int_a^q \xi_s''(u)(\Gamma_s(u)-u)\,du \geqslant 0$; toward an atom at $a > q$, it is $-\frac{1}{2}\int_q^a \xi_s''(u)(\Gamma_s(u)-u)\,du \geqslant 0$. The convexity/first-variation characterization of the Parisi functional therefore gives

$$\lim_{N\to\infty} \phi_N(s) = P_s.$$

At $s = 0$ this is also immediate from the product formula. The fixed point, AT margin, and pointwise sign persist for small changes of the deterministic field, by the same implicit-function and compactness argument as Section 3.6. They persist uniformly in $s \in [0,1]$, including when that field is changed while keeping the covariance ξ_s fixed. The limiting pressure is differentiable in the deterministic field, with derivative m at the base parameter. Convergence of the expected pressures is uniform on the resulting compact parameter set: their deterministic-field derivatives are bounded by one, and Gaussian interpolation bounds differences of expected pressures by a constant times the uniform differences of their covariance functions. A finite-net argument then upgrades pointwise convergence to uniform convergence.

For any fixed $\eta > 0$, the usual exponential tilt in the deterministic field gives

$$\langle \mathbf{1}_{M \geqslant m+\eta}\rangle_{s,h} \leqslant \exp\{F_{N,s}(h+t) - F_{N,s}(h) - Nt(m+\eta)\}, \qquad t > 0.$$

Choose t sufficiently small that the limiting difference quotient is at most $m+\eta/2$, uniformly in the compact parameter family. Uniform convergence of the means and Gaussian concentration (both log partition functions have squared Gaussian Lipschitz constants at most CN) give

$$\mathbb{E}\langle \mathbf{1}_{M\geqslant m+\eta}\rangle_{s,h} \leqslant Ce^{-cN}.$$

The negative tilt treats $M \leqslant m-\eta$. Thus

$$\mathbb{E}\langle \mathbf{1}_{|M-m|>\eta}\rangle_{s,h} \leqslant Ce^{-cN}, \tag{3.12}$$

uniformly in s and nearby h. This is a qualitative-RS consequence; no finite-size RS error bound or overlap estimate has been used.

Choose a constant a with $1/2 < a < \inf m(h)$ on a sufficiently small field interval, and restrict the spin set to

$$\Sigma_+ = \{\sigma \in \{-1,1\}^N : M(\sigma) \geqslant a\}.$$

For any $\sigma,\tau \in \Sigma_+$, the Hamming triangle inequality gives

$$R(\sigma,\tau) \geqslant M(\sigma) + M(\tau) - 1 \geqslant 2a-1 > 0.$$

Run the finite-step interpolation of Section 3.2 on pairs in Σ_+^2, restricted further to mutual overlap v. Necessarily $v \geqslant 2a-1>0$, so all prescribed matrix entries $Q_{j,ab}$ are nonnegative. More importantly, every actual overlap between any two spin configurations appearing at every interpolation/cascade replica is also positive, because every such configuration belongs to Σ_+. The covariance remainder

$$\xi_s(x) - \xi_s(y) - \xi_s'(y)(x-y)$$

is consequently nonnegative, since $x,y \in [0,1]$ and ξ_s is convex there even for odd p. At the decoupled field endpoint, drop the constraint Σ_+ to obtain an upper bound by the same product-field recursion. Thus (3.5) holds for this *restricted* pair partition function.

All computations of Section 3.3 for $v \geqslant 0$ remain valid; they use only increasing ξ', increasing ξ'', and the sign criterion on $[0,1]$. They give the same uniform coercivity

$$\frac{1}{N}\,\mathbb{E}\log\left[2^{-2N}\sum_{\substack{\sigma,\tau\in\Sigma_+\\ R(\sigma,\tau)=v}} e^{H_{N,s}(\sigma)+H_{N,s}(\tau)}\right] \leqslant 2P_s - c(v-q)^2.$$

We already know the qualitative full pressure limit $\phi_N(s) \to P_s$ uniformly. Compare the last restricted log partition function with twice the full log partition function. Both are Gaussian-Lipschitz with squared constants at most CN. For a fixed $\delta>0$, whenever $|v-q| \geqslant \delta$, their mean difference is at most $-c\delta^2 N/2$ for all sufficiently large N. Gaussian concentration therefore bounds the expected full Gibbs probability of this restricted pair event by $C_\delta e^{-c_\delta N}$. Sum over the at most $N+1$ allowed values of v. The pair event that one spin lies outside Σ_+ has exponentially small Gibbs probability by (3.12). Together these facts prove (3.8) for the full odd-power model.

The exact cavity decomposition, Taylor estimates, linear system, and magnetic estimate in Sections 3.5 and 3.7 never require convexity at negative overlaps. They apply unchanged, and yield the optimal N^{-1} overlap and magnetic bounds. Finally the exact identity (3.7) still holds. Although its integrand need not have a sign for negative R, the global Taylor bound

$$|\xi(r) - \xi(q) - \xi'(q)(r-q)| \leqslant C(r-q)^2$$

and (3.10) show $|D_N'(s)| \leqslant C/N$. Since $D_N(0)=0$, integration proves the absolute $O(1)$ total-pressure error claimed above.

At the critical scalar contact, the identity $m = q_p > 3/4$ supplies the required positive magnetization. The scalar certificate is included below.

3.10. **Consequences for the critical auxiliary field.** For a deterministic field pattern $a \in \Sigma_N$, define

$$F_N^a(h) = \log \mathbb{E}_{\nu\sim\mu_N} \exp\{X_N(\nu) + h \sum_i a_i \nu_i\}.$$

Gauge invariance implies that the law and the mean of $F_N^a(h)$ do not depend on a. Denote this common mean by $\mathbb{E}\, F_N(h)$, and its thermodynamic limit divided by N by $f(h)$.

Proposition 3.2 (Quantitative RS field lemma)**.** *There is a fixed compact interval J containing t in its interior and a smooth RS overlap branch $q(h)$ on J, with $q(t) = q$, such that uniformly on J,*

$$\mathbb{E}\, F_N(h) = N f(h) + O(1), \tag{3.13}$$

$$f(h) = \mathbb{E} \log\cosh(h + \sqrt{\xi'(q(h))}Z) + \frac{1}{2}\{\xi(1) - \xi(q(h)) - (1 - q(h))\xi'(q(h))\}, \tag{3.14}$$

$$\mathbb{E}\langle (R_{12} - q(h))^2 \rangle_h \leqslant C/N, \tag{3.15}$$

$$\mathbb{E}\langle (M - m(h))^2 \rangle_h \leqslant C/N, \qquad m(h) = \mathbb{E} \tanh(h + \sqrt{\xi'(q(h))}Z), \tag{3.16}$$

$$\mathrm{Var}(F_N'(h)) \leqslant CN. \tag{3.17}$$

At $h = t$, $m(t) = q$.

Proposition 3.2 is the result proved in Section 3. Its applicability follows from $\lambda < 1$, $D(u) < 0$ for $u > q$, and the cosh-transform identity for the scalar channel. For odd powers, the argument above first obtains qualitative RS and positive magnetization, restricts spins to $M \geqslant a > 1/2$ in the two-replica interpolation, and then runs the same finite-cavity bootstrap. Lemma 2.2 supplies the needed $m(t) = q > 1/2$. The SK-only theorem of Kusuoka–Nakajima is not being invoked as if it already covered a pure higher-spin model.

Proposition 3.3 (Joint field CLT)**.** *There is a constant $v > 0$ such that, for independent uniform patterns $a, b \in \Sigma_N$, independent of the same Gaussian disorder,*

$$\left(\frac{F_N^a(t) - \mathbb{E}\, F_N(t)}{\sqrt{N}}, \frac{F_N^b(t) - \mathbb{E}\, F_N(t)}{\sqrt{N}} \right) \implies (\sqrt{v}Z_1, \sqrt{v}Z_2), \tag{3.18}$$

where Z_1, Z_2 are independent standard Gaussians.

Proposition 3.3 follows from Proposition 3.2 by the all-parity RS-cavity and Gaussian Stein argument proved in Section 4. The usual CDP theorem alone only states a one-dimensional result for even-spin mixtures and is insufficient as a bare citation for (3.18) for odd p.

Theorem 3.4 (Critical half limit)**.** *Using Propositions 3.2 and 3.3, for every fixed $p \geqslant 3$,*

$$\frac{Z_N(\beta_c)}{\mathbb{E}\, Z_N(\beta_c)} \longrightarrow \frac{1}{2} \quad \textit{in probability,}$$

and consequently

$$\log Z_N(\beta_c) - \log \mathbb{E}\, Z_N(\beta_c) \longrightarrow -\log 2 \quad \textit{in probability.}$$

The same assertions hold for the distinct-index model with variance N, and, after using its exact annealed expectation, for the convention with coefficient $\sqrt{p!/N^{p-1}}$.

4. Joint auxiliary-field fluctuations

This section proves Proposition 3.3 from the quantitative field estimate. The argument applies to every integer $p \geqslant 3$; no convexity assertion about a negative cross overlap is used. In the one-site bulk system the covariance polynomial changes by $O(N^{-1})$. We use the uniform-on-compact-families version of the field estimate in Section 3 for this perturbation. Its hypotheses persist: the implicit fixed point is smooth, the AT inequality and the gap away from the fixed point are strict, and, for odd p, the magnetization remains above $1/2$.

4.1. **Cross thermal concentration without a two-system variational bound.** Consider two marginal RS Gibbs measures G_1, G_2, each on n spins. Let their magnetization vectors and covariance matrices be m_j, C_j. The disorders may be arbitrarily coupled. If $\mathbb{E}\, G_j^{\otimes 2}(R_{12}-q)^2 \leqslant C/n$, then

$$\mathbb{E}\,\mathrm{Tr}(C_j^2) \leqslant Cn, \qquad \mathbb{E}\big(\|m_j\|^2/n - q\big)^2 \leqslant C/n. \tag{4.1}$$

For the first bound, use four independent replicas within system j and the identity

$$\frac{\mathrm{Tr}(C_j^2)}{n^2} = G_j^{\otimes 4}\left[\frac{R_{13}-R_{14}-R_{23}+R_{24}}{2}\right]^2 \leqslant 4G_j^{\otimes 2}(R_{12}-q)^2.$$

The second bound is Jensen's inequality. For the cross overlap, set $u = m_1 \cdot m_2/n$. Its conditional thermal variance is exactly

$$(G_1 \otimes G_2)(R-u)^2 = \frac{\mathrm{Tr}(C_1C_2) + m_1^T C_2 m_1 + m_2^T C_1 m_2}{n^2}. \tag{4.2}$$

Since the covariance matrices are positive semidefinite, $\mathrm{Tr}(C_1C_2) \leqslant \sqrt{\mathrm{Tr}(C_1^2)\,\mathrm{Tr}(C_2^2)}$, $\|C_j\|_{\mathrm{op}} \leqslant \sqrt{\mathrm{Tr}(C_j^2)}$, and $\|m_j\|^2 \leqslant n$. Consequently

$$\mathbb{E}(G_1 \otimes G_2)(R-u)^2 \leqslant Cn^{-1/2}, \qquad \mathbb{E}\,|u-\bar{u}|^2 \longrightarrow 0, \quad \bar{u} = \max(-q, \min(q,u)). \tag{4.3}$$

The clipping assertion follows from $|u| \leqslant \|m_1\|\|m_2\|/n$ and (4.1). The same bounds apply to every pair among finitely many replicas.

4.2. **Exact cavity split and a bound on every normalizer.** Couple two full disorders with covariance correlation $\rho \in [0,1]$. Write a spin as (σ, e), where $\sigma \in \Sigma_{N-1}$ and $e \in \{-1,1\}$. Splitting tensor terms according to the parity of the number of occurrences of the last site gives the exact independent decomposition in each system

$$X_N^j(\sigma, e) = B_N^j(\sigma) + e z_j(\sigma).$$

The pair of processes (B_N^1, B_N^2) is independent of (z_1, z_2). If $r = (N-1)^{-1}\sigma\cdot\tau$, then

$$\begin{aligned} K_N(r) &:= \mathbb{E}\, z_j(\sigma) z_j(\tau) \\ &= \frac{N}{2}\left[\xi\left(\frac{(N-1)r+1}{N}\right) - \xi\left(\frac{(N-1)r-1}{N}\right)\right] = \xi'(r) + O(N^{-1}), \end{aligned}$$

uniformly for $r \in [-1,1]$, and the cross covariance is $\rho K_N(r)$. Put $\kappa_N = K_N(1)$, so $\kappa_N \to \xi'(1) > t$.

Let π_j be the Gibbs measure on the first $N-1$ sites with Hamiltonian B_N^j and its corresponding field pattern. These are independent of all cavity Gaussian variables and of the last-site field signs. They satisfy the same self-overlap bound C/N. Indeed, the tensor terms with no last index form a standard $(N-1)$-spin Hamiltonian with covariance polynomial $\xi_N(r) = ((N-1)/N)^{p-1}\xi(r)$, and the additional even-parity last-index terms have self-variance $O(N^{-1})$. The compact-family field estimate handles ξ_N; Gaussian streaming through the

additional terms changes the expectation of any nonnegative fixed-replica observable by a multiplicative factor at most $e^{C/N}$. The fixed point changes by $O(N^{-1})$. This proves the asserted bulk self-overlap estimate at the fixed value q.

Conditional on the bulk disorder, put

$$u_N = \frac{m_1 \cdot m_2}{N-1}, \qquad \bar{u}_N = \max(-q, \min(q, u_N)), \qquad m_j = \pi_j(\sigma).$$

At any Gaussian interpolation of the cavity fields used below, the lifted Gibbs law is

$$\mathcal{G}_j(\sigma, e) = \frac{\pi_j(\sigma) e^{e(h_j + z_j(\sigma))}}{2D_j}, \qquad D_j = \sum_\sigma \pi_j(\sigma) \cosh(h_j + z_j(\sigma)) \geqslant 1, \tag{4.4}$$

where $h_j \in \{-t, t\}$. This deterministic lower bound controls every inverse normalizer. More precisely, for any fixed number of replicas and any nonnegative function f of their bulk configurations,

$$\mathbb{E}_z \, \mathcal{G}_1^{\otimes n_1} \mathcal{G}_2^{\otimes n_2}(f) \leqslant C_{n_1+n_2} \, \pi_1^{\otimes n_1} \pi_2^{\otimes n_2}(f). \tag{4.5}$$

To see this, omit the denominators, sum the last spins, and bound each cosh by $e^{t+|z|}$. The expectation of the resulting fixed product is uniformly bounded, since all Gaussian variances and covariances are bounded. The bound is conditional on the bulk laws and uniform over N, ρ and the interpolation parameter. It does not assume independence among the cavity fields evaluated at different configurations.

4.3. **The conditional Gaussian-mixture comparison.** Conditional on π_1, π_2, construct a centered Gaussian pair (Z_1, Z_2), independent of the last field signs, with

$$\operatorname{Var} Z_1 = \operatorname{Var} Z_2 = t, \qquad \operatorname{Cov}(Z_1, Z_2) = \rho \xi'(\bar{u}_N).$$

This covariance matrix is positive semidefinite, because $|\xi'(u)| \leqslant \xi'(q) = t$ on $[-q, q]$, also for odd p. For all j, σ, take independent residual variables $\eta_j(\sigma) \sim N(0, \kappa_N - t)$, independent of the common pair, and replace the cavity process by

$$z_j^*(\sigma) = Z_j + \eta_j(\sigma).$$

The covariance differences between the actual and replacement processes are

$$\Delta_{jj}(\sigma, \tau) = K_N(R(\sigma, \tau)) - t - (\kappa_N - t) \mathbb{1}_{\{\sigma = \tau\}},$$
$$\Delta_{12}(\sigma, \tau) = \rho \{K_N(R(\sigma, \tau)) - \xi'(\bar{u}_N)\}.$$

The diagonal entries $\Delta_{jj}(\sigma, \sigma)$ vanish exactly. The self-overlap estimate, (4.3), and bounded derivatives of ξ show

$$\sum_{j,k} \mathbb{E}(\pi_j \otimes \pi_k) |\Delta_{jk}| \longrightarrow 0. \tag{4.6}$$

For the diagonal atomic correction use

$$(1-q)^2 \sum_\sigma \pi_j(\sigma)^2 \leqslant \pi_j^{\otimes 2}(R_{12} - q)^2,$$

so its mean is $O(N^{-1})$.

Interpolate the two centered Gaussian cavity processes by the square-root interpolation. For any bounded observable of finitely many lifted replicas, Gaussian differentiation expresses the derivative of its mean as a sum of terms with one covariance difference Δ_{jk} and at most two additional replicas. All last spins have absolute value one. Thus (4.5) bounds the absolute derivative by $C \sum_{j,k} (\pi_j \otimes \pi_k) |\Delta_{jk}|$, conditional on the bulk disorder. Integration and (4.6) prove that the actual and replacement cavity expectations differ by $o(1)$ after averaging. The

argument also permits multiplication of the observable by any bounded function of the bulk disorder, such as $\bar{u}_N$.

In the replacement system, let $y_j = h_j + Z_j$. Its last-spin mean is

$$M_j^* = \frac{\sum_\sigma \pi_j(\sigma)\sinh(y_j + \eta_j(\sigma))}{\sum_\sigma \pi_j(\sigma)\cosh(y_j + \eta_j(\sigma))}.$$

The elementary identity

$$\sinh(y+\eta) - \tanh(y)\cosh(y+\eta) = \frac{\sinh\eta}{\cosh y}$$

and the denominator bound in (4.4) give, uniformly in y_j,

$$\mathbb{E}_\eta\,|M_j^* - \tanh y_j| \leqslant C\left(\sum_\sigma \pi_j(\sigma)^2\right)^{1/2}.$$

The residual variables are independent and centered after applying sinh; their variances are uniformly bounded. Hence the replacement last-spin product converges in mean to $\tanh(h_1 + Z_1)\tanh(h_2 + Z_2)$. This proves the conditional Gaussian-mixture comparison, including its normalization and atomic errors.

4.4. **Moment consistency and the scalar contractions.** For independent uniform field patterns, average over the two independent last-site signs and define

$$\Phi_\rho(u) = \mathbb{E}\tanh(t\epsilon_1 + Z_1)\tanh(t\epsilon_2 + Z_2), \quad \operatorname{Var} Z_j = t, \quad \operatorname{Cov}(Z_1, Z_2) = \rho\xi'(u).$$

For identical field patterns define instead

$$\Psi_\rho(u) = \mathbb{E}\tanh(t + Z_1)\tanh(t + Z_2).$$

The identical random last sign gives the same formula, by simultaneous reflection of the Gaussian pair. Let T_ρ stand for the appropriate map. The preceding comparison proves

$$\mathbb{E}\langle e_1 e_2\rangle - \mathbb{E}\,T_\rho(\bar{u}_N) \longrightarrow 0. \tag{4.7}$$

By (4.5) and (4.3), the bulk cross overlap r also satisfies $\mathbb{E}\langle(r - \bar{u}_N)^2\rangle \to 0$ under the full lifted system. Exchangeability over sites therefore gives

$$\mathbb{E}\langle R\rangle = \mathbb{E}\langle e_1 e_2\rangle,$$
$$\mathbb{E}\langle R^2\rangle = \frac{1}{N} + \frac{N-1}{N}\,\mathbb{E}\langle r e_1 e_2\rangle.$$

Combining these equalities, the bounded-observable version of the comparison, and $R - r = O(N^{-1})$, yields the two consistency identities

$$\mathbb{E}[\bar{u}_N - T_\rho(\bar{u}_N)] \to 0, \qquad \mathbb{E}[\bar{u}_N^2 - \bar{u}_N T_\rho(\bar{u}_N)] \to 0. \tag{4.8}$$

Write $b = \mathbb{E}\operatorname{sech}^4(t + \sqrt{t}Z)$, so $\lambda = \xi''(q)b < 1$. For independent signs the function $m_t(z) = \frac{1}{2}\{\tanh(t+z) + \tanh(-t+z)\}$ is odd. Gaussian covariance differentiation, Cauchy–Schwarz and Jensen give

$$\Phi_\rho(0) = 0, \qquad |\Phi_\rho(u)| \leqslant \rho b|\xi'(u)| \leqslant \frac{\lambda}{p-1}|u| \quad (|u| \leqslant q).$$

Indeed $\mathbb{E}\,m_t'(\sqrt{t}Z)^2 \leqslant b$, and $|\xi'(u)| \leqslant \xi'(q)|u|/q = \xi''(q)|u|/(p-1)$. The second identity in (4.8) now implies $\mathbb{E}\,\bar{u}_N^2 \to 0$. Thus

$$\mathbb{E}\langle R^2\rangle \longrightarrow 0 \quad \text{for independent field patterns, for every } \rho \in [0,1]. \tag{4.9}$$

For identical signs, Cauchy–Schwarz shows that Ψ_ρ maps $[-q,q]$ into itself, and covariance differentiation gives

$$|\Psi_\rho'(u)| \leqslant \rho|\xi''(u)|b \leqslant \rho\lambda < 1.$$

These estimates hold at the endpoints by continuity, even if the Gaussian covariance is singular. Let u_ρ be the unique fixed point. Since $\Psi_\rho(0) = m(t)^2 = q^2 > 0$, continuity and the same covariance derivative show $u_\rho \in [q^2, q]$. The elementary contraction inequality

$$(u-u_\rho)(u-\Psi_\rho(u)) \geqslant (1-\lambda)(u-u_\rho)^2$$

and the two consistency identities imply $\mathbb{E}(\bar{u}_N - u_\rho)^2 \to 0$, and hence

$$\mathbb{E}\langle (R-u_\rho)^2\rangle \longrightarrow 0 \quad \text{for identical field patterns.} \tag{4.10}$$

No sign of $\xi''(u)$ for negative u was required; only its absolute bound was used.

4.5. **Gaussian Stein interpolation and the joint limit.** For independent uniform patterns $a_1,\dots,a_k$, define $Y_j = (F_N^{a_j}(t) - \mathbb{E}\,F_N(t))/\sqrt{N}$. Condition first on the patterns, whose conditional disorder means are identical by gauge invariance. Represent the disorder by a standard Gaussian vector g, let g' be independent, and put $g_\rho = \rho g + \sqrt{1-\rho^2}g'$. Gaussian integration by parts, equivalently the Ornstein–Uhlenbeck covariance identity (see [10, 14]), gives the exact identity

$$\mathbb{E}[Y_i f(Y)] = \sum_j \mathbb{E}[\partial_j f(Y) T_{ij,N}], \quad T_{ij,N} = \frac{1}{N}\int_0^1 \mathbb{E}_{g'}[\nabla F_N^{a_i}(g_\rho)\cdot\nabla F_N^{a_j}(g)]\,d\rho. \tag{4.11}$$

The identity remains true after averaging the patterns. Since $\nabla F_N^a = \langle v_\sigma\rangle_a$, the integrand after division by N is the cross Gibbs expectation of $\xi(R)$. Equations (4.9) and (4.10), boundedness of ξ, Jensen's inequality and dominated convergence in ρ give

$$T_{ij,N} \longrightarrow v\mathbb{1}_{\{i=j\}} \quad \text{in } L^1, \qquad v = \int_0^1 \xi(u_\rho)\,d\rho > 0.$$

Strict positivity follows, for example, from $u_\rho \geqslant q^2 > 0$. Apply (4.11) to $f(y) = e^{is\cdot y}$ and integrate the resulting differential equation for the characteristic function along a fixed ray in s. The L^1 errors tend to zero, giving

$$\mathbb{E}\,e^{is\cdot Y} \longrightarrow e^{-v\|s\|^2/2}.$$

Thus every fixed collection of independent field patterns has independent centered Gaussian limits with common variance v. In particular Proposition 3.3 is proved for every integer $p \geqslant 3$.

4.6. **An explicit expression for the field variance.** The variance in the joint limit can equivalently be written

$$v = \mathrm{Var}\big(\log\cosh(t+\sqrt{t}Z)\big) - \theta(q).$$

To verify this, let $c(s)$ be the covariance of two copies of $\log\cosh(t+\sqrt{t}Z)$ when their Gaussian arguments have covariance $s \in [0,t]$. Gaussian differentiation gives $c'(s) = \mathbb{E}\tanh(t+Z_1)\tanh(t+Z_2)$. Set $s(\rho) = \rho\xi'(u_\rho)$. The fixed-point equation says $c'(s(\rho)) = u_\rho$. The strict contraction permits differentiation. Integration by parts, first in ρ and then in s, gives

$$\int_0^1 \xi(u_\rho)\,d\rho = \xi(q) - \int_{q^2}^q s(u)\,du = \xi(q) - tq + c(t).$$

Since $c(0) = 0$ and $c(t)$ is the one-variable variance, this proves the displayed identity. Positivity also follows directly from $v = \int_0^1 \xi(u_\rho)d\rho > 0$.

5. Planted bands and the critical half limit

5.1. **An exact planting identity.** Represent $X_N(\sigma) = g \cdot v_\sigma$, where g is a standard Gaussian vector and

$$v_\sigma \cdot v_\tau = N\xi(R(\sigma,\tau)).$$

Gaussian change of measure gives, for every bounded measurable A,

$$\mathbb{E}_g[W_a(g)A(g)] = \mathbb{E}_g\, A(g+v_a), \tag{5.1}$$

$$\mathbb{E}_g[W_a(g)W_b(g)A(g)] = e^{N\xi(R(a,b))}\,\mathbb{E}_g\, A(g+v_a+v_b). \tag{5.2}$$

These identities are exact and use the true annealed normalization.

5.2. **One planted informative band has a centered Gaussian logarithm.** Choose $\varepsilon > 0$ so small that

$$\varepsilon < q/2, \qquad \beta_c^2 \varepsilon^{p-2} < 1/2. \tag{5.3}$$

Define the broad band mass, as a function of the original unplanted disorder, by

$$C_a(g) = \mathbb{E}_{\nu\sim\mu_N}\, W_\nu(g)\mathbb{1}_{B(a,\nu)},$$

where

$$B(a,\nu) = \begin{cases} \{|R(a,\nu)| \geqslant \varepsilon\}, & p \text{ even}, \\ \{R(a,\nu) \geqslant \varepsilon\}, & p \text{ odd}. \end{cases}$$

For the δ in (2.7), reduced if necessary so that $\delta < q - \varepsilon$, define the narrow positive band

$$C_a^{\mathrm{near}}(g) = \mathbb{E}_\nu\, W_\nu(g)\mathbb{1}_{\{|R(a,\nu)-q|\leqslant\delta\}}.$$

The narrow band is always contained in the broad band.

Lemma 5.1 (One-plant band approximation)**.** *Uniformly in deterministic a, under the original Gaussian disorder,*

$$\log C_a(g+v_a) = F_N^a(t) - N\{\xi(1)/2 + \theta(q)\} + o_{\mathbb{P}}(\sqrt{N}), \tag{5.4}$$

$$\log C_a^{\mathrm{near}}(g+v_a) = F_N^a(t) - N\{\xi(1)/2 + \theta(q)\} + o_{\mathbb{P}}(\sqrt{N}). \tag{5.5}$$

In particular the two logarithms divided by $\sqrt{N}$ converge to $N(0,v)$.

Proof. Gauge a to the all-plus pattern and write $m = M(\nu)$. Under planting,

$$C_a(g+v_a) = \mathbb{E}_\nu\, e^{X_N(\nu)+N\xi(m)-N\xi(1)/2}\mathbb{1}_{B(a,\nu)}.$$

Put

$$B_N(u) = F_N^a(\xi'(u)) - N\{\xi(1)/2+\theta(u)\}, \qquad b(u) = f(\xi'(u)) - \xi(1)/2 - \theta(u).$$

The thermodynamic Parisi upper bound evaluated at the RS trial δ_u gives $b(u) \leqslant C(u)$. At $u = q$, Proposition 3.2 and the critical contact yield equality:

$$\begin{aligned} f(t) &= \Psi(t) + \tfrac{1}{2}\{\xi(1) - \xi(q) - (1-q)t\} \\ &= \xi(1)/2 + \theta(q), \end{aligned}$$

so $b(q) = 0$, whereas $b(u) < 0$ for every $u \in [\varepsilon, 1] \setminus \{q\}$.

For each allowed positive magnetization u, exact tangent linearization at that magnetization gives

$$\mathbb{E}_\nu\, e^{X_N(\nu)+N\xi(M(\nu))-N\xi(1)/2}\mathbb{1}_{\{M(\nu)=u\}} \leqslant e^{B_N(u)}.$$

Thus

$$C_a(g+v_a) \leqslant 2(N+1)\exp\{\max_{u\in[\varepsilon,1]} B_N(u)\}. \tag{5.6}$$

The factor two is only needed for even p; in that case spin-flip symmetry gives $F_N^a(-h) = F_N^a(h)$ for the same disorder. For odd p, only the positive magnetization band is present.

Let J_0 be a small fixed neighborhood of q whose image under ξ' is contained in J. On the compact complement of J_0, $b \leqslant -c < 0$. Uniform convergence of the mean pressures follows from their deterministic Lipschitz bounds and pointwise convergence. Gaussian concentration, a grid of polynomial size, and the same deterministic Lipschitz bounds then imply that

$$\max_{u \notin J_0} B_N(u) \leqslant -cN/2$$

with probability tending to one. This step needs only qualitative Parisi upper bounds outside the field neighborhood, including for odd p.

On J_0, write

$$B_N(u) = Nb(u) + \sqrt{N}\, Y_N(\xi'(u)) + d_N(\xi'(u)), \quad Y_N(h) = \frac{F_N^a(h) - \mathbb{E}\, F_N(h)}{\sqrt{N}}, \quad \sup_{h \in J} |d_N(h)| \leqslant C.$$

Proposition 3.2 gives

$$\mathbb{E}\, |Y_N(h) - Y_N(k)|^2 \leqslant C|h-k|^2.$$

The one-dimensional marginal CLT in Proposition 3.3, and the standard one-parameter Kolmogorov tightness criterion, imply tightness of Y_N in $C(J)$. Since b has a unique maximum at q, any maximizing sequence in (5.6) converges in probability to q, and stochastic equicontinuity gives

$$\max_{u \in [\varepsilon, 1]} B_N(u) = B_N(q) + o_{\mathbb{P}}(\sqrt{N}).$$

This is a unique-maximizer argument; no N-dependent overlap large-deviation rate is assumed.

For the reverse inequality, let

$$d(m,q) = \xi(m) - \xi(q) - t(m-q).$$

For all positive m, convexity of ξ on $[0,1]$ gives $d(m,q) \geqslant 0$. Therefore

$$C_a^{\mathrm{near}}(g + v_a) \geqslant e^{B_N(q)} \langle \mathbb{1}_{\{|M-q| \leqslant \delta\}} \rangle_{h=t}.$$

By (3.16), the Gibbs probability on the right tends to one in probability. Since $C_a^{\mathrm{near}} \leqslant C_a$, the lower bound and (5.6) prove both approximations. Finally (3.13) and $b(q) = 0$ give $B_N(q) = F_N^a(t) - \mathbb{E}\, F_N(t) + O(1)$, proving the stated Gaussian limit. ∎

5.3. **Two almost orthogonal plants do not change the narrow band.**

Lemma 5.2 (Local second-plant perturbation)**.** *For independent uniform a, b, independent of g,*

$$\log C_a^{\mathrm{near}}(g + v_a + v_b) - \log C_a^{\mathrm{near}}(g + v_a) \longrightarrow 0 \quad \textit{in probability.}$$

The same holds with a, b exchanged. Consequently, under double planting,

$$\left(\frac{\log C_a^{\mathrm{near}}(g + v_a + v_b)}{\sqrt{N}}, \frac{\log C_b^{\mathrm{near}}(g + v_a + v_b)}{\sqrt{N}} \right) \Longrightarrow (\sqrt{v} Z_1, \sqrt{v} Z_2) \tag{5.7}$$

with independent coordinates.

Proof. For Ising vectors the exact Hamming geometry gives

$$|R(\nu, b)| + R(\nu, a) \leqslant 1 + |R(a,b)|. \tag{5.8}$$

For example, gauge $a = \mathbf{1}$, split sites according to $b_i = \pm 1$, and apply $|x + y| \leqslant |x| + |y|$; equivalently use the Hamming triangle inequality with both b and $-b$. On $|R(a,b)| \leqslant \gamma$, every ν in the narrow a-band therefore satisfies $|R(\nu, b)| \leqslant b$, with b as in (2.7).

Let $\pi_{g,a}$ be the probability measure obtained by normalizing the one-planted narrow-band weights. It depends on g, a, and is independent of the random pattern b. The ratio of the two band masses is exactly

$$\frac{C_a^{\mathrm{near}}(g+v_a+v_b)}{C_a^{\mathrm{near}}(g+v_a)} = \int e^{N\xi(R(\nu,b))}\, d\pi_{g,a}(\nu).$$

Use $|e^x-1| \leqslant e^{|x|}-1$, and average over the independent uniform b. On the good event $|R(a,b)| \leqslant \gamma$, the expected absolute difference of this ratio from one is bounded by

$$\sup_{\nu\in\Sigma_N} \mathbb{E}_b\left[(e^{N\beta_c^2|R(\nu,b)|^p}-1)\mathbb{1}_{\{|R(\nu,b)|\leqslant b\}}\right].$$

The supremum is redundant: for every fixed ν, the overlap has the distribution $S_N = N^{-1}\sum_{i=1}^N \epsilon_i$ of the mean of independent Rademachers. On $|S_N| \leqslant b$,

$$N\beta_c^2|S_N|^p \leqslant cNS_N^2, \qquad c = \beta_c^2 b^{p-2} < 1/2.$$

Gaussian linearization and $\cosh x \leqslant e^{x^2/2}$ give

$$\sup_N \mathbb{E}\, e^{c'NS_N^2} \leqslant (1-2c')^{-1/2} < \infty, \qquad c < c' < 1/2.$$

Since $N|S_N|^p \to 0$ in probability for $p > 2$, uniform integrability proves that the displayed expectation tends to zero. The omitted event $|R(a,b)| > \gamma$ has probability tending to zero. Thus the positive ratio tends to one in probability, and its logarithm tends to zero. Applying Lemma 5.1 to each one-plant narrow band, and then Proposition 3.3, proves (5.7). No assertion about the broad band under double planting is needed. ∎

5.4. **Critical overlap vanishing and the slowly vanishing cutoff.**

Lemma 5.3. *At $\beta = \beta_c$,*

$$\mathbb{E}\, G_N^{\otimes 2}(|R_{12}| \geqslant \varepsilon) \longrightarrow 0.$$

The same holds for inverse temperatures $\beta_N \to \beta_c$.

Proof. The thermodynamic pressure $f_0(\beta) = \lim N^{-1}\,\mathbb{E}\log Z_N(\beta)$ is convex, lies below $\beta^2/2$, and agrees with it on $[0,\beta_c]$. Annealed contact and convexity force differentiability at β_c, with derivative β_c. Convergence of derivatives of convex functions therefore gives, also along $\beta_N \to \beta_c$,

$$\beta_N\{1-\mathbb{E}\langle R_{12}^p\rangle_{\beta_N}\} = \frac{d}{d\beta}\,\frac{1}{N}\,\mathbb{E}\log Z_N(\beta)\Big|_{\beta=\beta_N} \longrightarrow \beta_c.$$

For even p, this proves the claim immediately.

For odd p, first remove negative macroscopic overlaps. The mean of $\log L_N(\beta_N)$ is $o(N)$, and Gaussian concentration gives $\mathbb{P}(L_N < e^{-N\eta}) \leqslant e^{-c_\eta N}$ for every fixed $\eta > 0$, eventually in N. On the complementary event, the normalized pair partition is bounded by $e^{2N\eta}$ times its annealed value. Hence

$$\mathbb{E}\, G_N^{\otimes 2}(R_{12} \leqslant -\varepsilon) \leqslant e^{-c_\eta N} + e^{2N\eta}\,\mathbb{E}_{a,b}\, e^{N\beta_N^2 R(a,b)^p}\mathbb{1}_{\{R(a,b)\leqslant -\varepsilon\}}.$$

Choosing $\eta < \beta_c^2\varepsilon^p/4$ makes the second term exponentially small. Thus the negative part of $\mathbb{E}\, R_{12}^p$ vanishes, and its positive part must vanish as well. This proves the claim for odd p. ∎

Let

$$u_N = \mathbb{E}\, G_N^{\otimes 2}(B(a,b)), \qquad k_N = (u_N + N^{-1})^{1/4}, \qquad \ell_N^{\mathrm{cut}} = -\log k_N.$$

Changing finitely many initial values if necessary, $0 < k_N < 1$, $k_N \to 0$, $\ell_N^{\mathrm{cut}} \to \infty$, $\ell_N^{\mathrm{cut}} \leqslant \frac{1}{4}\log N = o(\sqrt{N})$, and

$$u_N/k_N \longrightarrow 0. \tag{5.9}$$

Set

$$A_a(g) = \{C_a(g) \leqslant k_N\}, \qquad \widetilde{L}_N(g) = \mathbb{E}_a\, W_a(g)\mathbb{1}_{A_a(g)}.$$

This cutoff is allowed to depend on the deterministic sequence u_N; no quantitative overlap estimate at the zero-field critical point is assumed.

5.5. **First and second moments of the truncated likelihood.** The exact one-plant identity and Lemma 5.1 give

$$\mathbb{E}\,\widetilde{L}_N = \mathbb{E}_{a,g}\,\mathbb{1}_{\{C_a(g+v_a)\leqslant k_N\}} \longrightarrow \frac{1}{2}, \tag{5.10}$$

since $\log k_N/\sqrt{N} \to 0$.

The dangerous macroscopic-overlap contribution to the second moment has a direct deterministic bound. For fixed g, a on A_a,

$$\mathbb{E}_b\, W_b(g)\mathbb{1}_{A_b(g)}\mathbb{1}_{B(a,b)} \leqslant C_a(g) \leqslant k_N.$$

Therefore

$$\mathbb{E}_g\,\mathbb{E}_{a,b}\, W_a W_b \mathbb{1}_{A_a}\mathbb{1}_{A_b}\mathbb{1}_{B(a,b)} \leqslant k_N\,\mathbb{E}\,\widetilde{L}_N = o(1). \tag{5.11}$$

There is no second-moment exponential-rate comparison to prove in this region.

For odd p, the untruncated contribution of $R(a,b) \leqslant -\varepsilon$ is at most $e^{-N\beta_c^2\varepsilon^p}$. The remaining contribution is $|R(a,b)| < \varepsilon$ for both parities. By (5.2), it equals

$$\mathbb{E}_{a,b,g}\, e^{N\xi(R(a,b))}\mathbb{1}_{\{|R(a,b)|<\varepsilon\}}\mathbb{1}_{A_a(g+v_a+v_b)}\mathbb{1}_{A_b(g+v_a+v_b)}.$$

The Rademacher uniform-integrability argument from Lemma 5.2, now using (5.3), shows that

$$\mathbb{E}_{a,b}\left|e^{N\xi(R(a,b))} - 1\right|\mathbb{1}_{\{|R(a,b)|<\varepsilon\}} \longrightarrow 0, \qquad \mathbb{P}(|R(a,b)| \geqslant \varepsilon) \to 0.$$

Since the broad band contains the narrow band,

$$A_a(g) \subseteq \{C_a^{\mathrm{near}}(g) \leqslant k_N\}.$$

Consequently (5.7) yields

$$\limsup_{N\to\infty} \mathbb{E}_{g,a,b}\, W_a W_b \mathbb{1}_{A_a}\mathbb{1}_{A_b}\mathbb{1}_{\{|R(a,b)|<\varepsilon\}} \leqslant \frac{1}{4}. \tag{5.12}$$

Notice the direction of this inequality: no joint CLT for the broad bands under double planting has been asserted or used.

Combining (5.10)–(5.12) gives

$$\limsup_N \mathbb{E}\,\widetilde{L}_N^2 \leqslant 1/4, \qquad \mathbb{E}\,\widetilde{L}_N \to 1/2.$$

Thus

$$\mathbb{E}(\widetilde{L}_N - 1/2)^2 \longrightarrow 0. \tag{5.13}$$

5.6. **The truncation retains the partition function in probability.** For a fixed disorder realization, write $G_N(a) = \mu_N(a)W_a/L_N$. Then

$$C_a = L_N G_N(B(a,\cdot)),$$

and Markov's inequality inside the Gibbs measure gives

$$\begin{aligned} 0 \leqslant L_N - \widetilde{L}_N &= L_N G_N(C_a > k_N) \\ &\leqslant k_N^{-1} L_N^2 G_N^{\otimes 2}(B(a,b)). \end{aligned}$$

For every fixed $K > 0$, $\mathbb{P}(L_N > K) \leqslant 1/K$, since $\mathbb{E}\, L_N = 1$. On $L_N \leqslant K$, the expectation of the final bound is at most $K^2 u_N/k_N \to 0$. It follows that

$$L_N - \widetilde{L}_N \longrightarrow 0 \quad \text{in probability.}$$

Together with (5.13), this proves the exact-covariance assertion of Theorem 3.4. The logarithmic statement follows by continuity of log at $1/2$.

5.7. **Distinct indices and exact annealed centering.** Let $M_N = \binom{N}{p}$ and

$$H_N^{\mathrm{d}}(\sigma) = \sqrt{N/M_N} \sum_{|e|=p} g_e \sigma_e, \qquad \operatorname{Var} H_N^{\mathrm{d}} = N.$$

The exact-covariance tensor Hamiltonian admits the independent decomposition

$$H_N^{\mathrm{ord}} = \sqrt{c_N}\, H_N^{\mathrm{d}} + K_N, \qquad c_N = (N)_p/N^p = 1 - \binom{p}{2}/N + O(N^{-2}),$$

where K_N consists of the repeated-index Walsh terms. Its variance is $w_N = N(1-c_N) = \binom{p}{2} + O(N^{-1})$, and its covariance $u_N(r)$ is uniformly bounded. Moreover $u_N(r_N) \to 0$ whenever $r_N \to 0$, for fixed $p \geqslant 3$. The latter follows by grouping repeated indices according to their parity: their leading covariance is $\binom{p}{2} r^{p-2}$, and all remaining terms are uniformly $O(N^{-1})$; a constant Walsh term is $O(N^{1-p/2})$, which tends to zero whenever it can occur here.

Put $\widehat{\beta}_N = \beta_c \sqrt{c_N}$. Conditional on the distinct-index disorder, the ratio contributed by K_N, after dividing by its annealed factor, is

$$T_N = \left\langle e^{\beta_c K_N(\sigma) - \beta_c^2 w_N/2} \right\rangle_{\mathrm{d},\widehat{\beta}_N}.$$

It has conditional mean one and conditional second moment

$$\mathbb{E}_{K_N}\, T_N^2 = \left\langle e^{\beta_c^2 u_N(R_{12})} \right\rangle_{\mathrm{d},\widehat{\beta}_N}.$$

The distinct-index thermodynamic pressure is the same, and the convex-derivative argument in Lemma 5.3 applies at $\widehat{\beta}_N \to \beta_c$. Its negative-overlap argument for odd p is unchanged by bounded covariance corrections. Thus the overlap tends to zero. Boundedness and the central vanishing of u_N imply $T_N \to 1$ in L^2.

It remains to replace $\widehat{\beta}_N$ by β_c in the distinct-index likelihood. Write $\ell_N^{\mathrm{d}}(\beta) = \log Z_N^{\mathrm{d}}(\beta) - \beta^2 N/2$. Since $|\widehat{\beta}_N - \beta_c| = O(N^{-1})$, convergence in probability of the random convex pressure derivatives to β_c, uniformly between these two endpoints by convexity, gives

$$\ell_N^{\mathrm{d}}(\widehat{\beta}_N) - \ell_N^{\mathrm{d}}(\beta_c) = o_{\mathbb{P}}(1).$$

Here random pressure convergence follows from Gaussian concentration and the thermodynamic limit, and differentiability at the annealed contact again supplies derivative convergence. The coupled identity

$$L_N^{\mathrm{ord}}(\beta_c) = L_N^{\mathrm{d}}(\widehat{\beta}_N) T_N$$

therefore transfers the half limit to the variance-N distinct-index model.

Finally, the convention $\sqrt{p!/N^{p-1}}\sum_{|e|=p} g_e\sigma_e$ equals $\sqrt{c_N}H_N^{\mathrm{d}}$. Its self-variance is Nc_N, so its exact annealed centering is $\beta_c^2 Nc_N/2$ for the normalized partition function. The same $O(N^{-1})$ temperature comparison proves the assertion under that convention.

6. Three-spin band moment estimates

Inputs and finite certificates. The field estimates and the critical scalar contact were established in Sections 3 and 4. This argument supplies the own-band overlap and magnetization moments needed for the subsequent cross-band argument. It uses finite rooted covariance systems, whose exact integer construction and rational invertibility certificate are provided in `pure_band_rooted_cavity.py` and `pure_p3_cavity_certificate.py`.

6.1. **Statement and localization.** Write $\xi(r) = \beta_c^2 r^3$, and let $q, t = \xi'(q)$ be the nonzero critical contact. For a fixed pattern a, gauge it to the all-plus pattern and use the Gibbs law

$$G_N^\delta(\sigma) \propto \exp\{X_N(\sigma) + N\xi(M(\sigma))\}\mathbb{1}_{\{|M(\sigma)-q|\leqslant\delta\}}, \qquad M(\sigma) = N^{-1}\sum_i \sigma_i.$$

The covariance of X_N is $N\xi(R)$. Denote disorder and replica expectation by ν.

Theorem 6.1. *For every fixed finite integer r_0, one can choose a fixed sufficiently small band width $\delta > 0$ such that*

$$\nu|M_1 - q|^{2r} + \nu|R_{12} - q|^{2r} \leqslant C_r N^{-r}, \qquad 1 \leqslant r \leqslant r_0.$$

The following finite-size uniformity is also valid. Fix $K < \infty$. Let $\widetilde{X}_N$ be any centered Gaussian Hamiltonian with a valid covariance kernel satisfying

$$\sup_{\sigma,\tau}\left|\mathbb{E}\,\widetilde{X}_N(\sigma)\widetilde{X}_N(\tau) - N\xi(R(\sigma,\tau))\right| \leqslant K,$$

and let its deterministic planting potential $\widetilde{V}_N$ satisfy

$$\inf_{c\in\mathbb{R}}\sup_{\sigma}\left|\widetilde{V}_N(\sigma) - N\xi(M(\sigma)) - c\right| \leqslant K.$$

The same moment bounds hold, with constants $C_{r,K}$, for the Gibbs law with Hamiltonian $\widetilde{X}_N + \widetilde{V}_N$ on $\{|M - q_N| \leqslant \delta_N\}$, whenever $|q_N - q| + |\delta_N - \delta| \leqslant K/N$. The centered variables in the moment bound may use either q or q_N. This perturbation assertion is deduced by the comparison lemma below only after the canonical critical-band bounds have been proved. In particular, it covers the full and one-site bulk models, the two distinct-index normalizations, and the finite-size coefficient changes used later. No uniform high-moment theorem on an N-independent parameter neighborhood is needed. In particular one may take $r_0 = 32$, giving moments through order 64.

Fix a small macro precision $\eta > 0$, to be chosen after the finite list of moment orders is specified. Under the ordinary RS field Gibbs law G_t, the field estimate gives constants $c_\eta, C_\eta > 0$ with

$$\mathbb{E}\,G_t^{\otimes 2}(|R_{12} - q| > \eta) + \mathbb{E}\,G_t(|M - q| > \eta) \leqslant C_\eta e^{-c_\eta N}.$$

For $|M - q| \leqslant \delta$, write

$$d(M,q) = \xi(M) - \xi(q) - t(M - q), \qquad 0 \leqslant d(M,q) \leqslant C\delta^2.$$

The band law is the field law reweighted by $e^{Nd(M,q)}\mathbb{1}_{\{|M-q|\leqslant\delta\}}$. Its denominator is at least $G_t(|M-q|\leqslant\delta)$, which is at least $1/2$ outside an event of probability $Ce^{-c_\delta N}$. Thus, for δ sufficiently small compared with η,

$$\nu\mathbb{1}_{\{|R_{12}-q|>\eta\}}\leqslant Ce^{-cN},\qquad |M-q|\leqslant\delta<\eta. \tag{6.1}$$

Indeed the reweighting of two replicas costs at most $4e^{2CN\delta^2}$, which is absorbed by the field macro gap. These bounds hold for any fixed finite set of replicas by a union bound.

Lemma 6.2 (Band edges and bounded finite-size perturbations)**.** *Fix the critical pure covariance $\xi(r)=\beta_c^2r^p$, its positive contact $q\in(0,1)$, and $t=\xi'(q)$. Put $\theta(u)=u\xi'(u)-\xi(u)$, and let*

$$F_n(h)=\log 2^{-n}\sum_{\sigma\in\{-1,1\}^n}\exp\{X_n(\sigma)+hnM(\sigma)\},\qquad \mathbb{E}\,X_n(\sigma)X_n(\tau)=n\xi(R(\sigma,\tau)).$$

The field and contact inputs used in this lemma are precisely these: $n^{-1}\,\mathbb{E}\,F_n(h)\to f(h)$ locally uniformly near t, the function f is differentiable at t with $f'(t)=q$, and, for some $\delta_0>0$ with $[q-2\delta_0,q+2\delta_0]\subset(0,1)$,

$$f(\xi'(u))-\theta(u)<f(t)-\theta(q),\qquad 0<|u-q|\leqslant 2\delta_0. \tag{6.2}$$

Shrink δ_0, if necessary, so that all the fields $\xi'(u)$ in this display lie in the neighborhood of local uniform pressure convergence.

Fix $0<\delta\leqslant\delta_0$ and constants $K_0,B_0<\infty$. For each n, consider any centered Gaussian Hamiltonian $\widetilde{X}_{n,\alpha}$ and deterministic potential $\widetilde{V}_{n,\alpha}$ such that

$$\sup_{\sigma,\tau}\left|\mathbb{E}\,\widetilde{X}_{n,\alpha}(\sigma)\widetilde{X}_{n,\alpha}(\tau)-n\xi(R(\sigma,\tau))\right|\leqslant K_0, \tag{6.3}$$

$$\inf_{c\in\mathbb{R}}\sup_{\sigma}\left|\widetilde{V}_{n,\alpha}(\sigma)-n\xi(M(\sigma))-c\right|\leqslant B_0. \tag{6.4}$$

All covariance kernels in this statement are assumed to be valid Gaussian covariance kernels on the indicated finite spin set. Let $D\subset\{-1,1\}^n$ be any deterministic spin set satisfying

$$\{|M-q|\leqslant\delta/2\}\subset D\subset\{|M-q|\leqslant 2\delta\}, \tag{6.5}$$

and let $\widetilde{G}_{n,\alpha,D}$ be the Gibbs law on D with Hamiltonian $\widetilde{X}_{n,\alpha}+\widetilde{V}_{n,\alpha}$. All assertions below are for sufficiently large n, so that the inner core in (6.5) is nonempty. Then there are $c,C>0$, independent of n,α,D, such that

$$\mathbb{E}\,\widetilde{G}_{n,\alpha,D}(|M-q|\geqslant\delta/2)\leqslant Ce^{-cn}. \tag{6.6}$$

For two supports D,D' satisfying (6.5), any fixed replica number ℓ, and the same realization of the Hamiltonian,

$$\mathbb{E}\sup_{|F|\leqslant1}\left|\widetilde{G}^{\otimes\ell}_{n,\alpha,D}(F)-\widetilde{G}^{\otimes\ell}_{n,\alpha,D'}(F)\right|\leqslant C_\ell e^{-cn}. \tag{6.7}$$

Here the supremum is over bounded functions of the replicas, and it may be taken separately for each disorder realization.

Write G_n^δ for the canonical critical pure planted law on $D_0=\{|M-q|\leqslant\delta\}$. If $F\geqslant0$ is a spin observable of ℓ replicas that does not depend on the Gaussian coordinates, then

$$\mathbb{E}\,\widetilde{G}^{\otimes\ell}_{n,\alpha,D}(F)\leqslant e^{(2\ell^2+\ell)K_0+2\ell B_0}\,\mathbb{E}(G_n^\delta)^{\otimes\ell}(F)+C_\ell\|F\|_\infty e^{-cn}. \tag{6.8}$$

In particular, once the canonical critical moment estimates have been proved, all of them transfer to this perturbation class. The bounds remain uniform for any coupling of a fixed

number of such disorders, including every correlation in $[-1,1]$*, when the observable is a union of marginal boundary events.*

Proof. We first prove the boundary bound for the canonical Gaussian Hamiltonian and potential $n\xi(M)$, uniformly over the supports in (6.5). This part uses no high-moment estimate.

Differentiability $f'(t)=q$ gives exponential magnetic localization in the ordinary field Gibbs law. Indeed, for a fixed $\varepsilon>0$, choose a sufficiently small $s>0$ so that

$$f(t+s)-f(t)\leqslant s(q+\varepsilon/2).$$

The exponential-tilt inequality is

$$G_{n,t}(M\geqslant q+\varepsilon)\leqslant \exp\{F_n(t+s)-F_n(t)-ns(q+\varepsilon)\}.$$

Both log partition functions have squared Gaussian Lipschitz constants at most $n\xi(1)$. Local uniform convergence of their means and Gaussian concentration imply $\mathbb{E}\,G_{n,t}(M\geqslant q+\varepsilon)\leqslant C_\varepsilon e^{-c_\varepsilon n}$. A negative tilt gives the other tail. In particular,

$$\mathbb{P}\{G_{n,t}(|M-q|\leqslant \delta/2)<1/2\}\leqslant Ce^{-cn}. \tag{6.9}$$

Put $a=f(t)-\theta(q)$. Continuity and (6.2) give a fixed $d>0$ such that

$$\sup_{\delta/2\leqslant|u-q|\leqslant 2\delta}\{f(\xi'(u))-\theta(u)\}\leqslant a-d.$$

On the slice $M=u$, tangent linearization is exact and gives

$$2^{-n}\sum_{M(\sigma)=u}e^{X_n(\sigma)+n\xi(M(\sigma))}\leqslant \exp\{F_n(\xi'(u))-n\theta(u)\}.$$

There are at most $n+1$ such slices. Uniform convergence of the mean pressures and Gaussian concentration therefore show, outside an event of probability Ce^{-cn}, that their total mass on $\{\delta/2\leqslant|M-q|\leqslant 2\delta\}$ is at most

$$(n+1)e^{n(a-3d/4)}.$$

Since ξ is convex on this positive magnetization interval, $\xi(m)\geqslant tm-\theta(q)$. Thus every allowed support has partition function at least

$$e^{F_n(t)-n\theta(q)}G_{n,t}(|M-q|\leqslant\delta/2).$$

Concentration of $F_n(t)$, its limiting mean, and (6.9) bound this denominator below by $\frac{1}{2}e^{n(a-d/4)}$ outside another exponentially small event. The ratio is at most $2(n+1)e^{-nd/2}$ there and at most one everywhere. This proves the canonical version of (6.6).

We next give the comparison used for perturbations. Keep a fixed support D. Interpolate two centered Gaussian Hamiltonians by independent copies and interpolate their deterministic potentials linearly. For a nonnegative Gaussian-coordinate-independent observable F of ℓ replicas, Gaussian differentiation gives

$$\begin{aligned}\partial_s\nu_s(F)=\frac{1}{2}\sum_{i,j\leqslant\ell}\nu_s(F\Delta K_{ij})-\ell\sum_{i\leqslant\ell}\nu_s(F\Delta K_{i,\ell+1})+\frac{\ell(\ell+1)}{2}\nu_s(F\Delta K_{\ell+1,\ell+2})\\ -\frac{\ell}{2}\nu_s(F\Delta K_{\ell+1,\ell+1})+\sum_{i\leqslant\ell}\nu_s(F\Delta V_i)-\ell\nu_s(F\Delta V_{\ell+1}).\end{aligned}$$

The bounds on the covariance and potential differences imply

$$|\partial_s\nu_s(F)|\leqslant\{(2\ell^2+\ell)K_0+2\ell B_0\}\nu_s(F).$$

Additive constants in the deterministic potential cancel. Gronwall's inequality gives the corresponding two-sided multiplicative comparison on the common support D. Apply it first to the boundary indicator. The canonical boundary bound proves (6.6) for every perturbation.

For the same Hamiltonian on two supports D, D', their symmetric difference lies in $\{|M-q| \geqslant \delta/2\}$. The total variation distance of the two normalized restrictions is bounded by the sum of their probabilities of this symmetric difference. Tensorizing to ℓ replicas and using (6.6) proves (6.7). The multiplicative comparison on D, followed by canonical support replacement from D to D_0, proves (6.8). Finally, a union bound over marginal boundary events uses no independence of the disorders, which proves the asserted coupling uniformity. ■

Verification of the field interface. The hypotheses of this lemma are supplied by the earlier field and contact statements at the single critical pure model. Local uniform convergence of $n^{-1}\,\mathbb{E}\,F_n(h)$ follows from (3.13), or from pointwise convergence and the deterministic Lipschitz bound in h. The smooth RS pressure has derivative $m(h)$, so $f'(t) = m(t) = q$. The RS trial bound used in Lemma 5.1 gives

$$f(\xi'(u)) - \theta(u) \leqslant \xi(1)/2 + C(u), \qquad f(t) - \theta(q) = \xi(1)/2.$$

The strict critical contact inequality $C(u) < 0$ for $u \neq q$ therefore gives (6.2). This verification uses neither planted-band high moments nor an RS theorem for an arbitrary perturbed odd mixture.

The finite-size models and cavity lifts covered by the lemma. All total covariance and deterministic-potential bounds in this paragraph are uniform for fixed p and a fixed number of removed sites. The ordered model, the unit-self-variance distinct model, and the standard-coefficient distinct model have covariance polynomials C_N, including the factor β_c^2, satisfying

$$\sup_{|r|\leqslant 1} |C_N(r) - N\xi(r)| \leqslant K, \qquad V_N(\sigma) = C_N(M(\sigma)).$$

For the distinct models this follows directly from the finite set-partition expansion of their covariance; multiplication by $c_{N,p} = 1 + O(N^{-1})$ preserves the bound. A coefficient or temperature change of size $O(N^{-1})$ has the same property. Adding or removing the independent constant Gaussian modes used in the four-spin argument changes the covariance by $O(N^{-1})$ and changes normalized Gibbs probabilities by exactly zero.

For d removed sites and $n = N - d$, the Gaussian component even in every removed spin has covariance

$$C_N^{\mathrm{bulk}}(r) = 2^{-d} \sum_{\varepsilon\in\{-1,1\}^d} C_N\left(\frac{nr + \sum_{j=1}^d \varepsilon_j}{N}\right).$$

The deterministic even planting potential is the same expression with $r = M_n$. If $S = \sum_j \varepsilon_j$, then

$$\left|N\xi\left(\frac{nr+S}{N}\right) - n\xi(r)\right| \leqslant d\{2\|\xi'\|_\infty + \|\xi\|_\infty\}.$$

Thus the bulk law satisfies (6.3)– (6.4) relative to the canonical critical n-spin law. This is a same-size comparison; it does not assume a high-moment bound in advance for the bulk law.

In particular, for $p = 3$, $c = \beta_c^2$, and $n = N - 1$, the exact formulas are

$$\mathbb{E}\,B_N(\sigma)B_N(\tau) = \frac{cn^3}{N^2}R_n(\sigma,\tau)^3 + \frac{3cn}{N^2}R_n(\sigma,\tau), \qquad V_N^{\mathrm{bulk}}(\sigma) = \frac{cn^3}{N^2}M_n(\sigma)^3 + \frac{3cn}{N^2}M_n(\sigma).$$

The additional linear Gaussian process has self-variance $3cn/N^2 = O(N^{-1})$, hence standard deviation $O(N^{-1/2})$ at a fixed spin configuration. The deterministic linear correction is uniformly $O(N^{-1})$.

For a full spin and its $n = N - d$ bulk restriction, $|M_N - M_n| \leqslant 2d/N$. Consequently the original full band and the constraint $\{|M_n - q| \leqslant \delta\}$ with all removed spins free both satisfy the support sandwich (6.5), viewed on the full spin set, for all sufficiently large N. Their symmetric difference is in the fixed boundary region. The same is true after changes of the center or either endpoint by $O(N^{-1})$. Equation (6.7) therefore justifies every such band-boundary deletion or free last-spin lift with error $O(e^{-cN})$ in bounded fixed-replica expectations.

The actual finite-site cavity fields have uniformly bounded Gaussian variances, and their deterministic fields are uniformly bounded. Every cavity interpolation used here changes only these fields and keeps their variances and deterministic magnitudes bounded. Relative to the full canonical model, its total covariance and deterministic potential therefore change by $O(1)$. The preceding boundary and same-support interpolation comparisons are uniform along these Gaussian paths. For a scalar-mixture replacement whose covariance depends on the bulk disorder, first condition on that bulk. The remaining cavity fields are Gaussian, their covariance differences are uniformly bounded, and the bulk weights and the freely lifted support are fixed during the interpolation. The same-support streaming comparison applies conditionally, with a constant independent of the bulk. Comparing to the actual cavity endpoint and then averaging transfers its boundary bound. This conditional argument does not assert that the bulk and the replacement cavity fields are jointly Gaussian. The separate conditional Gaussian-streaming and normalizer proof in the cross-overlap section establishes the more precise weighted mixture error used there; the unconditional lemma above is not a substitute for it.

6.2. **Exact streaming and the seven unrooted types.** For a nonempty thermal replica set A with $|A| \in \{1, 2\}$, put

$$T_A = N^{-1} \sum_{i=1}^{N} \left(\prod_{a \in A} \sigma_i^a - q \right), \qquad e_A = \prod_{a \in A} \sigma_N^a.$$

Thus $T_{\{a\}} = M_a - q$ and $T_{\{a,b\}} = R_{ab} - q$. Let T_A^- be the same centered sum over the first $N - 1$ sites, still divided by N. Then $T_A = T_A^- + (e_A - q)/N$.

The Gaussian even/odd last-site split has cavity covariance $\xi'(R^-) + O(N^{-2})$. The planted term splits into an even bulk term and a last-spin field $\xi'(M^-) + O(N^{-2})$. Interpolate the cavity Gaussian process to an independent common scalar Gaussian with variance t, and its deterministic field to t. The endpoint field at interpolation zero is $t + \sqrt{t}Z$, independent of the bulk. Boundary errors are exponentially small by the preceding paragraph.

The exact one-site bulk covariance and planting potential are the ones displayed after Lemma 6.2; their total differences from the canonical critical $(N - 1)$-spin law are uniformly bounded. During the proof of the canonical critical moment bound, we use this lemma only for macroscopic localization, boundary removal, and comparison of bounded fixed-replica observables along a finite-site cavity path. No high-moment estimate for the perturbed bulk law is assumed. The reference last-spin field remains $t + \sqrt{t}Z$, with the unperturbed critical q, t; the finite-size corrections enter the explicit $O(N^{-1})$ remainder. Endpoint moments along the path are compared back to the original canonical N-spin law, with exponentially small boundary errors. After this canonical bootstrap is complete for every sufficiently large size, the same-size comparison (6.8) supplies the bulk and finite-convention moment bounds used subsequently.

Set

$$k=\xi''(q),\quad s=\mathbb{E}\tanh^4(t+\sqrt{t}Z),\quad D_0=1-q^2,\quad D_1=q-q^2,\quad D_2=s-q^2.$$

Nishimori scalar identities give $\mathbb{E}\tanh^{2j-1}(t+\sqrt{t}Z)=\mathbb{E}\tanh^{2j}(t+\sqrt{t}Z)$. For $|A|,|C|\in\{1,2\}$, therefore,

$$d(A,C):=\mathbb{E}[(e_A-q)e_C]=\begin{cases}D_0, & A=C,\\ D_1, & |A\triangle C|\in\{1,2\},\\ D_2, & \text{otherwise.}\end{cases}\tag{6.10}$$

For a bounded observable f involving n replicas, the leading Gaussian streaming part is

$$k\left\{\sum_{\ell<j\leqslant n}\nu(fe_{\{\ell,j\}}T^-_{\{\ell,j\}})-n\sum_{\ell\leqslant n}\nu(fe_{\{\ell,n+1\}}T^-_{\{\ell,n+1\}})+\frac{n(n+1)}{2}\nu(fe_{\{n+1,n+2\}}T^-_{\{n+1,n+2\}})\right\}.$$

The leading deterministic planted-field part is

$$k\left\{\sum_{\ell\leqslant n}\nu(fe_{\{\ell\}}T^-_{\{\ell\}})-n\nu(fe_{\{n+1\}}T^-_{\{n+1\}})\right\}.$$

All remaining terms have either an extra centered overlap/magnetization factor or an explicit N^{-1} factor. The Taylor error in ξ' is quadratic in the centered variable; the covariance finite-size error is $O(N^{-1})$ or smaller in either bulk normalization.

Site exchangeability gives $\nu(T_AT_B)=\nu((e_A-q)T_B)$. At cavity zero the term with T_B^- vanishes. Using the streaming formula and (6.10) gives $X=k\mathcal{M}X+\mathcal{E}$, with

$$X=(\nu M_1^2,\nu M_1M_2,\nu M_1Q_{12},\nu M_1Q_{23},\nu Q_{12}^2,\nu Q_{12}Q_{13},\nu Q_{12}Q_{34})^T,$$

where M_a,Q_{ab} are centered in this display, and

$$\mathcal{M}=\begin{pmatrix}D_0 & -D_1 & -D_1 & D_2 & 0 & 0 & 0\\ D_1 & D_0-2D_1 & D_1-2D_2 & -2D_1+3D_2 & 0 & 0 & 0\\ 0 & 0 & D_0+D_1 & -2D_1 & D_1 & -2D_1-2D_2 & 3D_2\\ 0 & 0 & 2D_1 & D_0-3D_1 & D_2 & 2D_1-6D_2 & -3D_1+6D_2\\ 0 & 0 & 2D_1 & -2D_2 & D_0 & -4D_1 & 3D_2\\ 0 & 0 & D_1+D_2 & D_1-3D_2 & D_1 & D_0-2D_1-3D_2 & -3D_1+6D_2\\ 0 & 0 & 2D_2 & 2D_1-4D_2 & D_2 & 4D_1-8D_2 & D_0-8D_1+10D_2\end{pmatrix}.$$

Its characteristic factors are

$$\det(I-k\mathcal{M})=[1-k(1-q)]^2[1-kb]^3[1-k(2b-(1-q))]^2,\quad b=\mathbb{E}\,\mathrm{sech}^4(t+\sqrt{t}Z).$$

The AT factor is $kb=\lambda<1$; the last factor is positive since $1-2\lambda+k(1-q)\geqslant 1-\lambda>0$. The first factor is nonzero at the critical three-spin point, as certified below. One does not assume it has the same sign as the AT factor.

6.3. **Rooted systems and higher moments.** For a desired even order $2r$, use either of the nonnegative bulk spectator monomials

$$P=(T_{\{1\}})^{2r-2},\qquad\text{or}\qquad P=(T_{\{1,2\}})^{2r-2}.$$

Retain every covariance variable $\nu(T_AT_BP)$, with $|A|,|B|\in\{1,2\}$. The covariance variable itself is unchanged by interchanging A and B, but its streaming equation requires an oriented representative. We fix that orientation as follows, exactly as in the certificate program. For $j\in\{0,1,2\}$, let $\mathscr{R}_j=\{1,\dots,j\}$, and put $F=(A\cup B)\setminus\mathscr{R}_j$, $m=|F|$. Let ϕ range over all

bijections that permute $\mathscr{R}_j$ and map F bijectively onto $\{j+1,\dots,j+m\}$. For a finite set S, write $\iota(S)$ for its increasing tuple. Define

$$\mathrm{can}_j(A,B) = \operatorname*{lex\,min}_{\phi} \mathrm{sort}_{\mathrm{lex}}\big(\iota(\phi(A)),\iota(\phi(B))\big).$$

Both the sorting and the minimum use ordinary lexicographic tuple order, with a proper initial segment preceding the longer tuple. For $j=0$ there are no roots; for $j=1$ the root is fixed; and for $j=2$ both permutations of the two roots are allowed.

The row and column keys are all distinct values of $\mathrm{can}_j(A,B)$ obtained from nonempty singleton or pair subsets of $\{1,\dots,j+4\}$, listed in increasing lexicographic order. There are respectively $7, 17, 26$ keys. The displayed seven-dimensional matrix is the $j=0$ matrix after the indicated ordering of its seven covariance variables. For the first spectator use $j=1$, and for the second use $j=2$.

For each canonical row key (A,B), regard its first tuple as the distinguished set A in the last-spin factor $e_A - q$, and its second tuple as B; do not reorient the row afterward. Take

$$n = |\mathscr{R}_j \cup A \cup B| = \max\big(j, \max(A\cup B)\big).$$

This includes every spectator root, even when it is absent from $A\cup B$. Each term in the following streaming rule is added to the column key $\mathrm{can}_j(B,C)$.

$$\begin{aligned}
&d(A,\{\ell\}) && \text{for each } 1\leqslant \ell\leqslant n,\\
&-n\,d(A,\{n+1\}),\\
&d(A,\{\ell,\ell'\}) && \text{for each } 1\leqslant \ell<\ell'\leqslant n,\\
&-n\,d(A,\{\ell,n+1\}) && \text{for each } 1\leqslant \ell\leqslant n,\\
&\tfrac{n(n+1)}{2}d(A,\{n+1,n+2\}).
\end{aligned}$$

The singleton or pair shown in each term is C; columns are reduced to their specified rooted orbits. The program `pure_band_rooted_cavity.py` implements exactly this rule using integer coefficient triples for (D_0,D_1,D_2). Let these matrices be $\mathcal{M}_1,\mathcal{M}_2$.

Site exchangeability again gives $\nu(T_AT_BP)=\nu((e_A-q)T_BP)$. Replace the remaining factors by their bulk versions. Every replacement contributes at most $C_rN^{-1}S_{r-1}+C_rN^{-(2r-1)}$, where

$$S_r = \nu|M_1-q|^{2r}+\nu|R_{12}-q|^{2r}, \qquad S_0=2.$$

This follows from expansion of $T_A = T_A^- + O(N^{-1})$ and Hölder's inequality. At cavity zero the centered last-spin factor has mean zero. Expand the streaming derivative at that endpoint and use the same scalar coefficients. Replacing the cavity bulk moments by their full endpoint values introduces one additional centered factor. On the macro-good event in (6.1), this costs at most $C_r\eta S_r$; its complement contributes C_re^{-cN}. The Taylor errors in the deterministic and covariance fields obey the same bound. Thus each rooted vector satisfies

$$(I-k\mathcal{M}_j)X^{(r,j)} = \mathcal{E}^{(r,j)}, \qquad \|\mathcal{E}^{(r,j)}\|_\infty \leqslant \frac{C_r}{N}S_{r-1}+C_r\eta S_r+C_rN^{-(2r-1)}+C_re^{-cN}. \tag{6.11}$$

Finite-replica cavity domination makes all estimates uniform on the interpolation. The constants depend on r, which is why a maximal finite order is fixed first.

The matrices are independent of r. Their invertibility at the critical parameter, with a certified margin as shown below, bounds all entries of the rooted vectors by the right side of (6.11). One entry in the first vector is νM_1^{2r}; one in the second is νQ_{12}^{2r}. Choose η sufficiently small for all $1\leqslant r\leqslant r_0$, and then choose δ sufficiently small to obtain (6.1) and the boundary estimates. Absorb the ηS_r term. Induction gives $S_r\leqslant C_rN^{-r}$, proving Theorem 6.1.

Transfer after the canonical bootstrap. The preceding induction has proved the bounds for the canonical critical pure band at every sufficiently large size. Apply (6.8) at that same size to the nonnegative observables $|M_1 - q|^{2r}$ and $|R_{12} - q|^{2r}$. Their suprema are bounded for fixed r, so the exponentially small support-replacement term is harmless. This proves the stated moment bounds for all the bounded total covariance and potential perturbations in the theorem. If $|q_N - q| \leqslant K/N$, the elementary inequality

$$|x - q_N|^{2r} \leqslant 2^{2r-1}\big(|x - q|^{2r} + (K/N)^{2r}\big)$$

allows either center. The exact bulk models and finite covariance conventions listed after the lemma fall in this class. Only now are these transferred high-moment bounds used in the subsequent cross-overlap argument.

6.4. **Finite certificate and longitudinal stability.** The accompanying outward-rounded scalar program encloses the critical parameters in

$$2.33964156 < t < 2.33964159, \qquad .81351755 < q < .81351760,$$
$$.1164029 < b < .1164039, \qquad k = 2t/q.$$

The root is bracketed by opposite signs of $A_3(t) = 2\,\mathbb{E}\log\cosh(t + \sqrt{t}Z) - t - \frac{2}{3}t\Gamma(t)$. The integral enclosures use composite Simpson quadrature with mesh .0005 on $[-15, 15]$, a fourth-derivative bound 10000 and a Gaussian-tail bound 10^{-12}. The total error is less than $1.1 \cdot 10^{-10}$; elementary functions and arithmetic are outward rounded. The derivative bound $|b'(t)| \leqslant 10$ transports the sech-fourth enclosure across the root interval.

Here are explicit analytic bounds for the quadrature remainder. Put $u = \tanh x$, so $D_x = (1 - u^2)D_u$. For $f(x) = \operatorname{sech}^4 x = (1 - u^2)^2$, the supremum bounds for $|D_x^j f|$, $0 \leqslant j \leqslant 4$, may be taken as

$$(1, 16, 96, 704, 6144).$$

The first bound is $0 \leqslant f \leqslant 1$; the remaining four follow by summing the absolute coefficients of the corresponding derivative polynomials on $|u| \leqslant 1$. If $\varphi_t(x - t)$ is the density of $t + \sqrt{t}Z$, Fourier inversion gives

$$\begin{aligned}\sup_x |D_x^j \varphi_t(x - t)| &\leqslant \frac{1}{2\pi}\int_{\mathbb{R}} |z|^j e^{-tz^2/2}\,dz \\ &= \frac{\Gamma_{\mathrm{E}}((j+1)/2)}{2\pi}\left(\frac{2}{t}\right)^{(j+1)/2} < 1, \quad 0 \leqslant j \leqslant 4, \quad 2.3 \leqslant t \leqslant 2.4.\end{aligned}$$

where Γ_{E} denotes Euler's gamma function. Leibniz's rule therefore bounds the fourth derivative of $f(x)\varphi_t(x - t)$ by

$$1 + 4 \cdot 16 + 6 \cdot 96 + 4 \cdot 704 + 6144 = 9601 < 10000.$$

For $\tanh^2 x$, the analogous derivative bounds are $(1, 4, 16, 80, 512)$, giving the smaller product bound 945. On $[-15, 15]$, $\log\cosh x$ has derivative bounds $(15, 1, 1, 4, 16)$, giving 57. Thus composite Simpson's remainder is at most

$$\frac{30}{180}(.0005)^4\, 10000 < 1.042 \cdot 10^{-10}$$

for all three integrals. Outside $[-15, 15]$, the event $|t + \sqrt{t}Z| > 15$ implies $|Z| > 8$. The bounded integrands have tail at most $2\varphi(8)/8$; for the unbounded integrand use $\log\cosh x \leqslant |x|$ to obtain

$$\mathbb{E}[(2.4 + \sqrt{2.4}|Z|)\mathbb{1}_{\{|Z|>8\}}] \leqslant (0.6 + 2\sqrt{2.4})\varphi(8) < 10^{-12}.$$

These estimates justify the outward error $1.1\cdot10^{-10}$. Finally, with $f=\operatorname{sech}^4$, the Gaussian heat equation gives $b'(t)=\mathbb{E}[f'(t+\sqrt{t}Z)+\frac{1}{2}f''(t+\sqrt{t}Z)]$. Here $|f'|\leqslant 4$, and $f''=(1-u^2)^2(20u^2-4)$ has absolute value at most 4; hence $|b'(t)|\leqslant 6<10$, as required by the certificate.

At the exact rational center $q_0=.813517575$, $t_0=2.339641575$, $b_0=.1164034$, construct each matrix $A_0=I-k_0\mathcal{M}_j$ using rational arithmetic, and compute its exact rational inverse. The certificate checks

$$\|A_0^{-1}\|_\infty<4000, \qquad \|A_0^{-1}\|_\infty \sup_{(q,t,b)\text{ in the box}} \|A(q,t,b)-A_0\|_\infty<.6,$$

for the seven-, seventeen-, and twenty-six-dimensional systems. A Neumann series proves uniform invertibility throughout the box. The scalar and matrix output is stored in `pure_p3_cavity_certificate.txt`; floating-point values printed for convenience are not used for these comparisons.

For interpretation, the physical longitudinal denominator is also strictly positive. Put $c=1-q$, $d=c-b$. The ordinary RS field susceptibility is

$$m'(t)=c-\frac{2kd^2}{1-3kb+2kc}, \qquad 1-km'(t)=\frac{(1-kb)(1-2kb+kc)}{1-3kb+2kc}>0.$$

Thus the informative branch is longitudinally stable as well as AT stable. The rooted-system proof uses the verified full matrices instead of identifying this scalar susceptibility with every replica covariance mode.

7. Cross-band moment estimates

This section isolates the additional input needed to include $p=3$ in the empirical-CDF argument. All estimates here are proved from the one-band moment hypothesis stated below. In particular, no exponential two-system chaos theorem and no convexity at negative cross overlaps are invoked.

One-band input and finite required order. Fix an even integer M. For each independently uniform pattern a, let G_a be the positive narrow-band Gibbs measure with weight $\exp\{X_N(\sigma)+N\xi(R(\sigma,a))\}$, where $\xi(r)=\beta_c^2r^p$, and band centered at $q=q_*$. Assume that for every even $2\leqslant m\leqslant M$, uniformly in the full and one-site bulk systems used in the cavity construction,

$$\mathbb{E}\langle|R_{12}-q|^m\rangle+\mathbb{E}\langle|R(\sigma,a)-q|^m\rangle\leqslant C_mN^{-m/2}. \tag{7.1}$$

The one-site bulk systems include the bounded changes in total covariance and deterministic planting potential covered by Lemma 6.2; their normalized covariance polynomials change by $O(N^{-1})$. The exact odd last-spin planting field is retained. Assume also that deleting a fixed narrow strip at the two band edges costs at most Ce^{-cN} in averaged Gibbs probability; this allows both signs of the last spin in the cavity lift. These are one-band assertions, obtained from the separate rooted cavity systems. To obtain optimal cross moments through order ten it is enough to prove (7.1) through $M=64$.

The two full Gaussian disorders may have any fixed covariance correlation $\rho\in[-1,1]$. All assertions and constants below are uniform in ρ. The pattern averages remain independent.

Lemma 7.1 (Tensor comparison of cross thermal fluctuations)**.** *Let π_1,π_2 be arbitrary random probability measures on $\{-1,1\}^n$, coupled in any way, and let $m_j=\pi_j(\sigma)$, $u=n^{-1}m_1\cdot m_2$. If their self overlaps have m-th moments $\mathbb{E}\,\pi_j^{\otimes2}|R-q|^m\leqslant C_mn^{-m/2}$, for an even m, then*

$$\mathbb{E}(\pi_1\otimes\pi_2)|R-u|^m\leqslant C_mn^{-m/4}. \tag{7.2}$$

Also, for $\bar u=\max(-q,\min(q,u))$, $\mathbb{E}\,|u-\bar u|^m\leqslant C_mn^{-m/2}$.

Proof. Write $m = 2k$, and let $x = \sigma - m_j$ be a centered spin vector in system j. Conditional on its random law, define the order-$2k$ tensor $T_j = \pi_j(x^{\otimes 2k})$. Its Hilbert–Schmidt norm obeys

$$\|T_j\|_{\mathrm{HS}}^2 = \pi_j^{\otimes 2}(x \cdot x')^{2k}.$$

The centered scalar product divided by n is

$$R(\sigma,\sigma') - \pi_j(\tau)R(\sigma,\tau) - \pi_j(\tau)R(\tau,\sigma') + \pi_j^{\otimes 2}(R).$$

Insert and cancel q. Jensen and the elementary power inequality give

$$\mathbb{E}\,\|T_j\|_{\mathrm{HS}}^2 \leqslant C_k n^{2k}\,\mathbb{E}\,\pi_j^{\otimes 2}|R-q|^{2k} \leqslant C_k n^k.$$

For a deterministic vector v of norm at most $\sqrt{n}$, tensor Cauchy–Schwarz gives, conditionally on both measures,

$$\pi_j(x \cdot v)^{2k} \leqslant \|T_j\|_{\mathrm{HS}}\|v\|^{2k} \leqslant n^k \|T_j\|_{\mathrm{HS}}.$$

After disorder averaging and division by n^{2k}, this is $O(n^{-k/2})$. It remains valid for $v = m_{3-j}$, which may depend on the same disorder, because its norm is bounded deterministically. The centered bilinear term is smaller:

$$\mathbb{E}(\pi_1 \otimes \pi_2)(x \cdot y)^{2k} = \mathbb{E}\langle T_1, T_2\rangle \leqslant \{\mathbb{E}\,\|T_1\|_{\mathrm{HS}}^2\,\mathbb{E}\,\|T_2\|_{\mathrm{HS}}^2\}^{1/2} \leqslant C_k n^k.$$

Expand $n(R-u) = x \cdot y + x \cdot m_2 + m_1 \cdot y$ and apply the power inequality. This proves (7.2).

Finally, Jensen gives $\mathbb{E}\,|\|m_j\|^2/n - q|^m \leqslant C_m n^{-m/2}$. Since $|u| \leqslant \|m_1\|\|m_2\|/n \leqslant (\|m_1\|^2 + \|m_2\|^2)/(2n)$, the clipping error has the asserted bound. ∎

Lemma 7.2 (All fixed coarse cross moments)**.** *Under* (7.1) *and the boundary-removal input, for each even* $2 \leqslant m \leqslant M$*,*

$$\mathbb{E}_{a,b,g}\langle |R(\sigma,\tau)|^m\rangle_{a,b} \leqslant C_m N^{-m/8}. \tag{7.3}$$

Consequently, for every fixed $\eta > 0$*,* $\mathbb{E}\langle \mathbb{1}_{\{|R|>\eta\}}\rangle \leqslant C_{M,\eta} N^{-M/8}$*.*

Proof. Use the exact last-site decomposition and write the two independent last pattern signs as a_N, b_N. Deleting the boundary strips changes the bounded observables below by an exponentially small amount. In the resulting lifted system,

$$\mathcal{G}_j(\sigma,e) = \frac{\pi_j(\sigma)e^{e(a_N^j h_N(\sigma) + z_j(\sigma))}}{2D_j}, \qquad D_j = \sum_\sigma \pi_j(\sigma)\cosh(a_N^j h_N(\sigma) + z_j(\sigma)) \geqslant 1.$$

Here $a_N^1 = a_N$, $a_N^2 = b_N$, the bulk laws π_j are independent of the last signs and cavity Gaussians, and h_N is the odd last-spin part of the exact deterministic planting term. It is uniformly bounded, and (7.1) gives

$$\big\|\pi_j|h_N - t|\big\|_{L^m} \leqslant C_m N^{-1/2}, \qquad t = \xi'(q).$$

For any fixed number of replicas, the denominator lower bound and bounded Gaussian exponential moments give the conditional domination

$$\mathbb{E}_z\,\mathcal{G}_1^{\otimes n_1}\mathcal{G}_2^{\otimes n_2}(f) \leqslant C_{n_1+n_2}\pi_1^{\otimes n_1}\pi_2^{\otimes n_2}(f) \qquad (f \geqslant 0) \tag{7.4}$$

for bulk observables, uniformly along every cavity interpolation used below.

Set $u = (N-1)^{-1}m_1 \cdot m_2$, and clip it to $[-q,q]$ as in Lemma 7.1. Put $a_m = \|\bar{u}\|_{L^m}$ and $d_N = N^{-1/4}$. The tensor lemma and (7.4) show that, under the full lift,

$$\|R - \bar{u}\|_{L^m} \leqslant C_m d_N, \tag{7.5}$$

where changing from the bulk overlap to the full overlap adds only $O(N^{-1})$.

We quantify the conditional Gaussian-mixture comparison, retaining the power of $\bar{u}$ in every error. First replace the bulk-dependent fields $a_N^j h_N(\sigma)$ by $a_N^j t$. Differentiating the Gibbs expectation along this deterministic field interpolation and using (7.4) bounds the error for the observable $\bar{u}^{m-1}e_1e_2$ by

$$C_m \,\mathbb{E}\left[|\bar{u}|^{m-1}\sum_j \pi_j|h_N - t|\right] \leqslant C_m N^{-1/2} a_m^{m-1}.$$

The exact cavity covariance is $K_N(r) = \xi'(r) + O(N^{-1})$. Let $\kappa_N = K_N(1)$. Replace the two cavity processes by $z_j^*(\sigma) = Z_j + \eta_j(\sigma)$, where, conditionally on the bulk,

$$\operatorname{Var} Z_j = t, \qquad \operatorname{Cov}(Z_1, Z_2) = \rho\xi'(\bar{u}),$$

and the $\eta_j(\sigma)$ are independent Gaussians of variance $\kappa_N - t$, independent of (Z_1, Z_2). The covariance matrix of (Z_1, Z_2) is positive semidefinite because $|\xi'(\bar{u})| \leqslant \xi'(q)$. The residual variance is positive for all large N.

Write Δ_{ij} for the difference of actual and replacement cavity covariances, and $D = \sum_{i,j}(\pi_i \otimes \pi_j)|\Delta_{ij}|$. The covariance difference vanishes on every self diagonal. The self part is bounded by the self-overlap error, the cross part by $|R - \bar{u}| + O(N^{-1})$, and the atomic correction by $\sum_\sigma \pi_j(\sigma)^2$. In the latter term, use

$$(1-q)^m \sum_\sigma \pi_j(\sigma)^2 \leqslant \pi_j^{\otimes 2}|R-q|^m.$$

Since the sum of squared weights is at most one, its L^m norm and the L^m norm of its square root are both $O_m(N^{-1/2})$. Thus no moment beyond order m is needed for either atomic error. Jensen, the tensor lemma, and (7.1) therefore imply

$$\|D\|_{L^m} \leqslant C_m d_N. \tag{7.6}$$

Gaussian interpolation and (7.4), still retaining the fixed bulk factor $|\bar{u}|^{m-1}$, bound the corresponding error by

$$C_m \,\mathbb{E}(|\bar{u}|^{m-1} D) \leqslant C_m d_N a_m^{m-1}.$$

In the replacement system, the last-spin mean differs from $\tanh(a_N^j t + Z_j)$ in conditional mean by at most $C\sqrt{\sum_\sigma \pi_j(\sigma)^2}$. This follows from

$$\sinh(y+\eta) - \tanh(y)\cosh(y+\eta) = \sinh(\eta)/\cosh(y),$$

the independent residual variables, and the denominator bound. The weighted error is again at most $C_m N^{-1/2} a_m^{m-1}$, by the self-overlap moments and Hölder. We have proved the quantitative identity

$$\left|\mathbb{E}\langle \bar{u}^{m-1} e_1 e_2\rangle - \mathbb{E}\,\bar{u}^{m-1}\Phi_\rho(\bar{u})\right| \leqslant C_m d_N a_m^{m-1} + Ce^{-cN}, \tag{7.7}$$

where

$$\Phi_\rho(v) = \mathbb{E}\tanh(ta_N + Z_1)\tanh(tb_N + Z_2), \quad \operatorname{Cov}(Z_1, Z_2) = \rho\xi'(v).$$

The scalar contraction has a strict uniform margin. Let $b = \mathbb{E}\operatorname{sech}^4(t + \sqrt{t}Z)$ and $\lambda = \xi''(q)b < 1$. The function $m_t(z) = \{\tanh(t+z) + \tanh(-t+z)\}/2$ is odd, so $\Phi_\rho(0) = 0$. Gaussian covariance differentiation, Cauchy–Schwarz, and Jensen give

$$|\Phi_\rho(v)| \leqslant |\rho| b |\xi'(v)| \leqslant \frac{\lambda}{p-1}|v| =: c_0|v|, \qquad |v| \leqslant q, \qquad c_0 < 1.$$

This uses the absolute value of $\xi'(v)$, so it holds for odd p and negative v as well.

Finally use full-site exchangeability *before* the boundary deletion: $\mathbb{E}\langle R^m\rangle = \mathbb{E}\langle R^{m-1}e_1e_2\rangle$. By (7.5) and Hölder,

$$|\,\mathbb{E}\langle R^m\rangle - \mathbb{E}\,\bar{u}^m| \leqslant C_m(d_N a_m^{m-1} + d_N^m),$$
$$|\,\mathbb{E}\langle R^{m-1}e_1e_2\rangle - \mathbb{E}\langle \bar{u}^{m-1}e_1e_2\rangle| \leqslant C_m(d_N a_m^{m-2} + d_N^{m-1}).$$

Combine these bounds with (7.7) and the contraction. Since m is even and $a_m, d_N \leqslant 1$,

$$(1-c_0)a_m^m \leqslant C_m(d_N a_m^{m-2} + d_N^{m-1}) + Ce^{-cN}.$$

Young's inequality absorbs half the left side and gives $a_m^m \leqslant C_m d_N^{m/2} = C_m N^{-m/8}$. Equation (7.5) proves (7.3). Markov's inequality gives the macro tail assertion. All constants were uniform in ρ. ∎

Lemma 7.3 (From polynomial macro tails to optimal cross moments)**.** *Assume the preceding input with $M = 64$. For the exact ordered pure model with any fixed $p \geqslant 3$, and two independent uniform patterns, the positive narrow bands satisfy*

$$\mathbb{E}\langle |R|^{2r}\rangle \leqslant C_r N^{-r}, \qquad 1 \leqslant r \leqslant 5.$$

The same conclusion holds for the distinct-index pure three-spin model if its one-band inputs are supplied.

Proof. Use precisely the cross-only last-site interpolation in Lemma 8.4. It preserves both self laws, both hard bands, and both full deterministic planting terms. Its cavity covariance satisfies, globally,

$$|K_N(r)| \leqslant M_0,$$

for a constant M_0. Fix $\eta > 0$ sufficiently small that

$$4\beta_c^2 p\eta^{p-2}e^{4M_0} < 1.$$

On $|r| \leqslant \eta$, $|K_N(r)| \leqslant L_N|r| + C/N^2$, with $L_N \to \beta_c^2 p\eta^{p-2}$. For the distinct three-spin covariance the constant error is only C/N; this also suffices below.

For every nonnegative observable of one replica from each group, the exact four-term streaming formula gives multiplicative comparison by at most e^{4M_0} between any interpolation time and its full endpoint. Lemma 7.2 therefore bounds each macro cross-overlap tail occurring in the streaming formula by $C_\eta N^{-8}$, uniformly in time. Changing full overlaps to R^- alters the fixed cutoff only by $O(N^{-1})$. Each of the four possible cross replica pairs has the same marginal law, so this statement covers all terms.

Repeat the odd-observable calculation from Lemma 8.4, splitting each cavity covariance at $|R^-| \leqslant \eta$. On the small-overlap event use $L_N|R^-| + C/N^2$ and Hölder. On its complement bound the observable and covariance by constants and use the N^{-8} tail. The zero-interpolation expectation still vanishes by the last-pattern-sign involution. Consequently, for even $2 \leqslant m \leqslant 10$,

$$A_m \leqslant \{4L_N e^{4M_0} + o(1)\}A_m + \frac{C_m}{N}A_{m-2} + C_m N^{1-m} + C_m N^{-8}.$$

The $O(N^{-2})$ covariance remainder is handled by Young as before. Even an $O(N^{-1})$ remainder contributes only $C_m N^{-m}$, which is already dominated by N^{1-m}. Absorb the first term. Induction from $A_0 = 1$ gives $A_m \leqslant C_m N^{-m/2}$, since $m \leqslant 10$ and $N^{-8} \leqslant N^{-m/2}$. This proof uses no sign of a negative cross overlap. ∎

Combining Lemma 7.3 with the Hölder-exponent 3/2 version of Lemma 8.5 gives, for $p = 3$,

$$N \operatorname{Var}(A_N) \leqslant CN^{-1/8} + Ce^{-cN}, \qquad \sqrt{N}(A_N - \mathbb{E}\, A_N) \longrightarrow 0 \quad \text{in } L^2.$$

This last conclusion is conditional only on the explicit one-band moment and boundary-removal inputs at the start of the section.

8. Band local limits and the stable characteristic equation

Notation. Throughout, $X_N(\nu)$ is centered Gaussian with covariance

$$\mathbb{E}\, X_N(\nu)X_N(\mu) = N\beta_c^2 R(\nu,\mu)^p, \qquad R(\nu,\mu) = N^{-1}\nu\cdot\mu,$$

and μ_N is uniform on the Ising cube. The temperature is included in X_N. Let

$$W_\nu(g) = e^{X_N(\nu) - N\beta_c^2/2}, \qquad L_N(g) = \mathbb{E}_{\nu\sim\mu_N} W_\nu(g).$$

The planted shift $P_a g = g + v_a$ satisfies $X_N(\nu; P_a g) = X_N(\nu; g) + N\beta_c^2 R(\nu,a)^p$.

8.1. A scalar local CLT from a monotone convex Gaussian coordinate.

Lemma 8.1 (Scalar Gaussian smoothing)**.** *Let $G \sim N(0,1)$ be independent of W_N, and let $Y_N = f_N(W_N, G)$, where, for every W_N,*

$$0 < c \leqslant \partial_g f_N(W_N, g) \leqslant C < \infty, \qquad \partial_{gg} f_N(W_N, g) \geqslant 0$$

for all $g \in \mathbb{R}$. The constants do not depend on N, W_N, g. If Y_N converges weakly to a law with continuous density p vanishing at infinity, then Y_N has a continuous density p_N and

$$\sup_{y\in\mathbb{R}} |p_N(y) - p(y)| \longrightarrow 0.$$

For local uniform convergence it is enough that p be continuous locally.

Proof. For fixed $W_N = w$, the map $g \mapsto f_N(w,g)$ is an increasing bijection of $\mathbb{R}$. Writing its inverse as $g_w(y)$ and the standard Gaussian density as φ, the conditional density is

$$p_{N,w}(y) = \frac{\varphi(g_w(y))}{f_N'(w, g_w(y))}.$$

It is bounded above by $\|\varphi\|_\infty/c$, and differentiation gives

$$p_{N,w}'(y) = \frac{\varphi'(g_w(y))}{f_N'(w,g_w(y))^2} - \frac{\varphi(g_w(y)) f_N''(w,g_w(y))}{f_N'(w,g_w(y))^3} \leqslant \|\varphi'\|_\infty/c^2 =: K.$$

Thus the averaged density satisfies the uniform one-sided Lipschitz inequality $p_N(y) - p_N(x) \leqslant K(y-x)$ for $y > x$. Suppose there were a fixed positive excess $p_N(x_N) - p(x_N) \geqslant \eta$ on a compact set. Pass to a subsequence with $x_N \to x$. The one-sided inequality propagates at least half of the excess over a fixed interval immediately to the left of x_N. Continuity of p and weak convergence of interval masses contradict this. A negative discrepancy propagates to the right in exactly the same way. This proves local uniform convergence. A density peak of fixed height arbitrarily far in the tails would likewise force a fixed positive mass in a nearby fixed-length interval, contradicting tightness. Since p vanishes at infinity, convergence is uniform on $\mathbb{R}$. ∎

For a planted positive band $\{R(\nu,a) \geqslant \varepsilon\}$, isolate the Gaussian direction $G = X_N(a)/(\beta_c\sqrt{N})$. For its logarithmic mass U_N,

$$\partial_G(U_N/\sqrt{N}) = \beta_c\langle R(\nu,a)^p\rangle \in [\beta_c\varepsilon^p, \beta_c], \qquad \partial_{GG}(U_N/\sqrt{N}) \geqslant 0.$$

The same statement holds for an absolute-overlap band when p is even. Arbitrary deterministic planted terms and deterministic centering do not affect this argument. Hence a weak Gaussian

CLT for such a logarithmic band mass automatically gives the scalar local CLT, including at logarithmic displacements from zero.

8.2. **Global joint densities from narrow-band geometry.** Fix k, and patterns $a_1, \dots, a_k \in \{-1,1\}^N$. Put

$$S_{ij} = R(a_i, a_j), \qquad C_{ij} = S_{ij}^p.$$

Assume $\max_{i\neq j} |S_{ij}| \to 0$. The matrices S, C are positive definite for all sufficiently large N. Let

$$G = C^{-1/2} \left(\frac{X_N(a_i)}{\beta_c \sqrt{N}} \right)_{i=1}^{k} \sim N(0, I_k).$$

Gaussian projection gives the exact decomposition

$$X_N(\nu) = X_N^{\perp}(\nu) + \beta_c \sqrt{N}\, (R(\nu, a_j)^p)_{j=1}^k C^{-1/2} G,$$

with $X_N^{\perp}$ independent of G.

Let $\mathcal{B}_i$ be any fixed nonempty band satisfying

$$R(\nu, a_i) \geqslant a > 1/\sqrt{2} \qquad (\nu \in \mathcal{B}_i).$$

An absolute-overlap condition works as well for even p. Consider logarithmic masses

$$U_i = \log \sum_{\nu \in \mathcal{B}_i} w_i(\nu; W) \exp\{\beta_c \sqrt{N}\, (R(\nu, a_j)^p)_j C^{-1/2} G\},$$

where the weights are positive and independent of G. They may include residual Gaussian disorder, arbitrary deterministic planting terms, and normalization factors. Put $Y_i = (U_i - b_{i,N})/\sqrt{N}$.

Lemma 8.2 (Global diffeomorphism and density bounds)**.** *There are constants $c, C > 0$, uniform for all sufficiently nearly orthogonal pattern tuples, such that the map $G \mapsto Y$, conditional on W, is a global smooth diffeomorphism, with*

$$\|J\| \leqslant C, \qquad \|J^{-1}\| \leqslant C, \qquad c \leqslant \det J \leqslant C, \quad J = \partial_G Y.$$

Consequently the conditional and unconditional densities of Y are bounded above by C. For $k = 2$, the density f_N of the unscaled vector (U_1, U_2) satisfies

$$\sup_{x_1, x_2} \big(|f_N| + |\partial_1 f_N| + |\partial_2 f_N| + |\partial_{12} f_N| \big) \leqslant C/N. \tag{8.1}$$

For one band, its unscaled density and first derivative are bounded by $C/\sqrt{N}$.

Proof. For $r_j = R(\nu, a_j)$, Bessel's inequality in the span of the normalized patterns gives

$$r^T S^{-1} r \leqslant 1, \qquad \sum_j r_j^2 \leqslant \lambda_{\max}(S) = 1 + o(1).$$

Since $|r_i| \geqslant a$,

$$\sum_{j\neq i} |r_j|^p \leqslant \left(\sum_{j\neq i} r_j^2 \right)^{p/2} \leqslant (1 + o(1) - a^2)^{p/2}.$$

The derivative matrix has rows

$$J_{i\bullet} = \beta_c \langle (r_j^p)_j \rangle_i C^{-1/2}.$$

Therefore

$$J_{ii} \geqslant \beta_c a^p - o(1), \qquad \sum_{j\neq i} |J_{ij}| \leqslant \beta_c (1 - a^2)^{p/2} + o(1).$$

As $a > 1/\sqrt{2}$, every row has a fixed strictly positive diagonal-dominance gap. This is a deterministic assertion, valid for every realization of all disorder coordinates.

For g, g', choose i with $|g_i - g'_i| = \|g - g'\|_\infty$. Integrating the i-th Jacobian row along the segment between them gives

$$\|Y(g) - Y(g')\|_\infty \geqslant c\|g - g'\|_\infty.$$

This proves injectivity and properness. The same row estimate makes each Jacobian invertible with uniformly bounded inverse. The image is open by the inverse function theorem, closed by properness, and nonempty; hence it is all of $\mathbb{R}^k$. The determinant is positive by homotopy from the positive diagonal matrix, preserving strict diagonal dominance. The conditional density is therefore

$$p_{N,W}(y) = \frac{\varphi_k(g_W(y))}{\det J(g_W(y))},$$

giving the global bound.

For the derivative bounds, write $T = \partial_G U = \sqrt{N}J$. The log-sum-exp representation and bounded slopes give, uniformly in G, W,

$$\|T^{-1}\| \leqslant CN^{-1/2}, \qquad \|\partial_G^2 U_i\| \leqslant CN, \qquad \|\partial_G^3 U_i\| \leqslant CN^{3/2}.$$

For $k = 2$, $\det T \asymp N$. The conditional density is $f = \varphi_2(g)/(\det T(g))$. Its first g-derivatives are $O(N^{-1/2})$, and its second g-derivatives are $O(1)$: this follows by differentiating the determinant using $\partial \log\det T = \operatorname{tr}(T^{-1}\partial T)$, and the corresponding second derivative. The derivatives of the inverse map satisfy

$$\partial_x g = T^{-1} = O(N^{-1/2}), \qquad \partial_x^2 g = -T^{-1}(\partial_G^2 U)[T^{-1}, T^{-1}] = O(N^{-1/2}).$$

The chain rule consequently gives $f = O(N^{-1})$, $\partial_x f = O(N^{-1})$, and $\partial_x^2 f = O(N^{-1})$, including the mixed derivative. The constants include the bounded Gaussian density and its first two derivatives. Averaging over W preserves the estimates. The scalar calculation is identical with $T \asymp \sqrt{N}$. ∎

8.3. **A joint local central limit theorem.**

Theorem 8.3 (Joint narrow-band Gaussian smoothing)**.** *In the setting of Lemma 8.2, suppose for fixed $q_i > 0$ that*

$$\mathbb{E}\langle |R(\nu, a_i)^p - q_i^p| \rangle_i \to 0, \qquad \mathbb{E}\langle |R(\nu, a_j)|^p \rangle_i \to 0 \quad (i \neq j). \tag{8.2}$$

If $Y \Rightarrow N(0, \Sigma)$, then its density converges uniformly on $\mathbb{R}^k$ to the density of $N(0, \Sigma)$. In particular, for fixed bounded boxes I and $x_N - b_N = o(\sqrt{N})$,

$$N^{k/2}\mathbb{P}(U \in x_N + I) \longrightarrow |I|\varphi_\Sigma(0).$$

The assertion includes $x_{i,N} = -\frac{1}{2}\log N + O(1)$ when $b_N = 0$.

Proof. Let $A = \operatorname{diag}(\beta_c q_i^p)$. By (8.2), $\mathbb{E}\,\|J - A\| \to 0$. We first justify, rather than assume, uniform convergence of the conditional Jacobian on fixed Gaussian-coordinate cubes.

Each component $Y_i(g)$ is convex in g, and all its gradients are uniformly bounded. On any fixed cube the Gaussian density is bounded below, so $\mathbb{E}_W \int \|J(g) - A\|\, dg \to 0$. For convex uniformly Lipschitz functions, convergence of the gradient in L^1 on a cube to a constant implies uniform convergence of the gradient on every strictly smaller cube. Here is a short deterministic verification: subtract the limiting affine function and normalize the value at zero. Uniform Lipschitz bounds give compactness in the uniform topology; the gradient's L^1 convergence makes every subsequential limit constant. Convex difference quotients then

bound each derivative in the smaller cube between two differences of the uniformly convergent functions. Thus every derivative converges uniformly to zero there. Applying this deterministic fact by the subsequence principle yields

$$\sup_{g\in[-K,K]^k} \|J(g) - A\| \longrightarrow 0 \quad \text{in probability over } W,$$

for every fixed K. Uniform boundedness upgrades this to convergence in mean.

Put $V_N = Y(W, 0)$, independent of G. On a fixed cube, $Y(W, G) = V_N + AG + o_{\mathbb{P}}(1)$, uniformly in G. Gaussian tails and bounded gradients imply this also without restricting G. Tightness of Y and the inequality $|V_N| \leqslant |Y(W, G)| + C|G|$ give tightness of V_N. For any subsequential limit V, independence gives

$$V + AG \sim N(0, \Sigma).$$

Deconvolution of characteristic functions identifies V uniquely as the possibly degenerate Gaussian with covariance $\Sigma - AA^T$. This covariance is necessarily nonnegative. Hence V_N converges to that law.

We next compare densities, not just distributions. On the event $\sup_{[-K,K]^k} \|J - A\| \leqslant \eta$, the global inverse-map bound in Lemma 8.2 and the affine approximation give

$$\sup_y \left| p_{N,W}(y) - \frac{\varphi_k(A^{-1}(y - V_N))}{\det A} \right| \leqslant C_K\eta + Ce^{-cK^2}.$$

Indeed, if an inverse image lies in the cube, its distance from the affine inverse is at most $C_K\eta$, and the determinant differs from $\det A$ by $O_K(\eta)$. If the inverse image lies outside the cube, the Gaussian density is exponentially small; the affine inverse cannot remain in the half-sized cube, by the same inverse-map estimate. On the exceptional event both densities are uniformly bounded. Average, let $N \to \infty$, then $\eta \downarrow 0$, then $K \to \infty$. This proves

$$\sup_y |p_N(y) - \mathbb{E}\,\varphi_A(y - V_N)| \to 0.$$

Weak convergence of V_N, convolved with the fixed nondegenerate Gaussian density φ_A, gives uniform convergence to φ_Σ. This completes the proof. ∎

Random patterns. For independent uniform patterns, condition on $\max_{i\neq j} |R(a_i, a_j)| \leqslant \gamma$, for a sufficiently small fixed γ. Its complement has probability $O(e^{-cN})$. The preceding density bounds hold uniformly on that event. Under the conditional law, the whitened Gaussian coordinates remain independent standard Gaussians, independent of the patterns and residual disorder. Therefore the theorem applies to averaged random-pattern laws whenever their joint weak CLT and (8.2) hold. The full unconditioned joint law can have a singular component from identical patterns; it has exponentially small mass and is kept separate, rather than incorrectly assigned a density.

Why localization holds for the critical narrow bands. Use $\mathcal{B}_a = \{|R(\nu, a) - q_*| \leqslant \delta\}$ with lower endpoint $a > 1/\sqrt{2}$, and put

$$\widehat{B}_a(g) = c_p\, \mathbb{E}_\nu\, W_\nu(P_a g)\mathbb{1}_{\mathcal{B}_a}, \qquad c_p = \begin{cases} 2, & p \text{ even}, \\ 1, & p \text{ odd}. \end{cases}$$

The same unique-maximizer argument that proves the one-band central limit theorem in Section 5 shows that the band Gibbs magnetization concentrates at q_*: for every fixed $\eta > 0$, all slices with $|R(\nu, a) - q_*| > \eta$ have a fixed negative limiting pressure gap, while the

denominator has logarithm $O_{\mathbb{P}}(\sqrt{N})$. Thus the first assertion of (8.2) follows by boundedness. For b an independent uniform pattern, independent of the one-planted band measure,

$$\mathbb{E}_b\langle|R(\nu,b)|^p\rangle_a = \mathbb{E}\,|N^{-1}\sum_{i=1}^{N}\epsilon_i|^p = O(N^{-p/2}).$$

This proves the off-diagonal assertion. The joint field CLT and the one-band approximation give the joint weak Gaussian limit with independent coordinates, all with variance $v > 0$. Consequently

$$N^{k/2} f_{\log \widehat{B}_{a_1},\dots,\log \widehat{B}_{a_k}}(x_1,\dots,x_k) \longrightarrow (2\pi v)^{-k/2} \tag{8.3}$$

uniformly when $x_i = -\frac{1}{2}\log N + O(1)$, on the good-pattern component. Set $\kappa = (2\pi v)^{-1/2}$.

8.4. **Threshold self-averaging from ordinary cross-overlap moments.** The needed moment estimate for the positive narrow bands is

$$\mathbb{E}_{a,b,g}\langle|R(\nu,\mu)|^{2r}\rangle_{a,b} \leqslant C_r N^{-r}, \qquad r \in \mathbb{N} \text{ fixed}. \tag{8.4}$$

Here each band has only its own planting term and both use the same original Gaussian disorder. For $p \geqslant 4$, this follows from the cross-only cavity argument: choose the lower band endpoint a_0 and a small pattern-overlap tolerance γ so that, with $d = 2(1-a_0)+\gamma$,

$$4\beta_c^2 p\, d^{p-2}\exp\{4\beta_c^2 p\, d^{p-1}\} < 1. \tag{8.5}$$

The certified bounds $q_p > 0.9313$ for $p \geqslant 4$ allow $a_0 > 0.92$, and $\beta_c^2 \leqslant 2\log 2$ makes (8.5) strict for a sufficiently small γ. For $p \geqslant 5$ there is much more margin.

Lemma 8.4 (Cross-only narrow-band cavity)**.** *Let a, b be independent uniform patterns, independent of the Gaussian disorder. For each pattern use the hard band $\{|R(\nu,a)-q_*| \leqslant \delta\}$, with lower endpoint $a_0 = q_* - \delta > 0$, and weight it by its own planting term $N\beta_c^2 R(\nu,a)^p$. If (8.5) holds for $d = 2(1-a_0)+\gamma < 1$, then (8.4) holds for all fixed r. The same assertion holds when the two full disorders have any correlation $\rho \in [-1,1]$, uniformly in ρ.*

Proof. We give the proof at full disorder correlation one; a correlation $|\rho| \leqslant 1$ only reduces every absolute cross-cavity bound. Write a spin as (σ,ϵ) and set $R^- = N^{-1}\sum_{i<N}\sigma_i\tau_i$. The exact Gaussian decomposition

$$X_N(\sigma,\epsilon) = B_N(\sigma) + \epsilon Z_N(\sigma)$$

has independent even and odd last-spin parts. Its cavity covariance is

$$K_N(x) = \frac{N\beta_c^2}{2}\{(x+N^{-1})^p - (x-N^{-1})^p\} = \beta_c^2 p x^{p-1} + O(N^{-2}).$$

For two disorder copies, their baseline pair is jointly independent of their cavity-field pair: every baseline–cavity cross covariance is zero by the same parity calculation. Keep the full baseline pair and both cavity self covariances unchanged, and interpolate only the cavity cross covariance from zero to K_N. This is a legitimate positive-semidefinite Gaussian interpolation: it can be realized with a common cavity process of weight $\sqrt{u}$ and independent processes of weights $\sqrt{1-u}$. Denote the averaged two-group Gibbs expectation by ν_u. Both hard bands and both complete planted terms are retained at every u.

Let

$$E_N = \left\{\left|N^{-1}\sum_{i<N} a_i b_i\right| \leqslant \gamma\right\}.$$

It depends only on the first $N-1$ pattern signs and has complement of probability at most Ce^{-cN}. For configurations in the two bands, the Hamming triangle inequality gives

$$|R(\nu,\mu)| \leqslant 2(1-a_0) + |R(a,b)|, \qquad |R^-(\nu,\mu)| \leqslant d + 2/N \quad \text{on } E_N.$$

Thus, on E_N,

$$|K_N(R^-)| \leqslant L_N|R^-| + C/N^2, \quad L_N = \beta_c^2 p(d+2/N)^{p-2},$$

and $\sup|K_N| \leqslant M_N = \beta_c^2 p(d+2/N)^{p-1} + C/N^2$.

For an observable f of one replica from each group, the cross-streaming identity has exactly four terms:

$$\begin{aligned}\partial_u \nu_u(f) = \nu_u\big[f(&\epsilon_1\eta_1 K_N(R^-_{11}) - \epsilon_1\eta_2 K_N(R^-_{12}) \\ &- \epsilon_2\eta_1 K_N(R^-_{21}) + \epsilon_2\eta_2 K_N(R^-_{22}))\big].\end{aligned}$$

Here subscripts denote independent Gibbs replicas within a group; ϵ is a last spin from the first group and η from the second. The same identity holds with the fixed pattern indicator $\mathbb{1}_{E_N}$ multiplying f.

Fix an even integer $m \geqslant 2$, and put

$$A_m = \nu_1(R_{11}^m), \qquad A_m^-(u) = \nu_u(\mathbb{1}_{E_N}|R^-_{11}|^m), \qquad A_0 = 1.$$

The derivative identity for the nonnegative moment gives

$$|(A_m^-)'(u)| \leqslant 4M_N A_m^-(u), \qquad A_m^-(u) \leqslant e^{4M_N} A_m^-(1).$$

At $u=0$,

$$\nu_0(\mathbb{1}_{E_N}(R^-_{11})^{m-1}\epsilon_1\eta_1) = 0.$$

Indeed, simultaneously flip a_N, all first-group last spins, and the entire first-group cavity field. The pattern law, zero-cross-covariance Gaussian law, hard band, and exact planted term are preserved; E_N, R^- are unchanged, while $\epsilon_1\eta_1$ changes sign. This argument does not delete or approximate any hard-band constraint.

Apply the four-term identity to this odd observable and use Hölder, exchanging Gibbs replicas within each group:

$$\begin{aligned}\big|\nu_1(\mathbb{1}_{E_N}(R^-_{11})^{m-1}\epsilon_1\eta_1)\big| &\leqslant 4L_N e^{4M_N} A_m^-(1) + CN^{-2}e^{4M_N}\{A_m^-(1)\}^{(m-1)/m} \\ &\leqslant (4L_N e^{4M_N} + \eta)A_m^-(1) + C_{m,\eta}N^{-2m}.\end{aligned}$$

Full site exchangeability is used *before* inserting E_N:

$$A_m = \nu_1(R_{11}^{m-1}\epsilon_1\eta_1).$$

Expand $R_{11} = R^-_{11} + \epsilon_1\eta_1/N$, then discard E_N^c. The elementary binomial/Young bound yields

$$A_m \leqslant \nu_1(\mathbb{1}_{E_N}(R^-_{11})^{m-1}\epsilon_1\eta_1) + \frac{C_m}{N}A_{m-2} + C_m N^{1-m} + Ce^{-cN}.$$

Also, for every fixed $\eta > 0$, $A_m^-(1) \leqslant (1+\eta)A_m + C_{m,\eta}N^{-m}$. Since $4L_N e^{4M_N} \to 4\beta_c^2 p d^{p-2} e^{4\beta_c^2 p d^{p-1}} < 1$, choose η small and then N large. Absorbing the resulting A_m term gives

$$A_m \leqslant \frac{C_m}{N}A_{m-2} + C_m N^{1-m} + Ce^{-cN}.$$

Induction on even m proves $A_m = O(N^{-m/2})$, since $1-m \leqslant -m/2$ for $m \geqslant 2$. The finitely many small N are absorbed in the constant. No self-RS estimate and no convexity at a negative cross overlap are required. ∎

Lemma 8.5 (Self-averaging of the threshold CDF)**.** *Assume* (8.4) *and the uniform joint density bound of Lemma 8.2. For any deterministic thresholds b_N, set*

$$A_N(g) = \mathbb{E}_a \mathbb{1}_{\{\log \widehat{B}_a(g) \leqslant b_N\}}, \qquad a_N = \mathbb{E}\, A_N.$$

For $p \geqslant 4$,

$$N \operatorname{Var}(A_N) \leqslant C_p N^{-(p-3)/3} + Ce^{-cN}.$$

In particular $\sqrt{N}(A_N - a_N) \to 0$ in L^2. For $p = 3$, if the cross-overlap moment hypothesis is supplied through order ten, the same conclusion holds with $N \operatorname{Var}(A_N) \leqslant CN^{-1/8} + Ce^{-cN}$.

Proof. Smooth the threshold on a width $\epsilon > 0$, using a function with derivative δ_ϵ supported in $[b_N - \epsilon, b_N + \epsilon]$ and bounded by C/ϵ. Denote the smoothed pattern average by $A_{N,\epsilon}$. For good nearly orthogonal pattern pairs, the joint logarithmic density is at most C/N, so

$$\mathbb{E}(A_N - A_{N,\epsilon})^2 \leqslant C\epsilon^2/N + Ce^{-cN}.$$

For the original Gaussian disorder coordinates,

$$\nabla \log \widehat{B}_a \cdot \nabla \log \widehat{B}_b = N\beta_c^2 \langle R(\nu,\mu)^p \rangle_{a,b}.$$

Gaussian Poincaré and Hölder therefore give

$$\begin{aligned}
\operatorname{Var} A_{N,\epsilon} &\leqslant N\beta_c^2\, \mathbb{E}[|\delta_\epsilon(U_a)\delta_\epsilon(U_b)|\langle |R|^p\rangle_{a,b}] \\
&\leqslant CN\{\mathbb{E}\,\delta_\epsilon(U_a)^2\delta_\epsilon(U_b)^2\}^{1/2}\{\mathbb{E}\langle |R|^{2p}\rangle_{a,b}\}^{1/2} + CN\epsilon^{-2}e^{-cN} \\
&\leqslant C\epsilon^{-1}N^{(1-p)/2} + CN\epsilon^{-2}e^{-cN}.
\end{aligned}$$

The second line uses Jensen for the square of the Gibbs expectation. In the last line, the density bound gives $\mathbb{E}\,\delta_\epsilon(U_a)^2\delta_\epsilon(U_b)^2 \leqslant C/(N\epsilon^2)$, and (8.4) is used with $r = p$. Hence

$$N \operatorname{Var} A_N \leqslant C\epsilon^2 + C\epsilon^{-1}N^{(3-p)/2} + CN^2\epsilon^{-2}e^{-cN}.$$

Choose $\epsilon = N^{-(p-3)/6}$.

For the conditional $p = 3$ assertion, use Hölder exponents $s = 3/2$ and $s' = 3$ instead of two and two. More generally, the same argument for $1 < s < 2$ gives

$$\operatorname{Var} A_{N,\epsilon} \leqslant CN^{1-p/2-1/s}\epsilon^{-2+2/s} + CN\epsilon^{-2}e^{-cN},$$

provided the absolute cross moment of order ps' has its Gaussian scale. Indeed, the density estimate gives $\mathbb{E}\,|\delta_\epsilon(U_a)\delta_\epsilon(U_b)|^s \leqslant CN^{-1}\epsilon^{2-2s}$, and Jensen and the overlap moment estimate bound the other Hölder factor by $CN^{-p/2}$. For $p = 3, s = 3/2$, the required ninth absolute moment follows from the tenth moment by Hölder, and therefore

$$N \operatorname{Var} A_N \leqslant C\epsilon^2 + CN^{-1/6}\epsilon^{-2/3} + CN^2\epsilon^{-2}e^{-cN}.$$

Choosing $\epsilon = N^{-1/16}$ proves the displayed rate. For $p = 3$, the required cross moments are supplied by the rooted-cavity and polynomial-tail argument above. ∎

8.5. **Oscillatory tails.** Put $Y_a = \sqrt{N}\widehat{B}_a > 0$ and

$$H_t(y) = e^{ity} - \mathbb{1}_{\{y \leqslant 1\}}, \qquad Q_N(t) = \sqrt{N}\,\mathbb{E}_a\, H_t(Y_a).$$

The density derivative bounds (8.1) imply, for every fixed $t \neq 0$,

$$\left\| \sqrt{N}\,\mathbb{E}_a\, e^{itY_a}\mathbb{1}_{\{Y_a > M\}} \right\|_2 \leqslant \frac{C}{|t|M} + C\sqrt{N}e^{-cN/2}. \tag{8.6}$$

Indeed, for a good pattern pair let $f_N(u,v)$ be the joint density of $(\log \widehat{B}_a, \log \widehat{B}_b)$, and set

$$K_N(y,z) = \frac{f_N(\log(y/\sqrt{N}), \log(z/\sqrt{N}))}{yz}.$$

Then (8.1) gives

$$|K_N| \leqslant \frac{C}{Nyz}, \quad |\partial_y K_N| \leqslant \frac{C}{Ny^2 z}, \quad |\partial_z K_N| \leqslant \frac{C}{Nyz^2}, \quad |\partial_{yz} K_N| \leqslant \frac{C}{Ny^2 z^2},$$

the last numerator being $\partial_{12} f_N - \partial_1 f_N - \partial_2 f_N + f_N$. Integrate $e^{it(y-z)} K_N(y,z)$ once in each variable on $[M,\infty)^2$. All boundary and integral terms have absolute sum at most $C/(Nt^2M^2)$; terms at infinity vanish by the displayed bounds. Average over patterns and multiply by N. Bad pattern pairs contribute at most Ne^{-cN}. The scalar version similarly gives a mean-tail bound $C/(|t|M)$.

At zero, $|H_t(y)| \leqslant |t|y$ for $y \leqslant 1$, and the global joint density bound gives

$$\left\| \sqrt{N}\, \mathbb{E}_a\, H_t(Y_a) \mathbb{1}_{\{Y_a < \eta\}} \right\|_2 \leqslant C|t|\eta + C\sqrt{N} e^{-cN/2}, \qquad \eta < 1. \tag{8.7}$$

On a fixed compact interval $\eta \leqslant y \leqslant M$, (8.3), for one and two patterns, gives convergence of the mean and second moment. Letting $N \to \infty$, then $\eta \downarrow 0$ and $M \to \infty$, yields

$$Q_N(t) \longrightarrow q(t) := \kappa \int_0^\infty \frac{e^{ity} - \mathbb{1}_{\{y \leqslant 1\}}}{y}\, dy \quad \text{in } L^2, \tag{8.8}$$

where the integral at infinity is oscillatory. The density and tail bounds also give the useful uniform estimate

$$\|Q_N(t)\|_2 \leqslant C(1 + |\log|t||), \qquad 0 < |t| \leqslant 1. \tag{8.9}$$

To see this, split at $M = 1/|t|$; on $(0,M]$, integrate the absolute value of $H_t(y)/y$ using the joint density bound, obtaining $C(1+|\log|t||)$. Apply (8.6) above M. The exponentially small pattern term is uniformly bounded.

8.6. **Replacing the broad Palm band by the narrow band.** Choose $\varepsilon > 0$ with $\beta_c^2 \varepsilon^{p-2} < 1/2$. Define the broad Palm band

$$B_a(g) = \mathbb{E}_\nu\, W_\nu(P_a g) \mathbb{1}_{\mathcal{B}_a^{\text{broad}}},$$

where the broad event is $R(\nu,a) \geqslant \varepsilon$ for odd p, and $|R(\nu,a)| \geqslant \varepsilon$ for even p. Choose the narrow band around q_* used above. For even p, the factor $c_p = 2$ accounts exactly for its negative counterpart by spin-flip symmetry. The strict negative variational gap outside the two critical neighborhoods (one for odd p) and Gaussian concentration yield constants $c, C > 0$ such that

$$\mathbb{P}(0 \leqslant B_a - \widehat{B}_a > e^{-cN}) \leqslant Ce^{-cN}. \tag{8.10}$$

This is an absolute-mass bound: for every excluded magnetization slice, the tangent field upper bound has mean at most $-cN$; there are at most $N+1$ slices, each with squared Gaussian Lipschitz constant CN.

Equation (8.10) and $|e^{ix} - e^{iy}| \leqslant |x-y|$ show that replacing $\widehat{B}_a$ by B_a in the oscillatory averages costs $o(1)$ in L^2, after the prefactor $\sqrt{N}$, uniformly for t in bounded intervals. For the threshold at $\sqrt{N}B = 1$, a discrepancy on the good event in (8.10) requires $\sqrt{N}\widehat{B}_a$ to be within $\sqrt{N}e^{-cN}$ below one. The scalar global density bound $C/\sqrt{N}$ for $\log \widehat{B}_a$ makes this probability exponentially small. Thus the threshold averages may also be replaced at $o(N^{-1/2})$ in L^2.

8.7. **Palm characteristic equation and the stable limit.** The exact Gaussian planting identity gives, for bounded measurable f,

$$\mathbb{E}[L_N f(L_N)] = \mathbb{E}_{a,g}\, f(L_N(P_a g)).$$

The exact decomposition is

$$L_N(P_a g) = L_N(g) + B_a(g) + D_a(g),$$

where

$$\mathbb{E}_{a,g}\,|D_a| = O(N^{1-p/2}).$$

To verify the last estimate, the removed unplanted broad-band mass has exponentially small mean. On $|R| \leqslant \varepsilon$, use $N\beta_c^2|R|^p \leqslant cNR^2$, $c < 1/2$, and the Rademacher overlap law to obtain

$$\mathbb{E}(e^{N\beta_c^2|R|^p} - 1)\mathbb{1}_{\{|R|\leqslant\varepsilon\}} \leqslant CN\,\mathbb{E}[|R|^p e^{cNR^2}] = O(N^{1-p/2}).$$

For odd p, the remaining large negative overlaps contribute only an exponentially small amount because $e^{N\beta_c^2 R^p} \leqslant 1$ there.

Set the deterministic truncation center

$$a_N = \mathbb{P}_{a,g}(\sqrt{N}\widehat{B}_a(g) \leqslant 1), \qquad Z_N = \sqrt{N}(L_N - a_N), \qquad \phi_N(t) = \mathbb{E}\, e^{itZ_N}. \tag{8.11}$$

The one-band weak CLT gives $a_N \to 1/2$. Write $A_N(g) = \mathbb{E}_a\, \mathbb{1}_{\{\sqrt{N}\widehat{B}_a(g)\leqslant 1\}}$. Differentiating ϕ_N, applying the Palm identity, and using the broad-to-narrow replacement gives

$$\phi_N'(t) = i\,\mathbb{E}\left[e^{itZ_N}\{\sqrt{N}(A_N - a_N) + Q_N(t)\}\right] + r_N(t),$$

where, on each bounded t-interval,

$$|r_N(t)| \leqslant |t|N\,\mathbb{E}\,|D_a| + o(1).$$

For $p \geqslant 5$, $N\,\mathbb{E}\,|D_a| = O(N^{2-p/2}) \to 0$. Lemma 8.5 and (8.8) therefore imply

$$\phi_N'(t) - iq(t)\phi_N(t) \longrightarrow 0 \qquad (t \neq 0).$$

There is no missing initial-condition or tightness step: (8.9), the CDF variance estimate, and the bounded Palm error give an integrable uniform bound $C(1 + |\log|t||)$ for $|\phi_N'(t)|$ near zero. Consequently every subsequential limit satisfies the integral equation with initial value $\phi(0) = 1$. Its unique solution is

$$\phi(t) = \exp\left\{\kappa \int_0^\infty \frac{e^{ity} - 1 - ity\mathbb{1}_{\{y\leqslant 1\}}}{y^2}\, dy\right\}.$$

Equivalently,

$$\phi(t) = \exp\left\{-\frac{\pi\kappa}{2}|t| + i\kappa t(1 - \gamma_{\mathrm{E}} - \log|t|)\right\}, \qquad \phi(0) = 1.$$

This is the totally right-skewed stable law of index one, with Lévy measure $\kappa y^{-2}\mathbb{1}_{y>0}\,dy$ and the displayed truncation convention. Hence

$$\sqrt{N}(L_N - a_N) \Rightarrow S_\kappa. \tag{8.12}$$

Since $a_N \to 1/2$, this also proves $L_N \to 1/2$ in probability and, by the delta method,

$$\sqrt{N}\left(\log Z_N^{\text{normalized}} - \frac{N\beta_c^2}{2} - \log a_N\right) = \sqrt{N}(\log L_N - \log a_N) \Rightarrow 2S_\kappa.$$

In the last display, $Z_N^{\text{normalized}} = \mathbb{E}_\nu\, e^{X_N(\nu)}$; this symbol is distinct from the centered fluctuation variable in (8.11). For the unnormalized counting partition function, add $N\log 2$ to the deterministic free-energy centering.

Interpretation of the centering. The center a_N in (8.11) is an exact deterministic finite-size probability. The proof establishes a complete nondegenerate limiting law with that truncation center. It does not assert an unproved expansion of a_N through order $N^{-1/2}$, nor replace it by an expectation center for the free energy. Different sufficiently narrow fixed critical bands change this center only by $o(N^{-1/2})$, by the same far-band argument.

9. Distinct-index covariance corrections

This section proves a direct extension of the band and Palm argument to distinct-index pure models. No transfer of the final fluctuation theorem by an unquantified partition-function ratio is used. All probabilities defining the truncation center below are computed in the stated finite-size model.

9.1. **Exact covariance and the two standard normalizations.** For fixed $p \geqslant 3$, write $(N)_p = N(N-1)\cdots(N-p+1)$ and $c_{N,p} = (N)_p/N^p$. First use the unit self-variance normalization

$$H_N^{\mathrm{d}}(\sigma) = \sqrt{\frac{N}{\binom{N}{p}}} \sum_{|I|=p} g_I \sigma_I, \qquad \sigma_I = \prod_{i\in I} \sigma_i.$$

Its covariance is

$$C_{N,p}(r) = \frac{N}{\binom{N}{p}} e_p(x_1,\dots,x_N), \qquad x_i = \sigma_i\tau_i, \quad r = N^{-1}\sum_i x_i, \qquad C_{N,p}(1) = N. \tag{9.1}$$

Here e_p is the elementary symmetric polynomial. The exact set-partition formula is

$$p!e_p(x) = \sum_{\pi\in\mathcal{P}_p} (-1)^{p-|\pi|} \prod_{B\in\pi} (|B|-1)!\, N^{|\pi|} r^{o(\pi)}, \tag{9.2}$$

where $o(\pi)$ counts the odd-sized blocks. Indeed, the usual power-sum formula for elementary symmetric polynomials has power sum N for an even block and Nr for an odd block. In particular, for every fixed number of derivatives,

$$\sup_{-1\leqslant r\leqslant 1} \left|\partial_r^j\{C_{N,p}(r)/N - r^p\}\right| \leqslant C_{p,j}/N.$$

Every term in (9.2), after multiplying by $N/(N)_p$, is a constant times $N^{1-p+k}r^o/c_{N,p}$, where $k = |\pi|$, $o = o(\pi)$, and

$$2k - o \leqslant p. \tag{9.3}$$

The inequality follows because each even block has at least two elements and each odd block at least one.

The other common convention is

$$H_N^{\mathrm{std}}(\sigma) = \sqrt{\frac{p!}{N^{p-1}}} \sum_{|I|=p} g_I\sigma_I = \sqrt{c_{N,p}}H_N^{\mathrm{d}}(\sigma).$$

Its covariance is $c_{N,p}C_{N,p}(r)$, with self variance $Nc_{N,p}$. Thus the exact annealed logarithmic center is $\beta_c^2 N/2$ for the first convention and $\beta_c^2 N c_{N,p}/2$ for the second. The limiting critical inverse temperature β_c, the critical overlap q_*, and the single-band variance v are the same for both conventions.

9.2. **Direct transfer of the field and band inputs.** We record precisely what an $O(1)$ covariance perturbation supplies. If two centered Gaussian Hamiltonians have self variances bounded by CN and their covariance functions differ uniformly by at most a fixed constant, Gaussian interpolation gives, for every nonnegative observable f of a fixed number of Gibbs replicas,

$$e^{-C_f}\,\mathbb{E}\langle f\rangle_0 \leqslant \mathbb{E}\langle f\rangle_1 \leqslant e^{C_f}\,\mathbb{E}\langle f\rangle_0. \tag{9.4}$$

The constants depend on the number of replicas and the covariance bound, not on the observable's value. To check this, interpolate the two Gaussian processes with independent copies. The Gaussian streaming identity expresses the derivative as a fixed finite sum of terms $\mathbb{E}\langle f\Delta C(\sigma^i,\sigma^j)\rangle$, and hence its absolute value is bounded by a constant times $\mathbb{E}\langle f\rangle$. Gronwall proves the display. Deterministic weights differing by a uniformly bounded additive term in the Hamiltonian are treated by the same elementary comparison. Gaussian interpolation also bounds the difference of expected total logarithmic partition functions by $O(1)$.

Apply this with (9.1). The uniform positive-field overlap and magnetization bounds

$$\mathbb{E}\langle (R-q(h))^2\rangle \leqslant C/N, \qquad \mathbb{E}\langle (M-m(h))^2\rangle \leqslant C/N$$

and the $O(1)$ expected free-energy error pass to the distinct model directly. They remain uniform over the compact field neighborhood and the interpolation parameters used in the field proof.

For completeness, the field and band *weak* CLTs also transfer under an explicit coupling. In the ordered-tensor model, the part with pairwise distinct indices has the law $\sqrt{c_{N,p}}H_N^{\mathrm{d}}$; its complement is an independent Gaussian process with constant self variance $N(1-c_{N,p})=O(1)$. Adding any independent Gaussian process of constant self variance $v_0=O(1)$ changes a logarithmic partition function by $O_{\mathbb{P}}(1)$, uniformly in arbitrary fixed positive weights: conditionally on the original weights,

$$0 \leqslant \mathbb{E}_K \log\langle e^K\rangle \leqslant v_0/2,$$

and this logarithm is $\sqrt{v_0}$-Lipschitz in the new standard Gaussian coordinates. Similarly, the process

$$\sqrt{c_{N,p}}H_N^{\mathrm{d}} + \sqrt{1-c_{N,p}}\widetilde{H}_N^{\mathrm{d}}$$

has the law of H_N^{d}, and the added process again has self variance $O(1)$. The two comparisons prove the required $O_{\mathbb{P}}(1)$ coupling error for each of any fixed number of bands. The exact planted terms differ by $O(1)$, by (9.1). Thus the joint $\sqrt{N}$-scale band CLT, the limiting variance v, and the exponential macroscopic magnetization gaps are unchanged.

The Gaussian projection used for density smoothing now has feature Gram matrix

$$\big(C_{N,p}(R(a_i,a_j))/N\big)_{i,j}.$$

The projection slopes are $\beta_c R(\nu,a_j)^p + O(N^{-1})$, uniformly over all spin configurations and the fixed pattern tuples. Consequently the global diagonal-dominance bound, the inverse-map derivative bounds, the convexity argument upgrading weak convergence to a joint local CLT, and the small- and large-mass tail bounds apply directly. The constant in the local density is still $\kappa=(2\pi v)^{-1/2}$.

9.3. **Cross-cavity moments and threshold self-averaging.** The exact last-site cavity covariance in the distinct model is

$$K_{N,p}^{\mathrm{d}}(r) = \beta_c^2\frac{N}{\binom{N}{p}}e_{p-1}(x_1,\dots,x_{N-1}), \qquad r=N^{-1}\sum_{i<N}x_i.$$

Applying the same finite partition formula, noting that every power has the parity of $p-1$, gives

$$\left|K^{\mathrm{d}}_{N,p}(r) - \beta_c^2 p r^{p-1}\right| \leqslant \frac{C}{N}|r| + CN^{-(p-1)/2}\mathbb{1}_{\{p \ modd\}}. \tag{9.5}$$

The coefficient correction in the leading term is $O(N^{-1})$. All nonconstant lower powers have coefficient $O(N^{-1})$ and are bounded by a constant times $|r|$ on $[-1,1]$. When a constant is present, its largest possible order is exactly the second term in the display. For $p \geqslant 5$ that term is $O(N^{-2})$; for $p=4$ it is absent. The hard-band cross-cavity proof therefore has the same limiting contraction coefficient

$$4\beta_c^2 p d^{p-2} e^{4\beta_c^2 p d^{p-1}} < 1.$$

All of its symmetry, four-term streaming, and exchangeability steps are exact for this Hamiltonian and its exact planting term. It gives all fixed even cross-overlap moments $O(N^{-r})$, for every $p \geqslant 4$, just as in Lemma 8.4.

The Poincaré identity now reads

$$\nabla U_a \cdot \nabla U_b = \beta_c^2 \langle C_{N,p}(R)\rangle_{a,b}.$$

For every fixed $s \geqslant 1$, the cross-overlap moments and (9.3) give

$$\left(\mathbb{E}\langle |C_{N,p}(R)|^s\rangle\right)^{1/s} \leqslant C_{p,s} N^{1-p/2}.$$

Indeed, each finite polynomial term has L^s norm at most $CN^{1-p+k-o/2} \leqslant CN^{1-p/2}$. Using $s=2$ in the smoothed-CDF Poincaré estimate gives precisely the same bound as Lemma 8.5. Hence $\sqrt{N}(A_N - a_N) \to 0$ in L^2 for $p \geqslant 4$.

9.4. **The direct Palm remainder for $p \geqslant 5$.** Normalize the weights using the model's exact self variance and define the broad and narrow bands by its exact planted shift. The Palm identity then holds without an approximation. On a sufficiently small fixed overlap interval, the finite polynomial expansion gives

$$|\beta_c^2 C_{N,p}(r)| \leqslant \frac{N\beta_c^2}{c_{N,p}}|r|^p + C \leqslant cNr^2 + C, \qquad c < 1/2.$$

For a uniform Rademacher overlap, every fixed moment satisfies $\mathbb{E}\,|R|^o e^{cNR^2} \leqslant C_o N^{-o/2}$. Therefore

$$\begin{aligned}\mathbb{E}\left|e^{\beta_c^2 C_{N,p}(R)} - 1\right|\mathbb{1}_{\{|R|\leqslant\varepsilon\}} &\leqslant C\,\mathbb{E}\,|C_{N,p}(R)|e^{cNR^2} \\ &\leqslant C\sum_{\pi\in\mathcal{P}_p} N^{1-p+|\pi|-o(\pi)/2} \leqslant CN^{1-p/2}.\end{aligned}$$

The removed unplanted broad mass has exponentially small mean. For odd p, $C_{N,p}(-r) = -C_{N,p}(r)$, and $C_{N,p}(r) > 0$ uniformly for $r \geqslant \varepsilon$ once N is large, so the unused large negative overlaps also give an exponentially small error. This proves the direct bound $N\,\mathbb{E}\,|D_a| \to 0$ for every $p \geqslant 5$. All hypotheses of the Palm characteristic equation now hold in the distinct model itself. With its own exact deterministic band-CDF center,

$$\sqrt{N}(L^{\mathrm{d}}_N - a^{\mathrm{d}}_N) \Rightarrow S_\kappa, \qquad \sqrt{N}(\log L^{\mathrm{d}}_N - \log a^{\mathrm{d}}_N) \Rightarrow 2S_\kappa.$$

Every preceding bound also holds after multiplying the covariance by $c_{N,p}$. This proves the identical assertions in the standard coefficient convention, using its own CDF center and exact annealed normalization $\beta_c^2 N c_{N,p}/2$.

9.5. **Distinct four-spin models: removal of an auxiliary Gaussian.** For $p = 4$, the finite covariance polynomial is

$$C_{N,4}(r) = \frac{N}{(N)_4}\left[N^4 r^4 - (6N^3 - 8N^2)r^2 + 3N^2 - 6N\right].$$

Its minimum on the real line is at least $-C/N$. The numerator's unconstrained quadratic minimum in r^2 is $-6N^2 + 18N - 16$, which proves the claim. Moreover $\mathbb{E}_a\, C_{N,4}(R(\nu, a)) = 0$ exactly.

The separately proved signed four-spin Gaussian-IBP lemma applies as follows. Add an independent constant Gaussian variable $u_N G$, where $u_N = \delta/\sqrt{N}$ and $\delta > 0$ is fixed, to the temperature-weighted Hamiltonian. The new normalized partition function is exactly

$$\widetilde{L}_N = e^{u_N G - u_N^2/2} L_N^{\mathrm{d}}.$$

The original first-order Palm term is $T_a = \beta_c^2\, \mathbb{E}_\nu\, W_\nu C_{N,4}(R(\nu, a))$. It has zero pattern mean and the deterministic lower bound $T_a \geqslant -CL_N^{\mathrm{d}}/N$. Walsh orthogonality gives

$$\mathbb{E}_a(T_a/L_N^{\mathrm{d}})^2 = \beta_c^4 \left(\frac{N}{\binom{N}{4}}\right)^2 \sum_{|I|=4} \langle \sigma_I \rangle^2 \leqslant \frac{C}{N^2}\langle R_{12}^2 \rangle.$$

The last inequality follows by expanding $(\sum_i \sigma_i^1 \sigma_i^2)^4$: all nonconstant Walsh-square coefficients are nonnegative. The low-overlap exponential Taylor remainder has L^1 norm $O(N^{-2})$, by (9.3) or directly by the displayed polynomial.

Here are the precise reasons these estimates close the signed Palm correction. Conditionally on all other disorder, every Taylor phase has the form $\exp\{it\sqrt{N} e^{u_N G - u_N^2/2} S\}$, with $S \geqslant L_N^{\mathrm{d}}/2$ for large N; the lower bound for T_a supplies this even though T_a can have either sign. One and two integrations by parts in G bound its expectations after multiplication by $e^{u_N G - u_N^2/2}$ and its square by $C_\delta/(|t|S)$ and $C_\delta/(t^2 S^2)$, respectively. The mean-zero first-order term is therefore bounded by $C_\delta\, \mathbb{E}\langle R_{12}^2 \rangle^{1/2} \to 0$, using the critical overlap limit proved in the half-partition argument. The second-order remainder is $O_\delta(N^{-1/2})$. The added constant mode contributes the characteristic-equation coefficient $-\delta^2 t/4$. Thus the already proved band conclusions and the signed Gaussian-IBP lemma give

$$\sqrt{N}(\widetilde{L}_N - \widetilde{a}_N) \Rightarrow S_\kappa + \frac{\delta}{2} G_0,$$

with independent summands.

Finally, the augmented planted narrow band is exactly $e^{u_N G + u_N^2/2} \widehat{B}_a^{\mathrm{d}}$. The scalar log-band density bound $C/\sqrt{N}$ therefore gives

$$|\widetilde{a}_N - a_N^{\mathrm{d}}| \leqslant \frac{C}{\sqrt{N}}\, \mathbb{E}\,|u_N G + u_N^2/2| = O_\delta(N^{-1}).$$

Since $L_N^{\mathrm{d}} \to 1/2$ in probability,

$$\sqrt{N}(\widetilde{L}_N - \widetilde{a}_N) = \sqrt{N}(L_N^{\mathrm{d}} - a_N^{\mathrm{d}}) + \frac{\delta}{2} G + o_{\mathbb{P}}(1).$$

The unaugmented fluctuation is independent of G. Divide the characteristic functions by the everywhere nonzero Gaussian factor $e^{-\delta^2 t^2/8}$; Lévy's continuity theorem gives

$$\sqrt{N}(L_N^{\mathrm{d}} - a_N^{\mathrm{d}}) \Rightarrow S_\kappa, \qquad \sqrt{N}(\log L_N^{\mathrm{d}} - \log a_N^{\mathrm{d}}) \Rightarrow 2S_\kappa.$$

The argument applies unchanged to the covariance $c_{N,4} C_{N,4}$. The auxiliary Gaussian has been removed exactly; it is not a remaining component of the distinct-index limiting law.

Scalar certificates used by the contraction. The only numerical input to the cross-cavity contraction is the rigorously enclosed scalar inequality $q_p > 0.9313$ for all $p \geqslant 4$, together with the analytic bound $\beta_c^2 \leqslant 2\log 2$. The outward-rounded quadrature certificate and the scalar monotonicity reducing all $p \geqslant 4$ to $p = 4$ are in Appendix A, which also proves $q_p > 0.7689$ for $p \geqslant 3$ and $q_p > 0.9781$ for $p \geqslant 5$. No unverified decimal approximation is used as an assumption here.

The additional three-spin band inputs. For $p = 3$, apply the bounded-covariance comparison (9.4) to the positive narrow-band self-overlap and magnetization moments proved by the rooted cavity systems. Both the covariance difference and the difference between the exact planted terms are uniformly $O(1)$. Hence every fixed nonnegative moment, and the exponentially small band-edge mass, retains its bound. This supplies the one-band inputs through order 64 for Lemma 7.2 in the distinct model. Its exact cavity covariance has an additional constant term of order N^{-1}; Lemma 7.3 explicitly allows this remainder and proves optimal cross moments through order ten. The Hölder-exponent 3/2 CDF estimate then applies as well. This paragraph establishes the distinct three-spin *band and CDF inputs*; the separate three-spin Palm argument is still required to identify its final fluctuation law.

10. Weighted overlap and band-density estimates

10.1. **A compactly weighted critical overlap bound.** Use the broad overlap event $B(a,\sigma)$, the unplanted mass $C_a = \mathbb{E}_\sigma W_\sigma \mathbb{1}_{B(a,\sigma)}$, and its planted version $B_a = C_a(P_a g)$. The global density bound for $\log B_a$ is $C/\sqrt{N}$, by own-pattern projection. This bound applies to the distinct covariance as well, since its derivative on the fixed broad band is positive for large N.

Corollary 10.1 (Compactly weighted annealed overlap bound)**.** *For every fixed $K < \infty$ and every fixed real $r \geqslant 1$, the exact-tensor and distinct-index critical models satisfy*

$$\mathbb{E}\left[L^2\langle |R_{12}|^r\rangle \mathbb{1}_{\{L\leqslant K\}}\right] \leqslant C_{K,r} N^{-1/2}.$$

This conclusion does not require a lower bound on L, or the half limit. It uses only the one-planted broad-band density bound stated above.

Proof. The exact planting identity and the submass inequality $B_a(g) = C_a(P_a g) \leqslant L(P_a g)$ give

$$\begin{aligned}\mathbb{E}\left[L^2\langle \mathbb{1}_B\rangle \mathbb{1}_{\{L\leqslant K\}}\right] &= \mathbb{E}_{g,a}\left[W_a(g) C_a(g) \mathbb{1}_{\{L(g)\leqslant K\}}\right] \\ &= \mathbb{E}_{g,a}\left[B_a(g)\mathbb{1}_{\{L(P_a g)\leqslant K\}}\right] \\ &\leqslant \mathbb{E}\left[B_a \mathbb{1}_{\{B_a\leqslant K\}}\right] \leqslant \frac{C}{\sqrt{N}} \int_{-\infty}^{\log K} e^u \, du = \frac{CK}{\sqrt{N}}.\end{aligned}$$

Thus no maximum-band estimate or lower bound on L is needed for this compactly weighted expectation.

On $\{|R| \leqslant \varepsilon\}$, the unweighted annealed numerator has the estimate

$$\mathbb{E}\left[L^2\langle |R|^r \mathbb{1}_{\{|R|\leqslant\varepsilon\}}\rangle\right] = \mathbb{E}_{a,b}\, |R(a,b)|^r e^{\mathrm{Cov}(X_a,X_b)} \mathbb{1}_{\{|R(a,b)|\leqslant\varepsilon\}} \leqslant C_r N^{-r/2}.$$

This is the same sub-Gaussian Rademacher estimate as in the lemma; the covariance is bounded above by $cNR^2 + C$, with $c < 1/2$. For odd p, the large negative-overlap numerator is exponentially small. Since $|R|^r \leqslant 1$ on the broad part and $N^{-r/2} \leqslant N^{-1/2}$ for $r \geqslant 1$, the result follows. ∎

10.2. **Weighted variation of the two-band logarithmic density.** Throughout this section the inverse temperature is included in the feature vector: $X_N(g,\sigma) = g \cdot v_\sigma$, and $C_N(\sigma,\tau) = v_\sigma \cdot v_\tau$. We consider either $C_N(\sigma,\tau) = N\beta^2 R(\sigma,\tau)^3$, or the distinct-index kernel

$$C_N(r) = \beta^2 \frac{N}{(N)_3}\big[(Nr)^3 - (3N-2)Nr\big].$$

Let a, b be independent uniform patterns. Their two Gaussian disorders may have correlation $\rho \in [0,1]$. Let B_a, B_b be the positive narrow planted-band masses and put $U_a = \log B_a$, $U_b = \log B_b$. Their magnetization interval has lower endpoint greater than $1/\sqrt{2}$, and the limiting magnetization is $q > 1/\sqrt{2}$. All estimates below are uniform in ρ. The one-band localization and cross-moment inputs are those already proved in the preceding sections; in particular, optimal cross-overlap moments are available through order ten.

Conditional coordinates. Fix a sufficiently small $\gamma > 0$ and call a pair good when $\mathcal{G} = \{|R(a,b)| \leqslant \gamma\}$. The Gaussian projection construction for the joint band density gives, conditional on the remaining Gaussian coordinates, a map

$$(U_a, U_b) = T(w), \qquad w \sim N(0, NI_2), \qquad J = D_w T, \qquad \|J\| + \|J^{-1}\| \leqslant C. \tag{10.1}$$

This map is a global smooth diffeomorphism; its determinant is positive. Constants in (10.1) are uniform in the good patterns, the residual disorder, and ρ. These are the same unscaled coordinates as in the joint-density construction. Each band logarithm is a log-sum-exp with a bounded two-dimensional slope vector $s_{i,\sigma}$, $i = 1, 2$. Localization gives

$$\sup_\rho \mathbb{E} \sum_{i=1}^{2} \langle |s_{i,\sigma} - \beta q^3 e_i| \rangle_i \longrightarrow 0. \tag{10.2}$$

The $O(N^{-1})$ correction in the distinct-index normalized slopes is harmless. The orthogonalization correction contains $R(a,b)^3$, whose moments have the required smaller orders.

The insertions needed below are

$$P_1 = C_N(a,b)K_{ab}, \qquad K_{ab} = \langle C_N(\nu,\mu)\rangle_{a,b}, \tag{10.3}$$

$$P_2 = K_{a|b}K_{b|a}, \qquad K_{a|b} = \langle C_N(a,\mu)\rangle_b, \quad K_{b|a} = \langle C_N(b,\nu)\rangle_a. \tag{10.4}$$

The bracket in K_{ab} uses one independent replica from each band. Products of brackets are always interpreted using additional independent replicas, conditional on the disorder.

Lemma 10.2 (Weighted logarithmic density variation)**.** *For either insertion $P = P_1, P_2$, define the conditional signed density*

$$h_P(u) = \frac{\phi_N(w(u))P(w(u))}{\det J(w(u))}, \qquad \phi_N(w) = \frac{1}{2\pi N}\exp\{-|w|^2/(2N)\}. \tag{10.5}$$

The expectation in the following statements includes the residual disorder and the patterns restricted to $\mathcal{G}$. Then

$$\sup_\rho \mathbb{E} \int_{\mathbb{R}^2} |\partial_{12} h_P(u)|\, du = o(N^{-1}). \tag{10.6}$$

The zeroth and first derivative total variations are $O(N^{-1})$. If the one-band magnetization moments through order 64 satisfy their optimal bounds, then for P_1 one has the quantitative

estimates

$$\begin{aligned}\sup_{\rho}\mathbb{E}\int|h_{P_1}| &= O(N^{-1}),\\ \sup_{\rho}\mathbb{E}\int(|\partial_1 h_{P_1}|+|\partial_2 h_{P_1}|) &= O(N^{-3/2}),\\ \sup_{\rho}\mathbb{E}\int|\partial_{12} h_{P_1}| &= O(N^{-2}).\end{aligned} \tag{10.7}$$

All bounds remain valid for complex insertions by applying them to real and imaginary parts.

Proof. Each covariance factor in (10.3)– (10.4) is $O(N^{-1/2})$ in the needed moments. For K_{ab} this is the optimal cross-overlap estimate; for a pattern–band covariance it follows by first averaging the independent uniform pattern. Jensen and Hölder therefore give

$$\sup_{N,\rho}\mathbb{E}\,|NP_j|^{5/4}<\infty, \qquad j=1,2. \tag{10.8}$$

For example, two covariance factors of order $5/2$ require overlap moments of order $15/2$, below the available order ten. This statement applies as well to the replica insertions arising when these Gibbs expectations are differentiated.

For any Gibbs observable F independent of w,

$$D_{w_\ell}\langle F\rangle = \langle F(s_\ell - \langle s_\ell\rangle)\rangle. \tag{10.9}$$

The second derivative is the corresponding third cumulant. A constant may be subtracted from each centered slope without changing it. Boundedness of all slopes and (10.2) consequently give a bounded random variable Δ_N, with $\sup_\rho \mathbb{E}\,\Delta_N \to 0$, such that

$$\|D_w J\| + \|D_w^2 J\| \leqslant C\Delta_N.$$

Every differentiated covariance insertion contains at least one slope deviation. Uniform integrability from (10.8), applied to the additional replica expressions, yields

$$\sup_{\rho} N\,\mathbb{E}\big(|D_w P_j| + |D_w^2 P_j|\big) \longrightarrow 0. \tag{10.10}$$

No order-one intra-band overlap is replaced by a cross-overlap in this argument: the covariance factors retain their original cross-band or pattern–band type throughout differentiation.

Here is the precise deterministic chain-rule estimate used to pass from these moments to density variation:

$$\begin{aligned}\int_{\mathbb{R}^2}|\partial_{12}h_P(u)|\,du \leqslant C\,\mathbb{E}_w\Bigg[&|D_w^2 P| + |D_w P|\left(\frac{|w|}{N}+\|D_w J\|\right)\\ &+|P|\left(\frac{1}{N}+\frac{|w|^2}{N^2}+\frac{|w|}{N}\|D_w J\|+\|D_w J\|^2+\|D_w^2 J\|\right)\Bigg].\end{aligned} \tag{10.11}$$

Its first derivative counterpart is

$$\int(|\partial_1 h_P|+|\partial_2 h_P|) \leqslant C\,\mathbb{E}_w\left[|D_w P| + |P|\left(\frac{|w|}{N}+\|D_w J\|\right)\right]. \tag{10.12}$$

The zeroth variation is exactly $\mathbb{E}_w|P|$. To verify these inequalities, use $D_u = J^{-T}D_w$, $D_w J^{-1} = -J^{-1}(D_w J)J^{-1}$, and differentiate $(\det J)^{-1}$ twice. The Gaussian score is $-w/N$, its derivative is $-I_2/N$, and the change of variables $du = (\det J)\,dw$ cancels the density denominator up to fixed constants. These operations produce precisely the terms displayed in (10.11).

Averaging these inequalities, using (10.8)–(10.10) and $w/\sqrt{N} \sim N(0, I_2)$, proves (10.6). In particular, $N|P|$ is uniformly integrable, while the bounded Jacobian derivative factors tend to zero in probability. Terms containing Gaussian scores have an extra factor $N^{-1/2}$ or N^{-1}.

For the quantitative assertion the optimal magnetization moments give, at every required fixed order,

$$s_{i,\sigma} - \beta q^3 e_i = O_{L^r}(N^{-1/2}), \qquad D_w J = O_{L^r}(N^{-1}), \qquad D_w^2 J = O_{L^r}(N^{-3/2}).$$

The cross-pattern slopes have smaller orders, by uniform-pattern moments. Centered-slope differentiation gives

$$P_1 = O_{L^{5/4}}(N^{-1}), \qquad D_w P_1 = O_{L^{5/4}}(N^{-3/2}), \qquad D_w^2 P_1 = O_{L^{5/4}}(N^{-2}). \tag{10.13}$$

The largest moment requirement occurs in the last expression. Use Hölder exponents 6 for the seed covariance, 3 for the band cross covariance, and 20/3 for each of the two slope deviations. Their reciprocal sum is 4/5. The band covariance thus needs only overlap moment nine; the slope moments are below 64. For factors involving a Jacobian derivative or a Gaussian score, the remaining Hölder room uses only fixed Gaussian moments and slope moments through order 20. Substitution in (10.11) and (10.12) proves (10.7). ■

Integration against bounded tests. For bounded smooth deterministic functions H_1, H_2, two integrations by parts give

$$\int H_1'(u)\overline{H_2'(v)} h_P(u,v)\, du\, dv = \int H_1(u)\overline{H_2(v)} \partial_{12} h_P(u,v)\, du\, dv. \tag{10.14}$$

One may first truncate in logarithmic coordinates. The bounded inverse Jacobian and Gaussian density make all boundary terms vanish; then the total-variation estimate justifies removal of truncation. The resulting bound depends only on $\|H_1\|_\infty \|H_2\|_\infty$, not on their derivatives. Bounded smooth approximation therefore extends bounds for the original expectation to bounded measurable, possibly N-dependent tests. Bad pattern pairs are separated in the original bounded expectation before interpolation; their contribution is polynomially bounded times e^{-cN}.

Gaussian divergence identity. If a smooth vector field V has square-integrable derivatives, its Gaussian divergence is $\delta V = \sum_j (g_j V_j - \partial_j V_j)$, and

$$\mathbb{E}\,|\delta V|^2 = \mathbb{E}\,\|V\|^2 + \mathbb{E} \sum_{i,j} (\partial_i V_j)\overline{(\partial_j V_i)}. \tag{10.15}$$

For $V = \mathbb{E}_a\, v_a H(U_a)$, the derivative term is an expectation of $H'(U_a)\overline{H'(U_b)} P_2$. The squared vector norm, after interpolating the two disorders, uses the same expression with P_1. At zero disorder correlation the marginal band laws are pattern independent. On the symmetric good-pair event their initial contribution vanishes exactly, because both three-spin kernels are odd in $R(a,b)$. These identities explain the two insertions above. In the compact Palm argument the ordinary Hilbert–Schmidt derivative norm instead uses P_1; derivatives through the additional scalar variables are treated separately there. Thus the density lemma itself makes no independence claim about those scalar variables.

11. The four-spin Palm correction

This section completes the four-spin Palm argument, using the band local limits, empirical threshold self-averaging, half limit, and vanishing critical overlap proved above.

11.1. **Established inputs and normalization.** Let the pure Ising four-spin Hamiltonian satisfy

$$\mathbb{E}H_N(\sigma)H_N(\tau) = NR(\sigma,\tau)^4, \qquad X_N = \beta H_N, \qquad W_\sigma = e^{X_N(\sigma)-N\beta^2/2}, \qquad L_N = \mathbb{E}_\sigma W_\sigma.$$

All pattern averages are uniform on the Ising cube. The temperature $\beta = \beta_c$ is fixed. The preceding critical-band results provide:

1. $L_N \to 1/2$ in probability and $\mathbb{E}_g\langle R_{12}^2\rangle \to 0$ for the full Gibbs measure.
2. Positive broad single-planted bands B_a, positive narrow bands $\widehat{B}_a$, and the exponential broad-to-narrow error estimates in Section 8.
3. With $Y_a = \sqrt{N}\widehat{B}_a$,

$$a_N = \mathbb{P}_{g,a}(Y_a \leqslant 1), \qquad A_N(g) = \mathbb{E}_a \mathbf{1}_{Y_a\leqslant 1}, \qquad Q_N(t) = \sqrt{N}\mathbb{E}_a(e^{itY_a} - \mathbf{1}_{Y_a\leqslant 1}),$$

we have $a_N \to 1/2$, $\sqrt{N}(A_N - a_N) \to 0$ in L^2, and for each $t \neq 0$,

$$Q_N(t) \longrightarrow q(t) = \kappa \int_0^\infty \frac{e^{ity} - \mathbf{1}_{y\leqslant 1}}{y}\, dy \quad \text{in } L^2.$$

We also have the uniform bound $\|Q_N(t)\|_2 \leqslant C(1 + |\log|t||)$ for $0 < |t| \leqslant 1$. These band statements are exactly the conclusions of the local-limit and cross-cavity arguments in Section 8, which applies to $p = 4$.

Write $Z_N = \sqrt{N}(L_N - a_N)$ and $\phi_N(t) = \mathbb{E}\exp(itZ_N)$.

11.2. **The exact constant mode.** The Gaussian random variable

$$A = \mathbb{E}_\sigma H_N(\sigma)$$

has variance

$$w_N = N\mathbb{E}_{\sigma,\tau}R(\sigma,\tau)^4 = \frac{3}{N} - \frac{2}{N^2}.$$

Moreover $\mathrm{Cov}(A, H_N(\sigma)) = w_N$ for every spin, so the process $H'_N = H_N - A$ is independent of A. Its covariance is $NR^4 - w_N$. Set

$$A = \sqrt{w_N}G, \quad G \sim N(0,1), \quad u_N = \beta\sqrt{w_N}, \quad v_N = \sqrt{N}u_N, \quad c_N(G) = e^{u_N G - u_N^2/2}.$$

For all sufficiently large N, v_N is bounded above and away from zero, and $v_N^2 \to 3\beta^2$. Define the normalized partition function of H'_N by

$$L'_N = \mathbb{E}_\sigma \exp\{\beta H'_N(\sigma) - (N\beta^2 - u_N^2)/2\}.$$

Then exactly

$$L_N = c_N(G)L'_N, \qquad \mathbb{E}L'_N = 1. \tag{11.1}$$

The full Gibbs probabilities are identical for H_N and H'_N, and thus are independent of G. Also $c_N \to 1$ in probability, so the assumed half limit implies $L'_N \to 1/2$ in probability.

There is an extra factor under planting that must be retained. If P'_a denotes planting in the residual process with covariance $N\beta^2R^4 - u_N^2$, and B'_a is its planted band, then the original broad planted band satisfies exactly

$$B_a(g) = c_N(G)e^{u_N^2}B'_a(g'). \tag{11.2}$$

Indeed planting translates G to $G + u_N$. We retain the factor $e^{u_N^2}$ throughout; none of the arguments below silently discard it.

11.3. **First-order overlap term and an L^1 remainder.** Choose a fixed $\epsilon > 0$ with $\beta^2\epsilon^2 < 1/2$, and let the broad band be $|R(\nu, a)| \geqslant \epsilon$. The exact Palm decomposition is

$$L_N(P_a g) = L_N + B_a + D_a.$$

Define

$$T_a = N\beta^2 \mathbb{E}_\nu W_\nu R(\nu, a)^4 \geqslant 0.$$

Then

$$D_a = T_a + \mathcal{R}_a, \qquad \mathbb{E}_{g,a}|\mathcal{R}_a| \leqslant CN^{-2}. \tag{11.3}$$

To prove this, put $x = N\beta^2 R^4$. The remainder is exactly

$$\mathcal{R}_a = \mathbb{E}_\nu W_\nu (e^x - 1 - x)\mathbf{1}_{|R|<\epsilon} - \mathbb{E}_\nu W_\nu (1 + x)\mathbf{1}_{|R|\geqslant\epsilon}.$$

Use $\mathbb{E}_g W_\nu = 1$, $0 \leqslant e^x - 1 - x \leqslant (x^2/2)e^x$, and $x \leqslant cNR^2$, where $c = \beta^2\epsilon^2 < 1/2$. The usual sub-Gaussian Rademacher overlap bound gives

$$\mathbb{E}R^8 e^{cNR^2} \leqslant CN^{-4}.$$

The low-overlap term is therefore at most $CN^2 N^{-4} = CN^{-2}$. The high-overlap term has exponentially small mean, even after the factor $1 + N\beta^2$. This proves (11.3).

Since $\mathbb{E}_a NR(\nu, a)^4 = w_N$, write

$$T_a = u_N^2 L_N + U_a, \qquad \mathbb{E}_a U_a = 0. \tag{11.4}$$

All three quantities L_N, T_a, U_a scale by the same positive factor $c_N(G)$. Below write $L_N = c_N x$, $T_a = c_N T'_a$, $U_a = c_N U'_a$, where $x = L'_N > 0$ and the primed quantities are independent of G.

11.4. **Exact Walsh bound for the mean-zero overlap term.** For any deterministic probability measure π on the Ising cube, let

$$m_S = \langle \prod_{i\in S} \nu_i \rangle_\pi, \qquad Q_2 = \langle R(\nu, \mu)^2 \rangle_{\pi\otimes\pi}.$$

The fourth-power identity is

$$\left(\sum_{i=1}^N x_i\right)^4 = 3N^2 - 2N + (12N - 16)\sum_{i<j} x_i x_j + 24 \sum_{i<j<k<\ell} x_i x_j x_k x_\ell, \qquad x_i \in \{-1, 1\}. \tag{11.5}$$

Consequently, by Walsh orthogonality in the uniform pattern a,

$$\mathbb{E}_a(U_a/L_N)^2 = \beta^4 \left[\frac{(12N - 16)^2}{N^6} \sum_{|S|=2} m_S^2 + \frac{24^2}{N^6} \sum_{|S|=4} m_S^2 \right]. \tag{11.6}$$

The first sum is at most $N^2 Q_2/2$. Apply (11.5) to $x_i = \nu_i\mu_i$ and average over $\pi\otimes\pi$. Because all Walsh-square terms are nonnegative and $12N - 16 > 0$,

$$24\sum_{|S|=4} m_S^2 \leqslant N^4\langle R^4\rangle \leqslant N^4 Q_2.$$

Thus, uniformly in the Gibbs measure,

$$\boxed{\mathbb{E}_a(U_a/L_N)^2 \leqslant \frac{C\beta^4}{N^2}Q_2, \qquad \mathbb{E}_a(T_a/L_N)^2 \leqslant \frac{C\beta^4}{N^2}.} \tag{11.7}$$

The same ratios hold with primed variables. This is stronger than the bound with an additional $O(1/N)$ inside the first parentheses. In particular,

$$N\mathbb{E}_{g',a}|U_a'|/x \leqslant C\mathbb{E}_{g'}\sqrt{Q_2} \longrightarrow 0. \tag{11.8}$$

11.5. **Two exact Gaussian integration-by-parts bounds.** Fix $S > 0$, independent of G, and $t \neq 0$. Let

$$F(G) = \exp\{it\sqrt{N}c_N(G)S\}.$$

Since $F' = itv_N S c_N F$, Gaussian integration by parts gives

$$\mathbb{E}_G c_N F = \frac{\mathbb{E}_G GF}{itv_N S}. \tag{11.9}$$

Differentiating once more and using $\mathbb{E}\,F'' = \mathbb{E}(G^2-1)F$ gives

$$\mathbb{E}_G c_N^2 F = \frac{\mathbb{E}_G(G^2 - 1 - u_N G)F}{(itv_N S)^2}. \tag{11.10}$$

Therefore, uniformly in sufficiently large N and $S > 0$,

$$\boxed{|\mathbb{E}_G c_N F| \leqslant \frac{C}{|t|S}, \qquad |\mathbb{E}_G c_N^2 F| \leqslant \frac{C}{t^2S^2}.} \tag{11.11}$$

These are oscillatory estimates; they require no moments of S, L_N', or its inverse. Their derivation is valid conditionally because all lognormal Gaussian moments are finite.

11.6. **Taylor expansion inside the Palm identity.** The exact planting identity and (11.3) give

$$\phi_N'(t) = i\mathbb{E}e^{itZ_N}[\sqrt{N}(A_N - a_N) + Q_N(t)] + \Delta_N(t) + O(|t|/N) + o(1), \tag{11.12}$$

where broad-to-narrow replacement produces the last term, uniformly on bounded t intervals, and

$$\Delta_N(t) = i\sqrt{N}\mathbb{E}_{g,a}e^{it\sqrt{N}(L_N+B_a-a_N)}(e^{it\sqrt{N}T_a} - 1).$$

The first-order term is

$$-tN\mathbb{E}_{g,a}T_a e^{it\sqrt{N}(L_N+B_a-a_N)}. \tag{11.13}$$

The second-order integral remainder has absolute value at most

$$|t|^2N^{3/2}\int_0^1(1-s)\left|\mathbb{E}_{g',a,G}(T_a')^2c_N^2 e^{it\sqrt{N}(c_N[x+e^{u^2_N}B_a'+sT_a']-a_N)}\right|ds.$$

Here $T_a' \geqslant 0$, so every phase coefficient

$$S_s = x + e^{u_N^2} B'_a + sT'_a \geqslant x.$$

Apply (11.11) conditionally and then (11.7). This bounds the remainder by

$$CN^{3/2}\mathbb{E}_{g',a}(T'_a/x)^2 \leqslant CN^{-1/2}. \tag{11.14}$$

The bound is uniform for all $t \neq 0$; at zero the remainder is exactly zero.

Split (11.13) using (11.4). For the mean-zero term, take $S = x + e^{u_N^2} B'_a \geqslant x$ in (11.11), then use (11.8):

$$|t|N \left| \mathbb{E}_{g,a} U_a e^{it\sqrt{N}(L_N + B_a - a_N)} \right| \leqslant CN\mathbb{E}_{g',a}|U'_a|/x = o(1). \tag{11.15}$$

This estimate also is uniform in $t \neq 0$. Hence

$$\Delta_N(t) = -tNu_N^2 \mathbb{E}_{g,a} L_N e^{it\sqrt{N}(L_N + B_a - a_N)} + o(1). \tag{11.16}$$

11.7. **The coefficient of the constant mode.** For each fixed $t \neq 0$,

$$\mathbb{E}_{g,a}(L_N - 1/2) e^{it\sqrt{N}(L_N + B_a - a_N)} \longrightarrow 0. \tag{11.17}$$

Here is the required uniform-integrability substitute. Fix $\eta \in (0, 1/4)$ and split on $|x - 1/2| \leqslant \eta$, where $x = L'_N$. On this event,

$$\mathbb{E}_G|c_N x - 1/2| \leqslant \eta + (1/2 + \eta)\mathbb{E}_G|c_N - 1| \leqslant \eta + O(N^{-1/2}).$$

On its complement, (11.11) with $S = x + e^{u_N^2} B'_a \geqslant x$ bounds the conditional expectation of the weighted L_N phase by $C/|t|$; the unweighted half-phase is bounded by $1/2$. Thus the contribution of this complement is at most

$$(C/|t| + 1/2)\mathbb{P}(|L'_N - 1/2| > \eta) \longrightarrow 0.$$

Let η tend to zero to prove (11.17). No assertion of uniform integrability of L_N itself is used.

Next, using the band decomposition and assumptions from Section 11.1,

$$\begin{aligned} \mathbb{E}_{g,a} e^{it\sqrt{N}(L_N + B_a - a_N)} &= \mathbb{E}_g e^{itZ_N} \left[a_N + \frac{1}{\sqrt{N}} \{ \sqrt{N}(A_N - a_N) + Q_N(t) \} \right] + o(1) \\ &= a_N \phi_N(t) + o(1). \end{aligned}$$

Combining this with (11.17) and $a_N \to 1/2$ gives

$$\mathbb{E}_{g,a} L_N e^{it\sqrt{N}(L_N + B_a - a_N)} = \frac{1}{4}\phi_N(t) + o(1). \tag{11.18}$$

Since $Nu_N^2 \to 3\beta^2$, equations (11.16) and (11.18) identify the correction:

$$\Delta_N(t) = -\frac{3\beta^2}{4}\, t\phi_N(t) + o(1). \tag{11.19}$$

11.8. **Characteristic function and the Gaussian-plus-stable law.** For every fixed $t \neq 0$,

$$\phi_N'(t) - \left[iq(t) - \frac{3\beta^2}{4}t\right]\phi_N(t) \longrightarrow 0. \tag{11.20}$$

The passage through $t = 0$ is legitimate. In (11.12), the band terms have the uniform integrable bound $C(1 + |\log|t||)$. The second-order remainder is $O(N^{-1/2})$, and the mean-zero first-order term is uniformly bounded by (11.15). The constant term can be bounded without oscillation by

$$|t|Nu_N^2\mathbb{E}L_N \leqslant C|t|.$$

The $\mathcal{R}$ error is $O(|t|/N)$. Therefore $|\phi_N'(t)| \leqslant C(1 + |\log|t||)$ on $0 < |t| \leqslant 1$, after adjusting the constant. The integral equation has initial value $\phi(0) = 1$, and its unique solution is

$$\boxed{\phi(t) = \exp\left\{-\frac{3\beta^2}{8}t^2 + \kappa\int_0^\infty \frac{e^{ity} - 1 - ity\mathbf{1}_{y\leqslant 1}}{y^2}\,dy\right\}.} \tag{11.21}$$

Equivalently,

$$\sqrt{N}(L_N - a_N) \implies S_\kappa + \frac{\sqrt{3}\,\beta}{2}G_0,$$

where S_κ has the same totally right-skewed index-one stable law and truncation convention as the $p \geqslant 5$ proof, and G_0 is an independent standard Gaussian. Since $a_N \to 1/2$, the logarithmic fluctuation is

$$\boxed{\sqrt{N}(\log L_N - \log a_N) \implies 2S_\kappa + \sqrt{3}\,\beta G_0.}$$

Thus the Gaussian contribution has variance $3\beta^2/4$ for the normalized partition-function fluctuation and $3\beta^2$ for the logarithmic free-energy fluctuation. The exact factor $e^{u_N^2}$ in planted bands was retained and creates no additional unaccounted drift.

11.9. **Critical overlap input.** The required vanishing of the full critical overlap was proved from differentiability at the annealed endpoint in the half-limit argument. Thus all inputs used in this four-spin step have now been supplied.

11.10. **Distinct-index four-spin normalization: an artificial mode and exact deconvolution.** The preceding Gaussian correction depends on the tensor convention. We now prove that it is absent for the standard distinct-index normalization

$$H_N^d(\sigma) = \sqrt{\frac{N}{M_N}}\sum_{S\subset[N],\,|S|=4} g_S\sigma_S, \qquad M_N = \binom{N}{4}.$$

Thus $\operatorname{Var} H_N^{\mathrm{d}}(\sigma) = N$, and the temperature-weighted covariance is

$$C_N^d(r) = \beta^2\frac{N}{(N)_4}\left[(Nr)^4 - (6N-8)(Nr)^2 + 3N(N-2)\right], \quad (N)_4 = N(N-1)(N-2)(N-3). \tag{11.22}$$

Its pattern average is exactly zero. Unlike the ordered-index Hamiltonian, this Gaussian process has no constant mode. Assume that the Section 11.1 band/half/overlap inputs have been established for this covariance (and for its addition of the constant covariance below).

11.10.1. *Uniform lower bound and low-overlap remainder.* The quadratic polynomial in $z = (Nr)^2$ appearing in (11.22) has minimum

$$3N(N-2) - (3N-4)^2 = -6N^2 + 18N - 16.$$

It follows that, for all sufficiently large N and all $r \in [-1,1]$,

$$C_N^d(r) \geqslant -\frac{C\beta^2}{N}. \tag{11.23}$$

Fix any constant $a > 0$, take an independent standard Gaussian G, and augment the model by

$$H_N^{\mathrm{aug}} = H_N^d + \sqrt{a/N}\, G, \qquad u_N = \beta\sqrt{a/N}, \quad v_N = \sqrt{N}u_N = \beta\sqrt{a}.$$

The covariance of the temperature-weighted augmented model is

$$C_N^{\mathrm{aug}}(r) = C_N^d(r) + u_N^2.$$

Write the exactly annealed-normalized weights and partition functions as

$$W_\nu^{\mathrm{aug}} = c_N(G) W_\nu^d, \qquad L_N^{\mathrm{aug}} = c_N(G) L_N^d, \qquad c_N = e^{u_N G - u_N^2/2}. \tag{11.24}$$

With the same fixed broad overlap cutoff, the augmented Palm decomposition satisfies

$$D_a^{\mathrm{aug}} = T_a^{\mathrm{aug}} + \mathcal{R}_a^{\mathrm{aug}}, \qquad T_a^{\mathrm{aug}} = \mathbb{E}_\nu W_\nu^{\mathrm{aug}} C_N^{\mathrm{aug}}(R(\nu,a)), \qquad \mathbb{E}|\mathcal{R}_a^{\mathrm{aug}}| \leqslant CN^{-2}. \tag{11.25}$$

To verify the last bound, use the same exact remainder formula as in Section 11.3, now with $x = C_N^{\mathrm{aug}}(R)$ of either sign. The universal inequality

$$|e^x - 1 - x| \leqslant \tfrac{1}{2} x^2 e^{|x|}$$

applies. From (11.22), on a sufficiently small fixed overlap interval,

$$|x| \leqslant C\left(NR^4 + R^2 + N^{-1}\right) \leqslant cNR^2 + C/N, \qquad c < 1/2,$$

where the leading coefficient tends to β^2, so the interval can be chosen to make $c < 1/2$. The Rademacher moments with the exponential tilt give

$$\mathbb{E} x^2 e^{|x|} \leqslant C\left[N^2 \mathbb{E} R^8 e^{cNR^2} + \mathbb{E} R^4 e^{cNR^2} + N^{-2}\mathbb{E} e^{cNR^2}\right] \leqslant CN^{-2}.$$

The removed high-overlap unplanted term again has exponentially small mean, since $|C_N^{\mathrm{aug}}(r)| \leqslant CN$ everywhere. This proves (11.25).

11.10.2. *The mean-zero Walsh bound.* Because $\mathbb{E}_a C_N^{\mathrm{d}}(R(\nu,a)) = 0$,

$$T_a^{\mathrm{aug}} = u_N^2 L_N^{\mathrm{aug}} + U_a^{\mathrm{aug}}, \qquad \mathbb{E}_a U_a^{\mathrm{aug}} = 0.$$

For any Gibbs measure, write $m_S = \langle \nu_S \rangle$ and put

$$Q_{4,N}^d = \frac{1}{M_N} \sum_{|S|=4} m_S^2 \geqslant 0.$$

Walsh orthogonality gives the exact identity

$$\mathbb{E}_a (U_a^{\mathrm{aug}} / L_N^{\mathrm{aug}})^2 = \frac{\beta^4 N^2}{M_N} Q_{4,N}^d. \tag{11.26}$$

Equation (11.22) also shows

$$Q^d_{4,N} = \frac{N^4\langle R^4\rangle - (6N-8)N^2\langle R^2\rangle + 3N(N-2)}{(N)_4} \leqslant C\langle R^4\rangle + C/N^2. \tag{11.27}$$

Consequently

$$\mathbb{E}_a(U_a^{\mathrm{aug}}/L_N^{\mathrm{aug}})^2 \leqslant \frac{C}{N^2}(\langle R^4\rangle + N^{-2}), \qquad \mathbb{E}_a(T_a^{\mathrm{aug}}/L_N^{\mathrm{aug}})^2 \leqslant C/N^2. \tag{11.28}$$

The first right-hand side is $o(N^{-2})$ after disorder averaging under vanishing critical overlap. The inequalities also hold for the ratios after conditioning out G.

11.10.3. *Control of the signed first-order term.* Here T_a^{aug} need not be nonnegative. Equation (11.23), however, gives

$$T_a^{\mathrm{aug}} \geqslant -CN^{-1}L_N^{\mathrm{aug}}. \tag{11.29}$$

After factoring out c_N, write $L_N^{\mathrm{aug}} = c_N x$, $B_a^{\mathrm{aug}} = c_N e^{u_N^2} B_a^{\mathrm{d}}$, and $T_a^{\mathrm{aug}} = c_N T'_a$. The coefficient in every second-order integral-remainder phase is therefore

$$S_s = x + e^{u_N^2}B_a^d + sT'_a \geqslant (1 - C/N)x \geqslant x/2$$

for all sufficiently large N. Thus the exact Gaussian identities (11.9)–(11.11) apply with the same constants up to a fixed factor. The entire proof of Sections 11.6–11.8 goes through, using (11.28) in place of (11.7). Since $Nu_N^2 = \beta^2 a$, the conclusion is

$$\sqrt{N}(L_N^{\mathrm{aug}} - a_N^{\mathrm{aug}}) \implies S_\kappa + \frac{\beta\sqrt{a}}{2}G_0. \tag{11.30}$$

The Gaussian variance in (11.30) is exactly the variance of the artificially added mode at half mass.

11.10.4. *Centers and deconvolution.* Let a_N^{d} use the distinct-model narrow planted band and the same threshold. Exactly,

$$\log \widehat{B}_a^{\mathrm{aug}} = \log \widehat{B}_a^d + u_N G + u_N^2/2. \tag{11.31}$$

The distinct-model scalar logarithmic density is globally bounded by $C/\sqrt{N}$: projecting onto its own Gaussian pattern gives a positive derivative bounded above and below by fixed multiples of $\sqrt{N}$ on the narrow band, since $C_N^{\mathrm{d}}(r) = N\beta^2 r^4 + O(1)$ there. Conditional inverse transformation therefore gives this density bound directly. As G is independent of the distinct-model band,

$$|a_N^{\mathrm{aug}} - a_N^d| \leqslant \frac{C}{\sqrt{N}}\mathbb{E}|u_N G + u_N^2/2| \leqslant C/N. \tag{11.32}$$

In particular their difference is negligible after multiplying by $\sqrt{N}$.

Set $Z_N^{\mathrm{d}} = \sqrt{N}(L_N^{\mathrm{d}} - a_N^{\mathrm{d}})$ and $Z_N^{\mathrm{aug}} = \sqrt{N}(L_N^{\mathrm{aug}} - a_N^{\mathrm{aug}})$. By the half limit for L_N^{d} and (11.24),

$$Z_N^{\mathrm{aug}} = Z_N^d + \frac{\beta\sqrt{a}}{2}G + o_{\mathbb{P}}(1). \tag{11.33}$$

The displayed Gaussian is independent of Z_N^{d} for each N. No preliminary tightness of Z_N^{d} is needed: the error in (11.33) follows from $\sqrt{N}(c_N-1)=\beta\sqrt{a}G+o_{\mathbb{P}}(1)$, $L_N^{\mathrm{d}}\to 1/2$, and (11.32).

Taking characteristic functions in (11.33), their boundedness makes the $o_{\mathbb{P}}(1)$ error negligible, and gives

$$\phi_N^{\mathrm{aug}}(t)-\phi_N^d(t)e^{-\beta^2 a t^2/8}\longrightarrow 0.$$

The augmented limit (11.30) contains exactly the same nonzero Gaussian factor, which can be divided out for every fixed t. Lévy's continuity theorem therefore proves

$$\boxed{\sqrt{N}(L_N^d-a_N^d)\Longrightarrow S_\kappa,\qquad \sqrt{N}(\log L_N^d-\log a_N^d)\Longrightarrow 2S_\kappa.} \tag{11.34}$$

This identifies the distinction between conventions: the ordered-index exact covariance NR^4 has the genuine constant Gaussian contribution from Section 11.8; the distinct-index covariance (11.22) has a purely stable limit after its corresponding deterministic truncation center. The auxiliary Gaussian in this section is completely removed by exact characteristic-function deconvolution.

12. The three-spin Palm equation

This section proves the critical Palm step for both standard Gaussian normalizations of the pure Ising three-spin model. The distinct-index argument is developed first; the final subsection supplies the additional weighted matrix estimates for the ordered-index covariance. The band and half-partition inputs are stated explicitly below. No auxiliary Gaussian mode is used.

12.1. Model and inputs.

Let $M=\binom{N}{3}$ and

$$X_N(\sigma)=\sum_{|S|=3} g_S v_{\sigma,S},\qquad v_{\sigma,S}=\sqrt{\lambda}\,\sigma_S,\qquad \lambda=\frac{\beta^2 N}{M}=O(N^{-2}).$$

Thus $C_N(\sigma,\tau)=v_\sigma\cdot v_\tau$, $C_N(\sigma,\sigma)=N\beta^2$, and

$$C_N(r)=\beta^2\frac{N}{(N)_3}[(Nr)^3-(3N-2)Nr].$$

Set $W_\sigma=\exp\{X_N(\sigma)-N\beta^2/2\}$ and $L=\mathbb{E}_\sigma W_\sigma$. Throughout, $\beta=\beta_c$ is the true thermodynamic critical inverse temperature, $\mathbb{E}_a$ denotes the uniform Ising-pattern average, and $P_a g=g+v_a$ is the Gaussian planting shift. A bracket without a band subscript denotes the Gibbs expectation for L. Use the following established results:

(1) $L\longrightarrow 1/2$ in probability.
(2) The full critical Gibbs overlap vanishes: $\mathbb{E}\langle R_{12}^2\rangle\longrightarrow 0$.
(3) Positive narrow planted bands have the proved one-/two-point local Gaussian limits, empirical threshold self-averaging, and the Fourier term limit

$$Q_N(t)=\sqrt{N}\mathbb{E}_a[e^{it\sqrt{N}B_a}-\mathbf{1}_{\sqrt{N}B_a\leqslant 1}]\longrightarrow q(t)=\kappa\int_0^\infty\frac{e^{ity}-\mathbf{1}_{y\leqslant 1}}{y}\,dy\quad\text{in } L^2,\quad t\neq 0.$$

Here B_a may be the broad band because its replacement error is exponential. Put $a_N=\mathbb{P}(\sqrt{N}B_a\leqslant 1)$, or equivalently use the narrow band; then $a_N\longrightarrow 1/2$.

(4) Uniformly near zero, $\|Q_N(t)\|_2 \leqslant C(1+|\log|t||)$. The logarithmic density and oscillatory tail bounds from the smoothing lemma hold.
(5) Narrow-band magnetization localization and optimal band cross moments through order ten hold. These imply the qualitative weighted joint-density variation estimate proved in the band-derivative argument below.

The conclusion under these inputs is

$$\sqrt{N}(L-a_N) \Longrightarrow S_\kappa + \frac{\sqrt{3}\,\beta^2}{2}G, \qquad \sqrt{N}(\log L - \log a_N) \Longrightarrow 2S_\kappa + \sqrt{3}\,\beta^2 G,$$

where G is an independent standard Gaussian and S_κ has Lévy measure $\kappa y^{-2}\mathbf{1}_{\{y>0\}}\,dy$ with compensation at $y \leqslant 1$.

12.2. **Exact low-overlap expansion.** The planting identity gives

$$L(P_a g) = L + B_a + T_a + V_a + \mathcal{R}_a,$$

where

$$T_a = \mathbb{E}_\nu W_\nu C_N(\nu,a) = v_a \cdot \nabla L, \qquad V_a = \frac{1}{2}\mathbb{E}_\nu W_\nu C_N(\nu,a)^2 \geqslant 0,$$

and

$$\mathbb{E}|\mathcal{R}_a| = O(N^{-3/2}). \tag{12.1}$$

Choose a sufficiently small fixed low-overlap cutoff. On it, use $|\exp(x) - 1 - x - x^2/2| \leqslant |x|^3\exp(|x|)/6$ and $|C_N(r)| \leqslant C(N|r|^3+|r|) \leqslant cNr^2 + O(1/N)$ with $c < 1/2$. The Rademacher overlap moments give a third absolute moment of order $N^{-3/2}$. Removed large positive overlaps are the broad planted band. Large negative overlaps have nonpositive covariance and contribute exponentially small errors, even after the polynomial first two terms are subtracted. This proves (12.1).

For every deterministic Gibbs measure and every fixed integer $r \geqslant 1$, Jensen and the uniform-pattern covariance moments give

$$\mathbb{E}_a|T_a/L|^{2r} \leqslant C_r N^{-r}, \qquad \mathbb{E}_a(V_a/L)^r \leqslant C_r N^{-r}. \tag{12.2}$$

These bounds are uniform in the disorder; they do not use a critical overlap rate.

12.3. **Compact characteristic observable.** Choose a smooth compactly supported $\chi : [0,\infty) \to [0,1]$ equal to one on a fixed neighborhood of $1/2$, with support in $[0,K]$. Define

$$f_{N,t}(x) = e^{it\sqrt{N}(x-a_N)}\chi(x), \qquad \phi_{N,\chi}(t) = \mathbb{E}f_{N,t}(L).$$

By the half limit, $\phi_{N,\chi}(t) - \mathbb{E}e^{it\sqrt{N}(L-a_N)} \longrightarrow 0$ uniformly in t. Also $\phi_{N,\chi}(0) \longrightarrow 1$. Exact planting gives

$$\phi'_{N,\chi}(t) = i\sqrt{N}\{\mathbb{E}_{a,g} f_{N,t}(L(P_a g)) - a_N \phi_{N,\chi}(t)\}. \tag{12.3}$$

The error in replacing the planted argument by $L+B+T+V$ is $O_T(N^{-1/2})$ on every bounded t interval, by (12.1) and the bound $\|f'_{N,t}\|_\infty \leqslant C_T\sqrt{N}$.

At this stage replace the broad band by the positive narrow band using its exponential absolute-mass/exceptional-probability estimate and the same Lipschitz bound. The resulting error in (12.3) is exponentially small. From this point onward, B_a always denotes that

positive narrow band, so every band derivative and Gaussian projection below has the required magnetization lower bound.

No relative cutoff is needed. Cauchy–Schwarz gives the exact inequality $V_a \geqslant T_a^2/(2L)$. Consequently, for every seed and every $\theta \in [0,1]$,

$$S_{a,\theta} = L+B_a+V_a+\theta T_a \geqslant B_a+(1-\theta^2/2)L+(T_a+\theta L)^2/(2L) \geqslant B_a+L/2, \qquad 0 \leqslant \theta \leqslant 1. \tag{12.4}$$

Thus every term involving $\chi(S)$ or its derivatives has $L \leqslant 2K$ and $B_a \leqslant K$; in particular all interpolation arguments remain positive. Fix another smooth cutoff $\chi_0(L)$, equal to one for $L \leqslant 2K$ and zero for $L \geqslant 3K$; it can be inserted in all such terms without changing them.

12.4. **The nonlinear odd term is negligible.** Duhamel's identity, rather than a second-order expansion in T, gives

$$f_{N,t}(L+B+V+T) - f_{N,t}(L+B+V) = \int_0^1 T\, f'_{N,t}(S_{a,\theta})\, d\theta.$$

The part involving $\chi'(S)$ is negligible after the Palm prefactor. Indeed Parseval gives exactly

$$\mathbb{E}_a T_a^2 = \lambda \|\nabla L\|^2, \qquad \|\nabla L\|^2 = L^2 \langle C_N(R_{12})\rangle,$$

and $\mathbb{E}\langle C_N(R)\rangle/(N\beta^2) \longrightarrow 0$ under vanishing overlap. Hence

$$\sqrt{N}\mathbb{E}|T_a|\mathbf{1}_{L\leqslant 2K} = o(1). \tag{12.5}$$

For the oscillatory part define

$$F_{a,\theta} = e^{it\sqrt{N}(B_a+V_a+\theta T_a)}\chi(S_{a,\theta})\chi_0(L), \qquad \mathcal{V}_\theta = \mathbb{E}_a v_a F_{a,\theta}.$$

Gaussian divergence δ is with respect to the original disorder. Since the phase $e^{it\sqrt{N}(L-a_N)}$ has gradient $it\sqrt{N}$ times itself times ∇L,

$$-tN\mathbb{E}_{g,a} T_a e^{it\sqrt{N}(L-a_N)} F_{a,\theta} = i\sqrt{N}\mathbb{E}_g e^{it\sqrt{N}(L-a_N)}\delta\mathcal{V}_\theta. \tag{12.6}$$

We prove, uniformly in $\theta \in [0,1]$ and bounded t intervals,

$$N\mathbb{E}|\delta\mathcal{V}_\theta|^2 \longrightarrow 0. \tag{12.7}$$

12.4.1. *Parseval and matrix bounds.* Because the a_S, $|S| = 3$, are orthonormal Walsh functions,

$$\|\mathbb{E}_a v_a h_a\|^2 \leqslant \lambda \mathbb{E}_a |h_a|^2. \tag{12.8}$$

Also, for any complex h_a with $|h_a| \leqslant C$,

$$\left\|\mathbb{E}_a v_a v_a^T h_a\right\|_{\mathrm{op}} \leqslant C\lambda. \tag{12.9}$$

The latter follows from Cauchy–Schwarz applied to the two linear forms in v_a. Thus $\mathbb{E}\|\mathcal{V}_\theta\|^2 \leqslant C\lambda = O(N^{-2})$.

Use the divergence isometry and the elementary matrix inequality

$$\mathbb{E}|\delta\mathcal{V}|^2 = \mathbb{E}\|\mathcal{V}\|^2 + \mathbb{E}\sum_{i,j} \partial_j \mathcal{V}_i\, \overline{\partial_i \mathcal{V}_j} \leqslant \mathbb{E}\|\mathcal{V}\|^2 + \mathbb{E}\|D\mathcal{V}\|_{\mathrm{HS}}^2,$$

Here the displayed conjugation is required for a complex vector field; $\|\cdot\|$ and $\|\cdot\|_{\mathrm{HS}}$ are the Hermitian Euclidean and Hilbert–Schmidt norms. Split the derivative into band, T, V, and direct-L parts. A fixed finite sum of their squared Hilbert–Schmidt norms bounds the total.

For the T part, use $\nabla T_a = (\operatorname{Hess} L)v_a$ and (12.9). Its squared norm is bounded by

$$CN\lambda^2 \|\operatorname{Hess} L\|_{\mathrm{HS}}^2 = CN\lambda^2 L^2 \langle C_N(R_{12})^2\rangle \leqslant \frac{C_K}{N}\langle k_N(R_{12})^2\rangle,$$

where $k_N = C_N/(N\beta^2)$. Its expectation is $o(N^{-1})$ by vanishing overlap. All factors are supported on $L \leqslant 3K$.

For the V part, the vector-valued version of Parseval and

$$\mathbb{E}_a\|\nabla V_a\|^2 \leqslant CL^2/N$$

give a bound $CN\lambda L^2/N = O_K(N^{-2})$. To verify this estimate, expand both gradients and bound the covariance between the two Gibbs replicas by $N\beta^2$. The remaining bound is

$$\mathbb{E}_a C_N(a,\nu)^2 C_N(a,\mu)^2 \leqslant C/N^2,$$

uniformly in the two spins.

Direct derivatives of $\chi(S)$ through L, and of $\chi_0(L)$, have squared norm at most

$$C\lambda\|\nabla L\|^2 \mathbf{1}_{L\leqslant 3K} \leqslant C_K N^{-1}\langle |k_N(R_{12})|\rangle = o_{L^1}(N^{-1}).$$

12.4.2. *Band part and the slow-coordinate commutators.* This is the only part needing the band-density input. Set $U_a = \log B_a$. If all other scalar variables are held fixed, the derivative of F_a through U_a is

$$\beta_a = B_a e^{it\sqrt{N}(B_a+V_a+\theta T_a)}[it\sqrt{N}\chi(S_a) + \chi'(S_a)]\chi_0(L).$$

The squared Hilbert–Schmidt norm of the band derivative is exactly

$$\mathbb{E}_{a,b}\beta_a\overline{\beta_b}P_{ab}, \qquad P_{ab} = C_N(a,b)\langle C_N(\nu,\mu)\rangle_{a,b}. \tag{12.10}$$

Here are the weighted-density definitions and estimates explicitly; they agree with the general weighted-band density lemma. Fix a good pattern pair $|R(a,b)| \leqslant \gamma$, with $\gamma > 0$ chosen as in the two-pattern smoothing lemma, and allow the two original disorders to have any correlation $\rho \in [0,1]$. In their doubled Gaussian space, project onto the two own-pattern directions and condition on the orthogonal residual. After Gram–Schmidt normalization, the two remaining coordinates are $w \sim N(0, NI_2)$, and the band logarithms are $(u,v) = T(w)$. The smoothing lemma gives a global diffeomorphism with Jacobian $J = D_w T$ satisfying $\|J\| + \|J^{-1}\| \leqslant C$, $\det J > 0$, uniformly in the residual, patterns, and ρ. Each component of T is a logarithmic sum of exponentials with bounded two-dimensional slopes $s_{i,\nu}$. Define the conditional weighted density by

$$h_P(u,v) = \frac{\varphi_N(w(u,v))P(w(u,v))}{\det J(w(u,v))}, \qquad \varphi_N(w) = \frac{e^{-|w|^2/(2N)}}{2\pi N}. \tag{12.11}$$

All expectations involving its variation below include the residual Gaussian variables and independent patterns, restricted to the good-pair event. The Jacobian bounds make this restriction harmless for the moment estimates.

The chain rule gives the following deterministic inequality, with $\mathbb{E}_w$ denoting the conditional Gaussian expectation:

$$\begin{aligned}\int_{\mathbb{R}^2} |\partial_{uv} h_P|\,du\,dv \leqslant C\mathbb{E}_w\Bigg[&|D_w^2 P| + |D_w P|\left(\frac{|w|}{N} + \|D_w J\|\right)\\ &+ |P|\left(\frac{1}{N} + \frac{|w|^2}{N^2} + \frac{|w|}{N}\|D_w J\| + \|D_w J\|^2 + \|D_w^2 J\|\right)\Bigg].\end{aligned} \tag{12.12}$$

The analogous first-derivative bound has right side $C\mathbb{E}_w[|D_wP|+|P|(|w|/N+\|D_wJ\|)]$. Indeed $D_u = J^{-T}D_w$, $D_wJ^{-1} = -J^{-1}(D_wJ)J^{-1}$, and the Gaussian logarithmic score is $-w/N$. Differentiating (12.11) twice and using $du\,dv = (\det J)dw$ gives exactly the terms in (12.12).

Weighted density bounds with random multipliers. The two bands in the following calculation use the same original disorder, as required when expanding the squared Hilbert–Schmidt norm. Put

$$\mathcal{G} = \{|R(a,b)| \leqslant \gamma\}, \qquad \mathbb{E}_{\mathcal{G}}X = \mathbb{E}[\mathbf{1}_{\mathcal{G}}X], \qquad \|X\|_{r,\mathcal{G}} = (\mathbb{E}_{\mathcal{G}}|X|^r)^{1/r}.$$

For expressions written in the conditional coordinates, the expectation includes the residual Gaussian variables, the patterns, and $w \sim N(0, NI_2)$. Thus it is the original joint expectation after the Gaussian change of coordinates. Quantities involving J^{-1} are only defined on $\mathcal{G}$.

Let $P = C_N(a,b)\langle C_N(\nu,\mu)\rangle_{a,b}$, as in the band-derivative calculation, and define the nonnegative envelopes

$$\begin{aligned}
\mathcal{A}_1 &= |D_wP| + |P|\left(\frac{|w|}{N} + \|D_wJ\|\right),\\
\mathcal{A}_2 &= |D_w^2P| + |D_wP|\left(\frac{|w|}{N} + \|D_wJ\|\right)\\
&\quad + |P|\left(\frac{1}{N} + \frac{|w|^2}{N^2} + \frac{|w|}{N}\|D_wJ\| + \|D_wJ\|^2 + \|D_w^2J\|\right).
\end{aligned}$$

The derivative norms here are ordinary Euclidean tensor norms; their dimension is two. The same chain-rule calculation used for the unweighted total variation gives the pointwise bounds

$$\begin{aligned}
|\partial_j h_P(T(w))|\det J(w) &\leqslant C\varphi_N(w)\mathcal{A}_1(w), \qquad j = 1,2,\\
|\partial_{12} h_P(T(w))|\det J(w) &\leqslant C\varphi_N(w)\mathcal{A}_2(w).
\end{aligned} \tag{12.13}$$

Indeed, $\partial_j = (J^{-1}e_j)\cdot D_w$, $D_wJ^{-1} = -J^{-1}(D_wJ)J^{-1}$, and $D_w\log\varphi_N = -w/N$. Differentiating $\varphi_N P/\det J$, and using the uniform bounds on J and J^{-1}, proves the display before any averaging.

Consequently, for an arbitrary random measurable multiplier $M(u,v)$, which may depend on the residual disorder and on the full partition function, change of variables and Hölder give

$$\begin{aligned}
\mathbb{E}_{\mathrm{res,a,b}}\mathbf{1}_{\mathcal{G}}\int |M|\,|h_P|\,du\,dv &\leqslant \|M\circ T\|_{5,\mathcal{G}}\|P\|_{5/4,\mathcal{G}},\\
\mathbb{E}_{\mathrm{res,a,b}}\mathbf{1}_{\mathcal{G}}\int |M|\,|\partial_j h_P|\,du\,dv &\leqslant C\|M\circ T\|_{5,\mathcal{G}}\|\mathcal{A}_1\|_{5/4,\mathcal{G}},\\
\mathbb{E}_{\mathrm{res,a,b}}\mathbf{1}_{\mathcal{G}}\int |M|\,|\partial_{12} h_P|\,du\,dv &\leqslant C\mathbb{E}_{\mathcal{G}}[|M\circ T|\mathcal{A}_2].
\end{aligned} \tag{12.14}$$

In particular, the last line is at most $C\|M\|_\infty\mathbb{E}_{\mathcal{G}}\mathcal{A}_2$ for bounded M. No independence of the multiplier and the logarithmic coordinates is used. The density is kept conditional until after (12.13) is applied.

The envelope moments and their allocation. The covariance insertion has the slightly stronger bound

$$\|P\|_{2,\mathcal{G}} \leqslant \|C_N(a,b)\|_6\|\langle|C_N(\nu,\mu)|\rangle_{a,b}\|_3 \leqslant CN^{-1}. \tag{12.15}$$

The exponents satisfy $1/6 + 1/3 = 1/2$. The first norm is a uniform-pattern moment. For the ordered covariance, Jensen gives

$$\|\langle|C_N(\nu,\mu)|\rangle_{a,b}\|_3 \leqslant \beta^2 N\big(\mathbb{E}\langle|R(\nu,\mu)|^9\rangle_{a,b}\big)^{1/3} = O(N^{-1/2}).$$

For the distinct covariance, the bound $|C_N(r)| \leqslant C(N|r|^3 + |r|)$ adds at most $C(\mathbb{E}\langle|R(\nu,\mu)|^3\rangle_{a,b})^{1/3} = O(N^{-1/2})$. Thus the available cross moments through order ten suffice.

With only qualitative slope localization, boundedness of the slopes and the replica differentiation formulas give

$$\|D_w P\|_{2,\mathcal{G}} + \|D_w^2 P\|_{2,\mathcal{G}} = O(N^{-1}), \qquad \mathbb{E}_{\mathcal{G}}|D_w^2 P| = o(N^{-1}),$$

while $D_w J, D_w^2 J$ are uniformly bounded and converge to zero in every finite moment. The little-oh assertion follows by expanding the Gibbs derivatives into finitely many replica expectations: each term retains one seed covariance and one cross-band covariance and has at least one slope deviation converging to zero in probability. Their covariance products are bounded in L^2 by the same allocation as (12.15). Since $\||w|/N\|_r = O_r(N^{-1/2})$, these estimates imply

$$\|\mathcal{A}_1\|_{5/4,\mathcal{G}} = O(N^{-1}), \qquad \mathbb{E}_{\mathcal{G}}\mathcal{A}_2 = o(N^{-1}). \tag{12.16}$$

For example,

$$\left\| P\frac{|w|}{N} \right\|_{5/4,\mathcal{G}} \leqslant \|P\|_{2,\mathcal{G}} \left\| \frac{|w|}{N} \right\|_{10/3} = O(N^{-3/2}),$$

because $1/2 + 3/10 = 4/5$. All other terms in the first bound are handled either by this allocation or by bounded Jacobian derivatives. In the second bound, terms containing only Gaussian scores gain $N^{-1/2}$ or N^{-1}, while every remaining Jacobian term contains a bounded factor tending to zero in a finite moment.

For the ordered-model quantitative argument, use the previously proved optimal one-band slope moments. A first derivative of P has one centered slope, and a second derivative has two, after expanding Gibbs means with additional replicas. The allocations

$$\frac{1}{6} + \frac{1}{3} + \frac{3}{10} = \frac{4}{5}, \qquad \frac{1}{6} + \frac{1}{3} + \frac{3}{20} + \frac{3}{20} = \frac{4}{5}$$

give, respectively,

$$\|D_w P\|_{5/4,\mathcal{G}} = O(N^{-3/2}), \qquad \|D_w^2 P\|_{5/4,\mathcal{G}} = O(N^{-2}). \tag{12.17}$$

Here the seed covariance uses its sixth moment, the cross-band covariance its third moment, and the one or two slope deviations their 10/3-th or 20/3-th moments. Also

$$\|D_w J\|_{r,\mathcal{G}} = O(N^{-1}), \qquad \|D_w^2 J\|_{r,\mathcal{G}} = O(N^{-3/2})$$

for the finite orders used below. Therefore

$$\|\mathcal{A}_1\|_{5/4,\mathcal{G}} = O(N^{-3/2}), \qquad \mathbb{E}_{\mathcal{G}}\mathcal{A}_2 = O(N^{-2}). \tag{12.18}$$

For clarity, the first assertion follows from

$$\|\mathcal{A}_1\|_{5/4,\mathcal{G}} \leqslant \|D_w P\|_{5/4,\mathcal{G}} + C\|P\|_{2,\mathcal{G}} \left(\left\| \frac{|w|}{N} \right\|_{10/3} + \|D_w J\|_{10/3,\mathcal{G}} \right).$$

For the second, pair $D_w P$ in $L^{5/4}$ with the score or $D_w J$ in L^5. Pair P in L^2 with the remaining parenthesis in L^2. The latter parenthesis is $O(N^{-1})$ in L^2: use the score in L^4, $D_w J$ in L^4, and $D_w^2 J$ in L^2. These bounds require slope moments through order ten, in addition to the displayed 20/3-th moments, all within the previously stated one-band moment range.

The full-partition multipliers have direct moment bounds. These estimates do not require applying a covariance-comparison lemma to a disorder-dependent observable. For every fixed $m \geqslant 1$, both three-spin covariance conventions satisfy

$$\sup_{\nu\in\Sigma_N} \mathbb{E}_a|C_N(a,\nu)|^m \leqslant C_m N^{-m/2}. \tag{12.19}$$

This follows directly from uniform Rademacher moments, using $|C_N(r)| \leqslant C(N|r|^3 + |r|)$. Condition on the original disorder g. Its unplanted Gibbs measure is then deterministic and independent of the auxiliary uniform patterns. Consequently, for any $c_1, \dots, c_k \in \{a, b\}$, including repetitions, Jensen and Hölder give

$$\mathbb{E}_{a,b}\left\langle \prod_{j=1}^{k} |C_N(c_j,\nu)|^r \right\rangle_g \leqslant C_{k,r} N^{-kr/2}. \tag{12.20}$$

In this argument the good-pair indicator is dropped after taking absolute values. No tilted law of the full partition function is introduced.

On $\mathcal{G}$, the projection coordinates satisfy

$$D_{w_\ell} g = N^{-1} \sum_{c\in\{a,b\}} b_{\ell c} v_c, \qquad |b_{\ell c}| \leqslant C.$$

The coefficients depend on the patterns but not on w. Hence the raw-integral derivatives are exactly

$$\begin{aligned} D_{w_\ell} L &= \frac{L}{N} \sum_c b_{\ell c} \langle C_N(c,\nu)\rangle_g, \\ D_{w_\ell} T_a &= \frac{L}{N} \sum_c b_{\ell c} \langle C_N(a,\nu) C_N(c,\nu)\rangle_g, \\ D_{w_\ell} V_a &= \frac{L}{2N} \sum_c b_{\ell c} \langle C_N(a,\nu)^2 C_N(c,\nu)\rangle_g. \end{aligned}$$

Equations (12.19)– (12.20) therefore prove, for every fixed finite r,

$$\begin{aligned} \|\mathbf{1}_{\{L\leqslant 3K\}} D_w L\|_{r,\mathcal{G}} &\leqslant C_{K,r} N^{-3/2}, \\ \|\mathbf{1}_{\{L\leqslant 3K\}} D_w T_a\|_{r,\mathcal{G}} &\leqslant C_{K,r} N^{-2}, \\ \|\mathbf{1}_{\{L\leqslant 3K\}} D_w V_a\|_{r,\mathcal{G}} &\leqslant C_{K,r} N^{-5/2}. \end{aligned} \tag{12.21}$$

Replacing D_w by ∂_u or ∂_v preserves these orders because J^{-1} is uniformly bounded. The high uniform-pattern moments in this calculation are elementary moments of an independent seed; they are not additional cross-band moment assumptions.

Write

$$E_a^{\mathrm{ph}} = e^{it\sqrt{N}(B_a + V_a + \theta T_a)}, \qquad S_a = L + B_a + V_a + \theta T_a.$$

The factor $\chi_0(L)$ in F_a equals one whenever $\chi(S_a)$ is nonzero, and $\chi(S_a)\chi_0'(L) = 0$, because $S_a \geqslant B_a + L/2$. Thus the slow part of the u-derivative of F_a is exactly

$$\begin{aligned} r_a := \partial_u F_a - \beta_a = E_a^{\mathrm{ph}} \chi_0(L) \big[& it\sqrt{N}\chi(S_a)\partial_u(V_a + \theta T_a) \\ & + \chi'(S_a)\partial_u(L + V_a + \theta T_a)\big]. \end{aligned} \tag{12.22}$$

The analogous formula holds for $r_b = \partial_v F_b - \beta_b$. Since $B_a = e^u$ is independent of the coordinate v, the same formula with ∂_v in place of ∂_u gives $q_a := \partial_v F_a$; similarly put $q_b := \partial_u F_b$. All

nonzero terms in these formulas have $L \leqslant 2K$. It follows from (12.21) that, uniformly for $\theta \in [0,1]$ and $|t| \leqslant T$,

$$\|r_a\|_{r,\mathcal{G}} + \|r_b\|_{r,\mathcal{G}} + \|q_a\|_{r,\mathcal{G}} + \|q_b\|_{r,\mathcal{G}} \leqslant C_{K,T,r} N^{-3/2}, \qquad 1 \leqslant r < \infty, \tag{12.23}$$

and $|\beta_a| + |\beta_b| \leqslant C_{K,T}\sqrt{N}$.

Explicit control of all four integration-by-parts terms. Put

$$I_N = \mathbb{E}_{\mathcal{G}}[\beta_a \overline{\beta_b} P], \qquad J_N = \mathbb{E}_{\mathrm{res,a,b}} \mathbf{1}_{\mathcal{G}} \int (\partial_u F_a)(\partial_v \overline{F}_b) h_P \, du \, dv.$$

Using $\partial_u F_a = \beta_a + r_a$ and $\partial_v F_b = \beta_b + r_b$, Hölder yields the explicit comparison

$$\begin{aligned} |I_N - J_N| &\leqslant C_{K,T}\sqrt{N} \|P\|_{5/4,\mathcal{G}} (\|r_a\|_{5,\mathcal{G}} + \|r_b\|_{5,\mathcal{G}}) \\ &\quad + \|P\|_{5/4,\mathcal{G}} \|r_a\|_{10,\mathcal{G}} \|r_b\|_{10,\mathcal{G}} = O_{K,T}(N^{-2}). \end{aligned} \tag{12.24}$$

The last product uses $1/(5/4) + 1/10 + 1/10 = 1$.

The exact two-coordinate integration-by-parts identity gives

$$\begin{aligned} J_N = \mathbb{E}_{\mathrm{res,a,b}} \mathbf{1}_{\mathcal{G}} \int \big[& F_a \overline{F}_b \, \partial_{uv} h_P + q_a \overline{F}_b \, \partial_u h_P \\ & + F_a \overline{q_b} \, \partial_v h_P + q_a \overline{q_b} \, h_P \big] \, du \, dv. \end{aligned}$$

Since $|F_a|, |F_b| \leqslant 1$, (12.14) bounds these four terms by

$$\begin{aligned} |J_N| &\leqslant C \mathbb{E}_{\mathcal{G}} \mathcal{A}_2 \\ &\quad + C(\|q_a\|_{5,\mathcal{G}} + \|q_b\|_{5,\mathcal{G}}) \|\mathcal{A}_1\|_{5/4,\mathcal{G}} \\ &\quad + \|q_a\|_{10,\mathcal{G}} \|q_b\|_{10,\mathcal{G}} \|P\|_{5/4,\mathcal{G}} \\ &\leqslant C \mathbb{E}_{\mathcal{G}} \mathcal{A}_2 + C_{K,T} N^{-3/2} \|\mathcal{A}_1\|_{5/4,\mathcal{G}} + C_{K,T} N^{-4}. \end{aligned} \tag{12.25}$$

One can first integrate over compact logarithmic-coordinate boxes. The inverse map is proper with bounded inverse Jacobian; the Gaussian density and the polynomial deterministic bounds on the cutoff derivatives make the boundary terms vanish. The displayed integrable majorants then justify removal of the boxes.

Combining (12.24)– (12.25) gives the single useful estimate

$$\mathbb{E} \|D^{(B)} \mathcal{V}_\theta\|_{\mathrm{HS}}^2 \leqslant C \mathbb{E}_{\mathcal{G}} \mathcal{A}_2 + C_{K,T} N^{-3/2} \|\mathcal{A}_1\|_{5/4,\mathcal{G}} + C_{K,T} N^{-2} + C_{K,T} N^3 e^{-cN}. \tag{12.26}$$

Here $D^{(B)}$ denotes differentiation through the band logarithm alone. The last term bounds the bad-pair part in the original Hilbert–Schmidt expression: $|\beta_a \beta_b| \leqslant CN$ and $|P| \leqslant CN^2$ pointwise. Bad pairs are removed before any coordinate change or integration by parts.

The qualitative envelope bounds make (12.26) $o(N^{-1})$, as needed for the distinct model. The quantitative bounds make it $O(N^{-2})$, as used for the ordered model before applying its deterministic matrix metric. This proves the compact multiplier assertion with all dependence on the full partition function retained.

Interface with the preceding comparison estimates. Only the band inputs are obtained from the preceding field, cavity, and bounded-covariance comparisons. Their observables are deterministic functions of finitely many spins and the fixed patterns: powers of magnetization deviations, cross overlaps, and the projection slopes. Centered Gibbs slopes are expanded with additional replicas, so these comparisons remain applicable to their moment bounds. Pattern-dependent projection coefficients are uniformly bounded on $\mathcal{G}$; the comparisons can be applied conditionally on the patterns.

In contrast, no such comparison is applied to $\mathbf{1}_{\{L \leqslant 3K\}}$, F_a, or a commutator containing the full L. Their moments follow directly from (12.19)– (12.23). Equations (12.13)– (12.14) then

combine those moments with the band-density bounds by Hölder, without an independence assumption and without a moment bound on the uncut partition function.

Equations (12.5)–(12.7) therefore remove the entire nonlinear odd term from the Palm equation.

12.5. **The quadratic term and its exact coefficient.** Its pattern average is

$$\mathbb{E}_a V_a = v_N L, \qquad v_N = \frac{\lambda^2 M}{2} = \frac{\beta^4 N^2}{2M} \sim \frac{3\beta^4}{N}.$$

Write $V_a = v_N L + \widetilde{V}_a$. Expanding the squared degree-three Walsh kernel shows

$$\mathbb{E}_a(\widetilde{V}_a/L)^2 \leqslant \frac{C}{N^2}\langle R_{12}^2\rangle. \tag{12.27}$$

Here is the coefficient count. The coefficient of a fixed Walsh monomial of degree $2r$, $r = 1, 2, 3$, in $C_N(a,\nu)^2$ is

$$\lambda^2 \binom{2r}{r}\binom{N-2r}{3-r}.$$

After the factor $1/2$, its orders are respectively N^{-2}, N^{-3}, and N^{-4}. Walsh orthogonality and $(2r)!\sum_{|S|=2r} m_S^2 \leqslant N^{2r}\langle R^{2r}\rangle$ give (12.27), because $\langle R^{2r}\rangle \leqslant \langle R^2\rangle$. All coefficients in the expansion of $(\sum x_i)^{2r}$ are nonnegative after using $x_i^2 = 1$, which justifies the displayed bound.

Expand only in V, which is nonnegative. Compact support forces $L \leqslant K$ along every interpolation phase. By (12.2), its second-order error in (12.3) is $O(N^{3/2}\mathbb{E}[V^2\mathbf{1}_{L\leqslant K}]) = O(N^{-1/2})$. The derivative of the smooth cutoff contributes $O(N^{-1/2})$. Equation (12.27) and vanishing overlap give

$$N\mathbb{E}|\widetilde{V}_a|\mathbf{1}_{L\leqslant K} = o(1).$$

The surviving correction is therefore

$$-tNv_N\mathbb{E}_{g,a}Le^{it\sqrt{N}(L+B_a-a_N)}\chi(L+B_a) + o(1). \tag{12.28}$$

Since L is bounded on this support and converges in probability to $1/2$, it can be replaced by $1/2$ in (12.28).

12.6. **Removing the cutoff shift in the base band term.** The extra base term is

$$E_N(t) = \sqrt{N}\mathbb{E}_{g,a}e^{it\sqrt{N}(L+B_a-a_N)}[\chi(L+B_a) - \chi(L)].$$

Use $\chi(L+B)-\chi(L) = \int_0^B \chi'(L+s)\,ds$ and Cauchy–Schwarz, keeping the possible dependence on L:

$$|E_N(t)| \leqslant \frac{C}{\sqrt{N}}\int_0^{K\sqrt{N}} F_N(t,r)\,dr, \quad F_N(t,r) = \left\|\sqrt{N}\mathbb{E}_a e^{itY_a}\mathbf{1}_{Y_a\geqslant r}\right\|_2, \quad Y_a = \sqrt{N}B_a. \tag{12.29}$$

The oscillatory tail lemma gives $F_N(t,r) \leqslant C/(|t|r) + \mathrm{poly}(N)e^{-cN}$. For small or intermediate r, combine $Q_N(t)$ with the interval density bound to get

$$F_N(t,r) \leqslant C(1 + |\log|t|| + |\log r|) \quad (0 < r \leqslant \max(1, 1/|t|)),$$

up to an exponentially small term. Consequently, for each fixed $t \neq 0$,

$$E_N(t) = O_t(\log N/\sqrt{N}) = o(1). \tag{12.30}$$

The same split at $r = 1/|t|$ gives the uniform integrable bound

$$|E_N(t)| \leqslant C_K(1 + |\log|t||), \qquad 0 < |t| \leqslant 1. \tag{12.31}$$

For the large-range part, write $x = |t|K\sqrt{N} > 1$; its contribution is bounded by $C_K \log(x)/x$, which is uniformly bounded. For the shorter range, its length divided by $\sqrt{N}$ is at most K, so the logarithmic bound suffices.

Thus the base Palm term has precisely the usual CDF/Fourier form multiplied by $\chi(L)$. It follows that

$$\mathbb{E}_{g,a} e^{it\sqrt{N}(L+B_a-a_N)}\chi(L + B_a) = a_N\phi_{N,\chi}(t) + o(1).$$

Combining with (12.28), $a_N \longrightarrow 1/2$, and $Nv_N \longrightarrow 3\beta^4$, the quadratic correction is

$$-\frac{3\beta^4}{4}\, t\phi_{N,\chi}(t) + o(1). \tag{12.32}$$

12.7. **The limiting characteristic equation.** Equations (12.3)–(12.32) and the band/CDF inputs give, for every $t \neq 0$,

$$\phi'_{N,\chi}(t) - \left[iq(t) - \frac{3\beta^4}{4}t\right]\phi_{N,\chi}(t) \longrightarrow 0.$$

All preceding estimates are uniform on bounded t intervals except for the explicitly integrable $1 + |\log|t||$ bound. In particular $|\phi'_{N,\chi}(t)| \leqslant C(1 + |\log|t||)$ near zero, after adjusting constants. Together with $\phi_{N,\chi}(0) \longrightarrow 1$, the integral equation has the unique limit

$$\boxed{\phi(t) = \exp\left\{-\frac{3\beta^4}{8}t^2 + \kappa\int_0^\infty \frac{e^{ity} - 1 - ity\mathbf{1}_{y\leqslant 1}}{y^2}\, dy\right\}.}$$

Since the compact cutoff changes the original characteristic function by $o(1)$, Lévy's continuity theorem gives the distributional limit stated in Section 12.1. The logarithmic limit follows because $a_N \longrightarrow 1/2$.

12.8. **Why the ordered model needs an additional estimate.** The decisive Parseval constant in (12.8) is $\lambda = O(N^{-2})$. For the ordered-index p=3 model, the repeated-index linear Gaussian modes instead have a Parseval constant of order N^{-1}. The preceding argument therefore completes the distinct-index statement under its listed band inputs. The next subsection proves the additional estimates required for the ordered-index statement; a change of the variance coefficient alone would be insufficient.

12.9. **The ordered-index model and the weighted matrix norm.** This section supplies the additional argument for the exact ordered covariance $C_N(r) = N\beta^2 r^3$. Beyond the inputs already listed, use the quantitative one-band self/magnetization moments through order 64, uniformly under the finite cavity perturbations, and optimal band cross moments through order ten uniformly over disorder correlations. These are the previously established one-band moment and cross-overlap estimates. The stronger band estimates permit a quantitative version of the weighted-density calculation.

12.9.1. *A direct compactly weighted overlap estimate.* For each fixed $K < \infty$ and real $s \geqslant 1$,

$$\mathbb{E}\left[L^2\langle |R_{12}|^s\rangle \mathbf{1}_{L\leqslant K}\right] \leqslant C_{K,s}N^{-1/2}. \tag{12.33}$$

This estimate needs no maximal-band or cluster-counting argument. Let C_a^{broad} denote the unplanted mass with $R(\nu,a) \geqslant \epsilon$, and let $B_a^{\mathrm{broad}} = C_a^{\mathrm{broad}}(P_a g)$. Exact planting gives

$$\begin{aligned}\mathbb{E}[L^2\langle \mathbf{1}_{R\geqslant\epsilon}\rangle \mathbf{1}_{L\leqslant K}] &= \mathbb{E}_{g,a}[B_a^{\mathrm{broad}}(g)\mathbf{1}_{L(P_a g)\leqslant K}] \\ &\leqslant \mathbb{E}[B_a^{\mathrm{broad}}\mathbf{1}_{B_a^{\mathrm{broad}}\leqslant K}] \leqslant CK/\sqrt{N}.\end{aligned} \tag{12.34}$$

The first inequality uses only that the broad band is a positive submass of the planted partition function. For the second, the logarithm of this broad band has density at most $C/\sqrt{N}$: projection onto the own-pattern Gaussian coordinate has derivative at least $\beta\sqrt{N}\epsilon^3$. Integrating e^u over $u \leqslant \log K$ proves the bound.

On $|R| \leqslant \epsilon$, the annealed two-replica numerator is $O(N^{-s/2})$ by the sub-Gaussian overlap estimate, after choosing ϵ sufficiently small. The large negative-overlap numerator is exponentially small, since its covariance is negative. This proves (12.33). The same proof works for the distinct covariance once N is large enough that its own-pattern derivative is positive on the fixed broad band.

12.9.2. *Quantitative weighted-density estimates.* For independent patterns, the bounded band slopes satisfy the quantitative localization

$$\|s_{i,\nu} - \beta q^3 e_i\|_{L^r} = O(N^{-1/2})$$

for every required fixed moment order. The cross-pattern slopes are smaller; the displayed bound is sufficient. Centered-slope cumulant formulas yield

$$D_w J = O_{L^r}(N^{-1}), \qquad D_w^2 J = O_{L^r}(N^{-3/2}).$$

For $P = C_N(a,b)\langle C_N(\nu,\mu)\rangle_{a,b}$, the same replica differentiation formulas used earlier give

$$P = O_{L^{5/4}}(N^{-1}), \qquad D_w P = O_{L^{5/4}}(N^{-3/2}), \qquad D_w^2 P = O_{L^{5/4}}(N^{-2}). \tag{12.35}$$

A second derivative introduces two centered slopes, not a single uncentered bounded slope. The moment budget is explicit: use Hölder exponents 6 for the seed covariance, 3 for the band cross covariance, and 20/3 for each of two slope deviations. Their reciprocal sum is 4/5, which yields the $L^{5/4}$ bound. The cross covariance uses only overlap moment nine; all slope moments are well below order 64. Gaussian-score and Jacobian terms use fixed moments within the same range.

Applying (12.12) and its first-derivative version gives, uniformly over the disorder correlation,

$$\mathbb{E}\int |\partial_{uv} h_P|\, du\, dv = O(N^{-2}), \qquad \mathbb{E}\int (|\partial_u h_P| + |\partial_v h_P|)\, du\, dv = O(N^{-3/2}). \tag{12.36}$$

The slow-coordinate commutators of Section 12.4.2 retain their previous rates. Thus the unweighted squared Hilbert–Schmidt norm of the compact band derivative is $O(N^{-2})$, with an actual rate rather than only $o(N^{-1})$.

12.9.3. *A uniform deterministic-band-test vector bound.* For any deterministic measurable $H : \mathbb{R} \to \mathbb{C}$ with $|H| \leqslant 1$,

$$\boxed{\mathbb{E}\,\|\mathbb{E}_a v_a H(\log B_a)\|^2 \leqslant CN^{-2}.} \tag{12.37}$$

The constant is uniform in H, including N-dependent tests. To prove this, expand the squared norm as

$$\mathbb{E}_{a,b} C_N(a,b)\mathbb{E}_g H(U_a)\overline{H(U_b)}.$$

First separate bad pattern pairs directly in this expression; their contribution is at most $CN\exp(-cN)$, because $|H| \leqslant 1$. For good pairs, interpolate the two original disorders from independent to identical. At the independent endpoint the marginal band laws are pattern independent by gauge invariance. The initial contribution over good pairs is exactly zero: $C_N(a,b)$ is odd in $R(a,b)$, while the good event $|R(a,b)| \leqslant \gamma$ and the uniform overlap law are symmetric.

For smooth tests, the interpolation derivative is $H'(U_a)\overline{H'(U_b)}P$. Integrate by parts in both logarithmic coordinates and apply (12.36). This bounds the good-pair contribution by C/N^2. Bad pairs were already removed before differentiation, so no derivative bound on the test is needed there. Bounded smooth approximation, applied to the original squared-norm expression, extends the result to every bounded measurable test. This proves (12.37) without a correlated-pattern central limit theorem.

12.9.4. *The compact vector itself is small.* Use the same compact phase $F_{a,\theta}$ as in Section 12.4, now with the ordered covariance. Let

$$F_{a,0}^{\mathrm{base}} = e^{it\sqrt{N}B_a}\chi(L + B_a).$$

Whenever either cutoff in $F_{a,\theta} - F_{a,0}^{\mathrm{base}}$ is nonzero, positivity $S_\theta \geqslant B_a + L/2$ gives $L \leqslant 2K$. Hence

$$\mathbb{E}|F_{a,\theta} - F_{a,0}^{\mathrm{base}}|^2 \leqslant C_T N\mathbb{E}[(T_a^2 + V_a^2)\mathbf{1}_{L\leqslant 2K}] = O(N^{-1/2}). \tag{12.38}$$

For the last equality, the exact odd Walsh expansion and (12.33) give $\mathbb{E}[T_a^2\mathbf{1}_{L\leqslant 2K}] = O(N^{-3/2})$; Jensen gives the $O(N^{-2})$ bound for V_a^2 on this compact event.

The maximum feature variance is $O(N^{-1})$, so vector-valued Parseval turns (12.38) into a squared vector error $O(N^{-3/2})$. For the base vector, use

$$\chi(L + B) = \chi(L) + \int_0^K \chi'(L + s)\mathbf{1}_{B\geqslant s}\,ds.$$

Apply (12.37) to the deterministic tests $e^{it\sqrt{N}e^u}$ and $e^{it\sqrt{N}e^u}\mathbf{1}_{e^u\geqslant s}$. Bounded random factors $\chi(L)$ and $\chi'(L + s)$ do not affect their L^2 bounds, and Minkowski's inequality handles the integral. Therefore the base vector has squared norm $O(N^{-2})$, and

$$\mathbb{E}\|\mathcal{V}_\theta\|^2 = O(N^{-3/2}), \tag{12.39}$$

uniformly in θ and bounded t intervals.

12.9.5. *The weighted matrix norm.* Collect the Gaussian coordinates into the independent degree-one and degree-three Walsh blocks. Their variances are exactly

$$\lambda_1 = \frac{\beta^2(3N-2)}{N^2}, \qquad \lambda_3 = \frac{6\beta^2}{N^2}.$$

Let Σ be the diagonal matrix of these feature variances, and choose the deterministic metric

$$D = \operatorname{diag}(N^{3/4} I_{\text{linear}}, I_{\text{triples}}).$$

For every complex matrix A, invariance of the transposed-conjugate trace under diagonal similarity and Cauchy–Schwarz give

$$\left| \sum_{ij} A_{ij} \overline{A_{ji}} \right| \leqslant \|D^{-1/2} A D^{1/2}\|_{\mathrm{HS}}^2. \tag{12.40}$$

Thus the divergence isometry can use this weighted derivative norm, while retaining the ordinary vector norm (12.39).

More generally write the linear metric exponent as $\alpha \in (1/2, 1)$. For a bounded seed multiplier,

$$M_h = \mathbb{E}_a v_a v_a^T h_a = \Sigma^{1/2} Q_h \Sigma^{1/2}, \qquad \|Q_h\|_{\mathrm{op}} \leqslant C.$$

The exact kernels needed for the Hessian calculation are, with $r = R(\sigma, \tau)$,

$$\begin{aligned} K_D(r) &= v_\sigma^T D v_\tau = \beta^2 \left[N r^3 + (N^\alpha - 1)(3 - 2/N) r \right], \\ K_\Sigma(r) &= v_\sigma^T \Sigma v_\tau = \frac{\beta^4}{N} \left[(3 - 2/N)(3 - 8/N) r + 6 r^3 \right]. \end{aligned} \tag{12.41}$$

Their product has nonnegative even-power coefficients for all sufficiently large N. Also,

$$\|D^{-1/2} \Sigma^{1/2}\|_{\mathrm{op}}^2 = O(N^{-1-\alpha}).$$

The T derivative therefore has weighted squared norm at most

$$C \left[N^{-\alpha} L^2 \langle R^4 + R^6 \rangle + N^{-1} L^2 \langle R^2 + R^4 \rangle \right] \mathbf{1}_{L \leqslant 2K}. \tag{12.42}$$

To obtain (12.42), bound the left factor in $D^{-1/2} M_h (\operatorname{Hess} L) D^{1/2}$ by its operator norm, and evaluate the remaining squared Hilbert–Schmidt norm as $L^2 \langle K_\Sigma(R) K_D(R) \rangle$. Include the factor N from the oscillatory derivative.

Direct L derivatives of the cutoff are bounded similarly by

$$C \left[N^{-\alpha} L^2 \langle |R|^3 \rangle + N^{-1} L^2 \langle |R| \rangle \right] \mathbf{1}_{L \leqslant 2K}. \tag{12.43}$$

The V derivative is $O_K(N^{-1-\alpha})$: apply weighted vector-valued Parseval, note $|K_D(r)| \leqslant CN$ from its diagonal bound, and use the uniform fourth covariance moment exactly as in Section 12.4.1.

Finally the band derivative's weighted squared norm is at most N^α times its unweighted norm, hence is $O(N^{-2+\alpha})$ by (12.36) and the slow-coordinate commutator bounds. All bad-pair contributions remain exponentially small.

With $\alpha = 3/4$, (12.33), (12.39), and (12.42)–(12.43) give

$$\mathbb{E}|\delta \mathcal{V}_\theta|^2 = O(N^{-5/4}), \qquad \sqrt{N} \|\delta \mathcal{V}_\theta\|_2 = O(N^{-1/8}). \tag{12.44}$$

This proves nonlinear odd-term removal in the ordered model, uniformly in θ and bounded t intervals. The term containing the derivative $\chi'(S)$ without the oscillatory factor is $O(N^{-1/4})$ by the compact T^2 estimate used in (12.38).

12.9.6. *Ordered quadratic coefficient and final law.* For the exact kernel $C_N(r) = N\beta^2 r^3$, the quadratic mean is

$$v_N = \frac{\beta^4 N^2}{2}\mathbb{E}R^6 = \frac{15\beta^4}{2N} - \frac{15\beta^4}{N^2} + \frac{8\beta^4}{N^3}. \tag{12.45}$$

The nonconstant Walsh coefficients of V_a/L have orders N^{-2}, N^{-3}, and N^{-4} in degrees two, four, and six. Explicitly, before the common factor $\beta^4/(2N^4)$, the coefficients in $(\sum_i x_i)^6$ are

$$90N^2 - 300N + 272, \qquad 360N - 960, \qquad 720.$$

Therefore the same centered variance bound holds:

$$\mathbb{E}_a[(V_a - v_N L)/L]^2 \leqslant CN^{-2}\langle R_{12}^2\rangle.$$

The compact estimate (12.33) makes its first-order Palm error $O(N^{-1/4})$; the second-order V error remains $O(N^{-1/2})$. All cutoff and near-zero arguments in Sections 12.5–12.7 are unchanged. Since $Nv_N \longrightarrow 15\beta^4/2$, the limiting equation now has Gaussian drift $-(15\beta^4/8)t\phi(t)$, and

$$\boxed{\sqrt{N}(L - a_N) \Longrightarrow S_\kappa + \sqrt{\frac{15}{8}}\,\beta^2 G, \qquad \sqrt{N}(\log L - \log a_N) \Longrightarrow 2S_\kappa + \sqrt{\frac{15}{2}}\,\beta^2 G.}$$

The Gaussian is independent of the stable component, as identified by the product characteristic function. This ordered-index conclusion uses the additional quantitative band-moment inputs explicitly stated at the start of this section.

13. Completion of the main theorem

The scalar, field, band, local-limit, CDF, and weighted-density results above supply every input to the Palm equations. The direct $p \geqslant 5$ equation gives (1.9) with $\tau_p = 0$. The four-spin argument gives the ordered constant-mode variance and removes the auxiliary mode in the distinct model. The compact three-spin argument gives the stated quadratic-mode variances in both conventions. All estimates are uniform under the $O(N^{-1})$ coefficient changes in H_N^{std}; the finite covariance and its own band center are used throughout. Thus (1.9) holds in every case in Theorem 1.1.

Since $a_N \to 1/2$ and $\sqrt{N}(L_N - a_N)$ is tight, Taylor expansion of the logarithm on a fixed neighborhood of $1/2$ gives $\sqrt{N}(\log L_N - \log a_N) = a_N^{-1}\sqrt{N}(L_N - a_N) + o_{\mathbb{P}}(1)$. This proves (1.10). The exact annealed factors follow from the self-variances in Section 1.1. Corollary 1.2 then gives the median formulation.

Appendix A. A reproducible scalar integral certificate

For $Y \sim N(t,t)$, write φ_t for its density. The following elementary global bounds are useful:

$$\|\varphi_t\|_\infty \leqslant (2\pi t)^{-1/2}, \quad \|\varphi_t'\|_\infty \leqslant (t\sqrt{2\pi e})^{-1}, \quad \|\varphi_t''\|_\infty \leqslant (\sqrt{2\pi}\,t^{3/2})^{-1}.$$

For $f_1(x) = \tanh^2 x$, the bounds $|f_1| \leqslant 1$, $|f_1'| \leqslant 1$, $|f_1''| \leqslant 2$ hold. For $f_2(x) = \log\cosh x$, we have $|f_2'|, |f_2''| \leqslant 1$ and $|f_2(x)| \leqslant A$ on $[-A, A]$. Consequently, if M_i bound the three density derivatives above,

$$\|(f_1\varphi_t)''\|_{[-A,A]} \leqslant 2M_0 + 2M_1 + M_2, \qquad \|(f_2\varphi_t)''\|_{[-A,A]} \leqslant M_0 + 2M_1 + AM_2.$$

The following rational choices suffice:

t	A	M_0	M_1	M_2	$\|(f_1\varphi_t)''\|$	$\|(f_2\varphi_t)''\|$
2	12	.283	.122	.142	.952	2.231
4	20	.200	.061	.050	.572	1.322
6	30	.164	.041	.028	.438	1.086

Use the midpoint grid $x_j = -A + (j + 1/2)/1000$, $0 \leqslant j < 2000A$. The midpoint error of an integrand with second derivative bounded by M is at most $2AM/(24 \cdot 10^6)$. For all entries in the table it is below $2.716 \cdot 10^{-6}$. The elementary Gaussian tail formula gives $\mathbb{E}[|Y|\mathbb{1}_{\{|Y|>A\}}] < 10^{-10}$ for all three pairs (t, A). Thus $3 \cdot 10^{-6}$ bounds the sum of midpoint and tail errors for either integral.

The accompanying standard-library Python program `pure_scalar_certificate.py` evaluates each finite midpoint sum with outward-rounded decimal intervals at precision 45. It encloses π independently using the alternating rational sums in Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$, using 80 terms for each arctangent. Arithmetic interval endpoints use directed rounding; each correctly rounded decimal exponential, logarithm or square root is expanded by one adjacent decimal on either side. The program therefore accounts for arithmetic roundoff as well as the analytic quadrature error. Its output is stored in `pure_scalar_certificate.txt`.

The much wider rational enclosures used in the proof are

t	$\Gamma(t) >$	$\Psi(t) <$	consequence
2	.7689	1.5001	$A_3(2) < -.025$
4	.9313	3.3674	$A_4(4) < -.0591$
6	.9781	5.3267	$A_5(6) < -.04148$

The uniqueness and monotonicity argument in Lemma 2.2 now gives

$$q_p > .7689 \quad (p \geqslant 3), \qquad q_p > .9313 \quad (p \geqslant 4), \qquad q_p > .9781 \quad (p \geqslant 5).$$

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